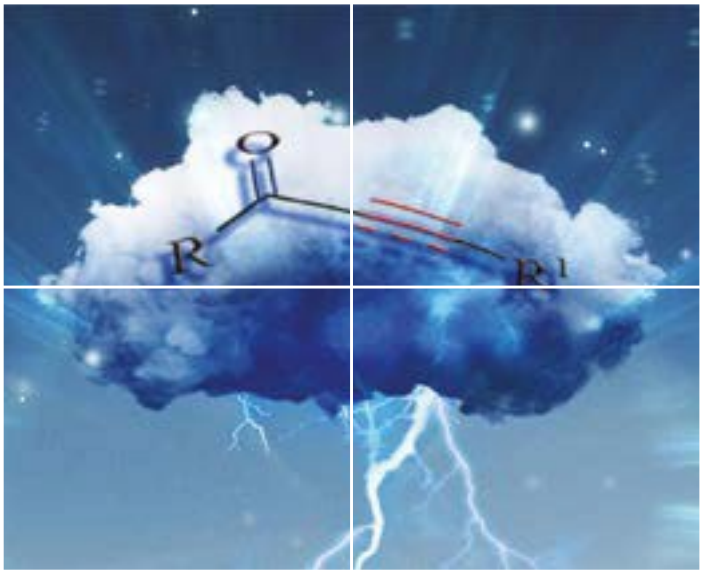
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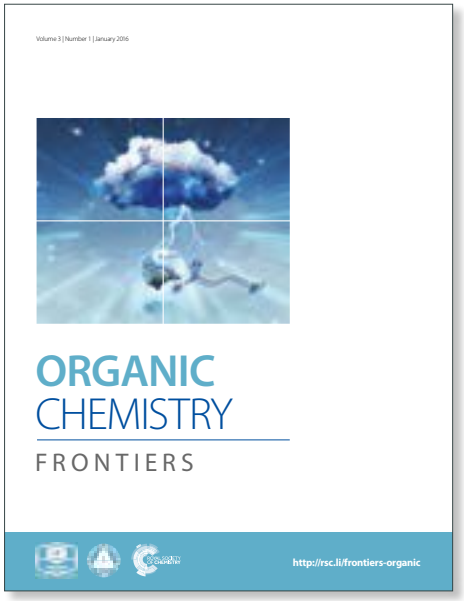
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Palladium-Catalyzed Intermolecular Dearomatic Allenylation of Hydrocycloalk[*b*]indoles with 2,3-Allenyl Carbonates

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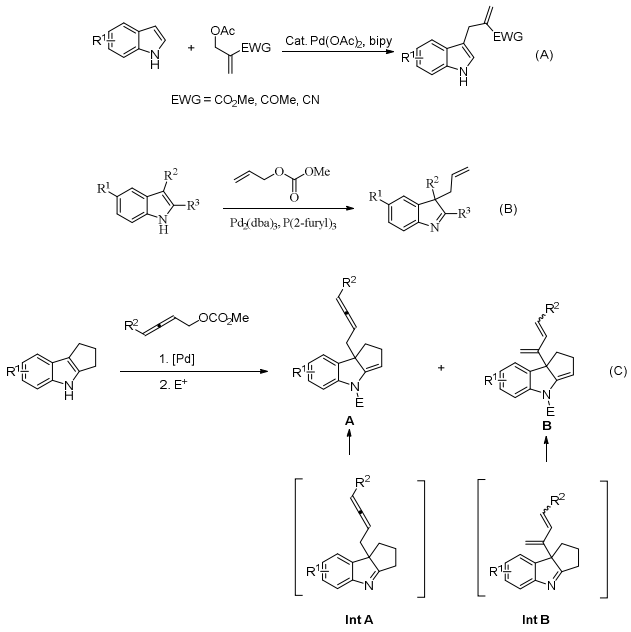
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Run-duo Gao,[a,b†] Yizhan Zhai,[a,b†] Shu-Li You,*[a,b] and Shengming Ma*[a,c]

A palladium-catalyzed intermolecular dearomatic allenylation of hydrocycloalk[*b*]indoles with 2,3-allenyl carbonates has been developed, providing an access to functionalized allenes containing an indole unit under the optimized conditions. Both terminal and non-terminal allenes could react smoothly with an exclusive chemoselectivity and good yields by applying different reaction parameters.

Indoles are attractive and fundamental structural motifs in numerous natural products and pharmaceuticals.[1] In addition, nucleophility of C-3 of indoles leads to numerous transformation.[2] In 2004, Ma's group developed a palladium-catalyzed functionalization of 2-electron-deficient group-substituted acetates with indoles (Scheme 1 A).[2e] Starting from 2008, Rawal (Scheme 1 B), [3a] Du,[3b] and You,[3c, 3d] reported that 2,3-disubstituted indoles could react with allylic carbonates in the presence of palladium catalyst. On the other hand, allenes are playing an increasingly important role in organic chemistry and medicinal chemistry.[4,5] Much attention has been paid to the reaction of 2,3-allenol derivatives with different nucleophiles.[6-8] Herein, we wish to provide an efficient method for synthesizing functionalized allenes via the dearomatization of hydrocycloalk[*b*]indoles (Scheme 1 C).



Scheme 1 (A) Palladium-catalyzed functionalization of electron-deficient alkenes with indoles. (B) Dearomatization of indoles derivatives with allylic carbonate. (C) Dearomatization of hydrocycloalk [*b*] indoles with 2,3-allenyl carbonates.

In principle, there are two types of reactivities for such coupling reaction of 2,3-allenyl carbonates, generating two different products, allenes **A** and 1,3-dienes **B**, upon the treatment with an electrophile. We initiated our study with 1,2,3,4-tetrahydrocyclopentaindole **1a**[9] and buta-2,3-dienyl methyl carbonate **2a**[10a-c] followed by an immediate trapping of the possible imine product **Int A** or **Int B** with methyl chloroformate. The formation of enamine **4aaa** was observed, albeit in <10% yield using P, N ligand **L1** or monophosphine ligand **L2**. Bisphosphines with the difference of a length of carbon chain had very limited influence on the reaction.

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Electronic supplementary information (ESI) available: Preparation and characterisation data as well as ^{1}H and ^{13}C NMR spectra of all compounds. See DOI: 10.1039/x0xx00000x

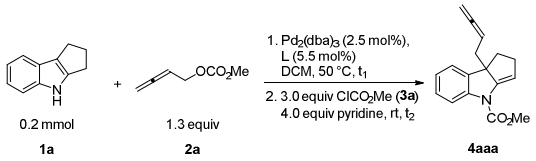
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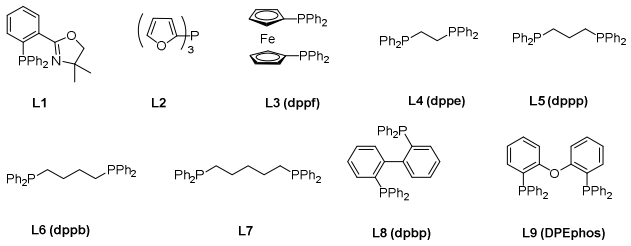
Interestingly, decent levels of reaction were observed by using commercially available dpbp **L8** and DPEphos **L9**.

Table 1. Identifying the ligand.[a]



0.2 mmol **1a** | 1.3 equiv **2a** | **4aaa**

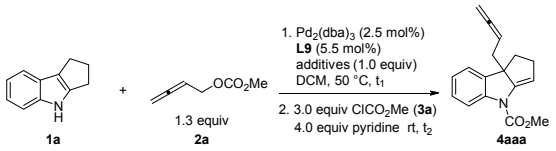
Entry	Ligand	t_1/t_2 (h)	NMR Yield of **4aaa**[b] (%)
1[c]	**L1**	6.5/10	6
2[c]	**L2**	9.7/11.3	5
3	**L3**	6.5/10	10
4	**L4**	6.5/10	NR
5	**L5**	9.7/11.3	8
6	**L6**	9.7/11.3	26
7	**L7**	9/13	8
8	**L8**	32.7/overnight	53
9	**L9**	8.7/overnight	64



[a] Reaction conditions: the reactions were carried out using 0.2 mmol of **1a**, 0.26 mmol of **2a**, 2.5 mol% of $Pd_2(dba)_3$, 5.5 mol% of **L** and 2 mL of DCM at 50 °C. After the reactions were complete, 0.8 mmol of pyridine and 0.6 mmol of $ClCO_2Me$ were added at room temperature. [b] Determined by using CH_2Br_2 as the internal standard. [c] 11 mol% of **L** was added.

Further optimization of reaction conditions indicated that increasing the amount of substrate **2a** to 2.0 equiv has no influence on the reaction. Elevating the reaction temperature even led to a worse result (entry 3, Table 2). Adding molecular sieves had slight influence on the reaction (entries 4 and 5, Table 2). Interestingly, there is a profound base effect: with *N,O*-bis-(trimethylsilyl)-acetamide (BSA) as the base, the reaction afforded the best result (entries 6-9, Table 2). By reducing the concentration, the yield could be further improved (entry 10, Table 2).

Table 2. Optimization of reaction conditions: reaction of 1,2,3,4-tetrahydro-cyclopenta[*b*]indole 1a and 2,3-butadienyl carbonate 2a.[a]



1a | 1.3 equiv **2a** | **4aaa**

Entry	Additives	t_1/t_2 (h)	Yield of **4aaa**[b] (%)
1	--	8.7/overnight	64
2[c]	--	11.5/15	64
3[d]	--	3.5/13.5	56
4[e, f]	3Å MS	8/overnight	66
5[e, f]	4Å MS	8/overnight	58
6[f]	KO^tBu	8.5/13.5	13
7[f]	K_3PO_4	7.5/15.5	61
8[f]	$MgSO_4$	9.6/12.5	59
9	BSA	6.5/15.5	83
10[g]	BSA	10/3	87 (80[h])
11[i]	BSA	6.7/16.8	83

[a] Reaction conditions: Unless otherwise specified, the reactions were carried out using 0.2 mmol of **1a**, 1.3 equiv of **2a**, 1.0 equiv of additive, 2.5 mol% of $Pd_2(dba)_3$, and 5.5 mol% of **L9** in 2.0 mL of DCM at 50 °C. After the reactions were complete, 0.8 mmol of pyridine and 0.6 mmol of $ClCO_2Me$ were added at room temperature. [b] The yield was determined by NMR analysis of the crude product. [c] The reaction was carried out with 2.0 equiv of **2a**. [d] The reaction was carried out at 80 °C. [e] 100 mg of additive was added. [f] The reactions were carried out using 0.2 mmol of **1a**, 1.3 equiv of **2a**, 1.0 equiv of additive, 2.5 mol% of $Pd_2(dba)_3$, and 5.5 mol% of **L9** in 2.0 mL of DCM at 50 °C. After the reactions were complete, the additive was filtered from the solution. 2 mL of DCM, 0.8 mmol of pyridine and 0.6 mmol of $ClCO_2Me$ were added at room temperature. [g] The reaction was carried out in 4 mL of DCM. [h] Isolated yield. [i] The reaction was carried out in 1 mL of DCM.

The scope for the electrophilic trapping reagent was also studied. In addition to chloroformate, different acyl chlorides such as

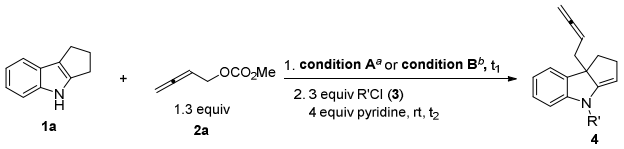
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propionyl chloride, isobutyryl chloride, acetyl chloride, chloroacetyl chloride, benzoyl chloride, cinnamoyl chloride, and benzenesulfonyl chloride may be applied (Table 3).

Table 3. The scope of the electrophilic trapping reagents for the reaction of **1a** and **2a**.

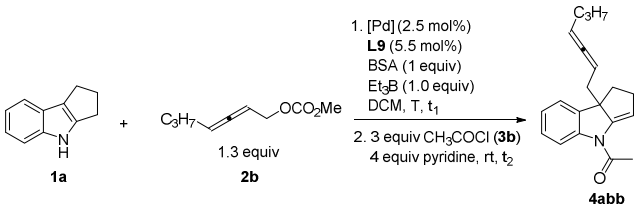


Entry	R′	Conditions (**A**/**B**)	t_1/t_2 (h)	Yield of **4**[c] (%)
1	(**3b**)	**A**	6.5/21	79 (**4aab**)
2	(**3c**)	**A**	6.3/16.8	86 (**4aac**)
3	(**3d**)	**B**	8/3	91 (**4aad**)
4	(**3e**)	**B**	8/3	88 (**4aae**)
5	(**3f**)	**A**	6.4/ 21.3	63 (**4aaf**)
6	(**3g**)	**A**	6.3/29.8	58 (**4aag**)
7	(**3h**)	**B**	8/3	58 (**4aah**)

[a] **Conditions A**: The reaction was carried out using 0.2 mmol of **1a**, 1.3 equiv of **2a**, 1.0 equiv of BSA , 2.5 mol% of $Pd_2(dba)_3$, and 5.5 mol% of **L9** in 4.0 mL of DCM at 50 °C and monitored by TLC or LC-MS. After the reaction was complete, 4.0 equiv of pyridine and 3.0 equiv of corresponding **3** were added at room temperature. [b] **Conditions B**: The reaction was carried out using 0.2 mmol of **1a**, 1.3 equiv of **2a**, 1 equiv of BSA, 1 equiv of Et_3B, 4 mol% of $Pd(acac)_2$, and 6 mol% of **L9** in 4.0 mL of DCM at room temperature and monitored by TLC or LC-MS. After the reaction was complete, 4.0 equiv of pyridine and 3.0 equiv of corresponding **3** were added at room temperature. [c] Isolated yield.

However, when non-terminal hepta-2,3-dienyl methyl carbonate **2b**[10c-d] was applied under **conditions A**, only 38% yield of **4abb** was obtained (entry 1, Table 4). Thus, we commenced to optimize the reaction conditions for the substituted allenylic carbonates. With $ZnCl_2$, no product was generated (entry 2, Table 4). There was a slight increase of yield when 1 equiv Et_3B was added since Et_3B may further activate indoles by forming N-B bond [11] (entry 3, Table 4).

Table 4. Optimization with hepta-2,3-dienyl methyl carbonte **2b** with **1a**.[a]



Entry	[Pd]	T (°C)	t_1/t_2 (h)	Yield of **4abb**[b] (%)
1[c]	$Pd_2(dba)_3$	50	9/12	38
2[d]	$Pd_2(dba)_3$	50	7.8/--	--
3	$Pd_2(dba)_3$	50	8/8.3	41
4	$Pd_2(dba)_3$	45	9/11.2	45
5[e]	$Pd_2(dba)_3$	45	9/11.2	37
6[f]	$Pd_2(dba)_3$	45	9/11.2	26
7	$Pd_2(dba)_3$	40	8/8	60
8	$Pd_2(dba)_3$	35	8/11.3	81
9	$Pd_2(dba)_3$	rt	9/11.3	91
10[g]	$Pd(OAc)_2$	rt	9/10.8	77
11[g]	$Pd(cod)Cl_2$	rt	9/11.3	68
12[g]	$Pd(acac)_2$	rt	8/12.3	92

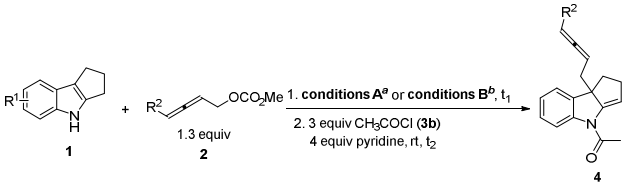
[a] Reaction conditions: the reaction was carried out using 0.2 mmol of **1a**, 1.3 equiv of **2b**, 1 equiv of BSA, 1 equiv of Et_3B, 2.5 mol% of [Pd] and 5.5 mol% of **L9** in 4.0 mL of DCM at room temperature and monitored by TLC or LC-MS. After the reaction was complete, 4 equiv of pyridine and 3 equiv of CH_3COCl (**3b**) were added at room temperature. [b] The yield was determined by NMR analysis. [c] No Et_3B was added. [d] The reaction was carried out with 2 equiv of $ZnCl_2$ instead of Et_3B. [e] 0.5 equiv of Et_3B was added. [f] 0.1 equiv of Et_3B was added. [g] 4 mol% [Pd] and 6 mol% of **L9** were added.

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Table 5. The scope for the reaction of 1,2,3,4-tetrahydrocyclopenta[*b*]indole and 2,3-allenylic carbonate derivative



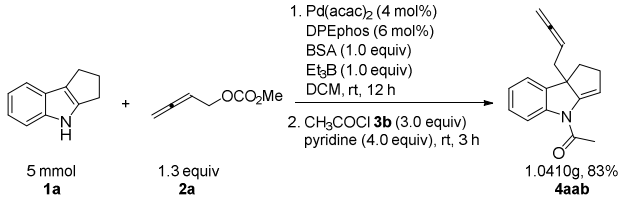
Entry	R^1	R^2	Conditions (A/B)	t_1/t_2 (h)	Yield of **4**[c] (%)	*dr*[d]
1	7-Me (**1b**)	H (**2a**)	**A**	5.6/21.8	83 (**4bab**)	--
2	6-Me, 8-Me (**1c**)	H (**2a**)	**B**	8/12.3	81 (**4cab**)	--
3	7-MeO (**1d**)	H (**2a**)	**A**	4.5/20	88 (**4dab**)	--
4	7-Ph (**1e**)	H (**2a**)	**A**	4.5/19	89 (**4eab**)	--
5	7-F (**1f**)	H (**2a**)	**A**	4.5/17.4	85 (**4fab**)	--
6	7-Cl (**1g**)	H (**2a**)	**A**	4.5/16.5	85 (**4gab**)	--
7	7-Br (**1h**)	H (**2a**)	**A**	5.3/22	78 (**4hab**)	--
8	6-Br (**1i**)	H (**2a**)	**B**	10/11.5	80 (**4iab**)	--
9	H (**1a**)	*n*-C_3H_7 (**2b**)	**B**	8/13	68 (**4abb**)	1.2/1
10	H (**1a**)	*n*-C_7H_{15} (**2c**)	**B**	10/8	60 (**4acb**)	1.4/1
11	H (**1a**)	 (**2d**)	**B**	10/13	48 (**4adb**)	1.1/1
12	H (**1a**)	 (**2e**)	**B**	8/11	68 (**4aeb**)	1.2/1
13	H (**1a**)	 (**2f**)	**B**	8/9	78 (**4afb**)	1.4/1

[a] **Conditions A**: The reaction was carried out using 0.2 mmol of **1**, 1.3 equiv of **2**, 1 equiv of BSA , 2.5 mol% of $Pd_2(dba)_3$, 5.5 mol% of **L9** in 4.0 mL of DCM at 50 °C and monitored by TLC or LC-MS. After the reaction was complete, 4.0 equiv of pyridine and 3.0 equiv of CH_3COCl (**3b**) were added at room temperature. [b] **Conditions B**: The reaction was carried out using 1.0 mmol of **1**, 1.3 equiv of **2**, 1 equiv of BSA, 1 equiv of Et_3B, 4 mol% of $Pd(acac)_2$, 6 mol% of **L9** in 20 mL of DCM at room temperature and monitored by TLC or LC-MS. After the reaction was complete, 4.0 equiv of pyridine and 3.0 equiv of CH_3COCl (**3b**) were added at room temperature. [c] Isolated yield. [d] Determined by quantitative ^{13}C NMR experiment.

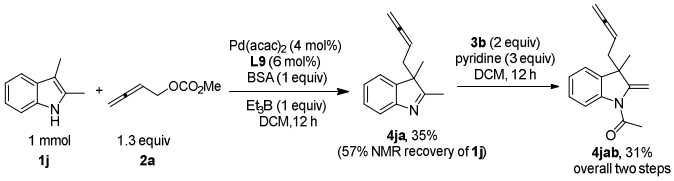
When we reduced the amount of Et_3B, the yield decreased (entries 4, 5, and 6, Table 4). To our delight, when we carried out the reaction at room temperature, 91% yield could be obtained (entry 9, Table 9). Unfortunately, the product could not be purified due to the contamination of the dba from the catalyst. Therefore other Pd catalysts were tested (entries 10, 11 and 12, Table 4). Finally, $Pd(acac)_2$, DPEphos, BSA and Et_3B in DCM at room temperature were defined as the optimized reaction conditions for further study (**conditions B**).

With conditions A and B in hand, we began to investigate the scope of this reaction. Satisfactory yields could be observed regardless of the electronic properties on the indole moiety (entries 2-9, Table 4). It is worth to note that MeO-, F-, Cl-, and Br- groups could also be tolerated. For allenylic carbonates, R^2 may be C_3H_7 and C_7H_{15}, alkynyl, vinyl and phenethyl group (entries 10 , 11, 12 , 13, and 14, Table 4).

The reaction may be easily conducted on 5 mmol scale of 1a, affording **4aab** in 83% yield.



For acyclic indoles, the dearomatization reaction could also be conducted smoothly to afford indoline with an exocyclic C-C double bond **4jab**.



Scheme 2. Dearomatic reaction of acyclic indole **1j**.

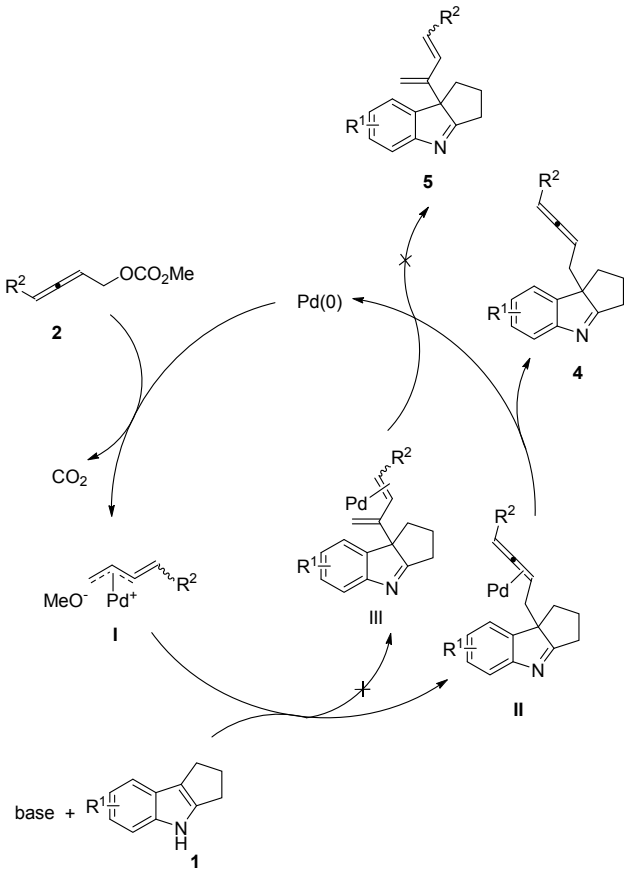
A possible mechanism is proposed. Oxidative addition of Pd (0) with 2,3-allenyl carbonates would yield, after releasing CO_2, methyleneallylic palladium intermediate **I**, which could react with the indole to yield the product **4** via intermediate **II** by releasing the catalytically active Pd(0) to finish the catalytic cycle. Since 1,3-diene product **5** was not detected in this reaction, the C3-carbon atom of the indole is acting as a soft nucelophile attacking the C1-atom in allenyl carbonates **2**.[12]

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Scheme 3. A Proposed Mechanism

In summary, a palladium-catalyzed intermolecular dearomatization of hydrocycloalk[*b*] indoles with 2,3-allenyl carbonates was first developed with a high chemoselectivity affording functionalized allene-substituted hydrocycloalk[*b*]indoles. Both mono-substituted and 1,3-disubstituted allenes could be generated smoothly with nucleophilic indoles. Further exploration of synthesizing optically active allenes with various nucleophiles is ongoing in our laboratory.

Acknowledgement

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Conflicts of interest

There are no conflicts to declare.

Notes and references

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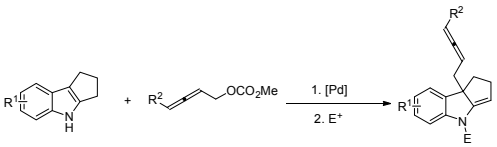
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Supporting information

Palladium-Catalyzed Intermolecular Dearomatic Allenylaion of Hydrocycloalk[*b*]indoles with 2,3-Allenyl Carbonates

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Experimental Details and Analytical Data	S3-S25
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^{1}H, ^{13}C, and ^{19}F NMR spectra of all the new compounds	S28-S86

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General Information. All reactions were carried out under the atmosphere of Ar in oven-dried Schlenk tubes. All of hydrocycloalk[*b*]indoles **1a-1i** were synthesized according to literature method.[1] 2,3-Dimethyl indole **1j** was purchased from Adamas Reagent. Terminal 2,3-butadienol was synthesized according to literature.[2] All of non-terminal alka-2,3-dienols were prepared as reported.[3] $Pd_2(dba)_3$ was synthesized according to the reported method.[4] $Pd(acac)_2$ was purchased from 9dingchem and stored in a glove box. DPEphos was purchased from Energy Chemical. BSA was purchased from Sigma-Aldrich and distilled before use. Et_3B was purchased from TCI. CH_2Cl_2 was dried over calcium hydride and distilled before use. Other reagents were used without further treatment. Petroleum ether (60 ^oC - 90 ^oC) was used for silica gel column chromatography. All the temperatures were referred to the oil baths used. Tetramethylsilane (TMS) was used as the internal standard for the ^1H NMR analysis; ^{13}C NMR experiments were measured in relative to the signal of $CDCl_3$ (77.00 ppm). Run-Duo Gao is responsible for compounds related to **Typical Procedure II**, and Yizhan Zhai is responsible for compounds related to **Typical Procedure I, III, IV,** and the preparation of **4jb**.

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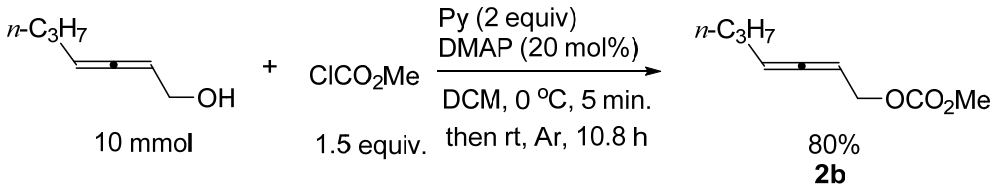
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Synthesis of methyl carbonate of alka-2,3-dienols

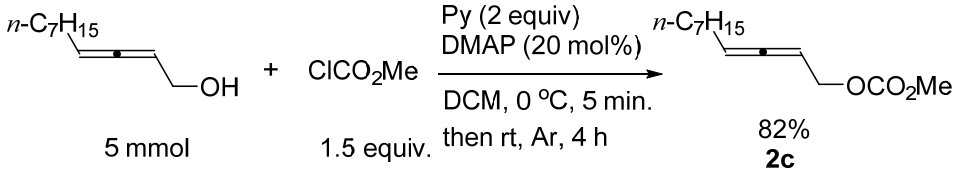
Compound **2a** was prepared according to literature.[5]

1. Synthesis of hepta-2,3-dienyl methyl carbonte (2b, zyz-2-34)



Typical Procedure I: hepta-2,3-dien-1-ol (1.1628 g, 10 mmol), DMAP (245.9 mg, 2 mmol), pyridine (1.6 mL, d = 0.9819 g/cm^3, 1.5710 g, 20 mmol), and DCM (30 mL) were added sequentially into a flask under argon at room temperature. Then methyl chloroformate (1.2 mL, d = 1.223 g/cm^3, 1.4676 g, 15 mmol) was added dropwise within 5 min at 0 ºC. The resulting mixture was allowed to stir at room temperature for 10.8 h as monitored by TLC. Upon completion, DCM (30 mL) and 1 M HCl (30 mL) were added and the organic phase was separated. The aqueous phase was extracted with DCM (30 mL × 2). The combined organic phase was washed with a saturated aqueous solution of $NaHCO_3$ (30 mL) and brine (30 mL) and dried over anhydrous Na_2SO_4. After filtration and evaporation, the residue was purified by silica gel column chromatography to afford **2b** (1.4074 g, 80%) (eluent: petroleum ether / ethyl acetate = 20/1) as a liquid: ^{1}H NMR (400 MHz, $CDCl_3$) δ 5.32-5.24 (m, 2 H, HC=C=CH), 4.64-4.55 (m, 2 H, OCH_2), 3.78 (s, 3 H, OCH_3), 2.05-1.97 (m, 2 H, CH_2), 1.44 (sext, J = 7.4 Hz, 2 H, CH_2), 0.93 (t, J = 7.2 Hz, 3 H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ 205.7, 155.5, 92.8, 86.2, 66.2, 54.6, 30.2, 22.0, 13.4; MS (DART) *m/z* 188 $(M+NH_4)^+$, 171 $(M+H)^+$; IR (neat, cm^{-1}): 2959, 2933, 2873, 1965, 1746, 1444, 1368, 1247; HRMS (DART) Calcd for $C_9H_{15}O_3$ $[(M+H)^+]$: 171.1016, Found: 171.1015.

2. Synthesis of undeca-2,3-dienyl methyl carbonate (2c, zyz-2-111)



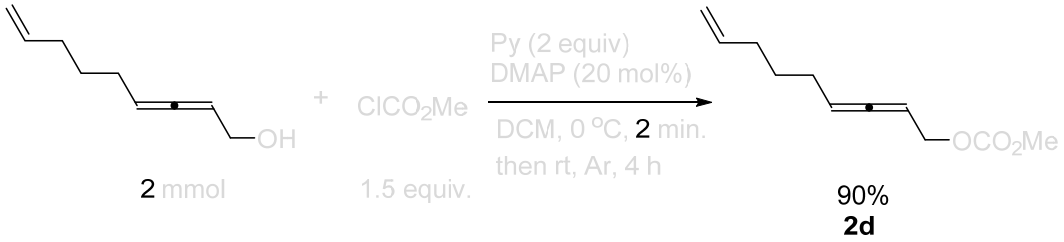
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According to **Typical Procedure I**, the reaction of undeca-2,3-dienol (821.9 mg, 5 mmol), DMAP (121.1 mg, 1 mmol), pyridine (0.85 mL, d = 0.9819 g/cm^3, 834.6 mg, 10 mmol), DCM (20 mL), and methyl chloroformate (0.6 mL, d = 1.223 g/cm^3, 733.8 mg, 7.5 mmol) afforded **2c** (908.0 mg, 82%) (eluent: petroleum ether / ethyl acetate = 10/1) as a liquid: ^{1}H NMR (400 MHz, $CDCl_3$) δ 5.33-5.22 (m, 2 H, HC=C=CH), 4.65-4.56 (m, 2 H, OCH_2), 3.79 (s, 3 H, OCH_3), 2.06-1.93 (m, 2 H, CH_2), 1.43-1.24 (m, 10 H, 5×CH_2), 0.88 (t, *J* = 6.4 Hz, 3 H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ 205.7, 155.6, 93.1, 86.4, 66.4, 54.7, 31.8, 29.04, 28.95, 28.9, 28.2, 22.6, 14.0; MS (DART) *m/z* 244 $(M+NH_4)^+$, 227 $(M+H)^+$; IR (neat, cm^{-1}): 2956, 2925, 2855, 1966, 1748, 1444, 1367, 1251; HRMS (DART) Calcd for $C_{13}H_{23}O_3$ $(M+H)^+$: 227.1642, Found: 227.1640.

3. Synthesis of nona-2,3,8-trienyl methyl carbonate (2d, zyz-2-127)

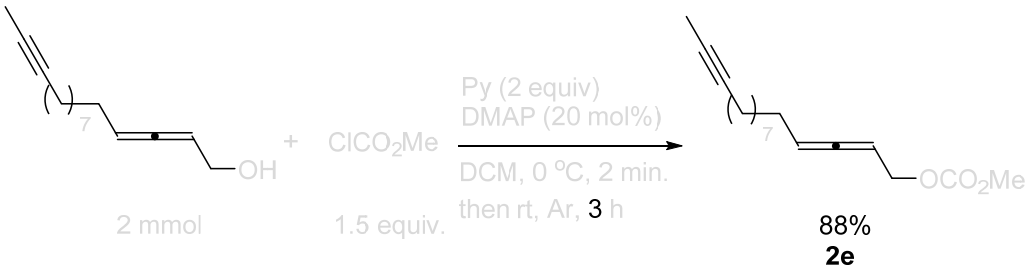


According to **Typical Procedure I**, the reaction of nona-2,3,8-trienol (278.9 mg, 2 mmol), DMAP (49.7 mg, 0.4 mmol), pyridine (0.32 mL, d = 0.9819 g/cm^3, 314.2 mg, 4 mmol), DCM (8 mL), and methyl chloroformate (0.23 mL, d = 1.223 g/cm^3, 281.3 mg, 3 mmol) afforded **2d** (354.6 mg, 90%) (eluent: petroleum ether / ethyl acetate = 10/1) as a liquid: ^{1}H NMR (400 MHz, $CDCl_3$) δ 5.87-5.72 (m, 1 H, =CH), 5.33-5.23 (m, 2 H, HC=C=CH), 5.05-4.92 (m, 2 H, $=CH_2$), 4.63-4.57 (m, 2 H, OCH_2), 3.78 (s, 3 H, OCH_3), 2.13-1.99 (m, 4 H, 2×CH_2), 1.51 (quint, *J* = 7.5 Hz, 2 H, CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 205.7, 155.5, 138.3, 114.7, 92.8, 86.5, 66.2, 54.7, 33.0, 28.0, 27.5; MS (DART) *m/z*: 214 $(M+NH_4)^+$; IR (neat, cm^{-1}): 3078, 2930, 2856, 1966, 1747, 1640, 1443,1368, 1251; HRMS (DART) Calcd for $C_{11}H_{17}O_3$ $(M+H)^+$: 197.1172, Found: 197.1172.

4. Synthesis of pentadeca-2,3-dien-13-ynyl methyl carbonate (2e, zyz-2-131)

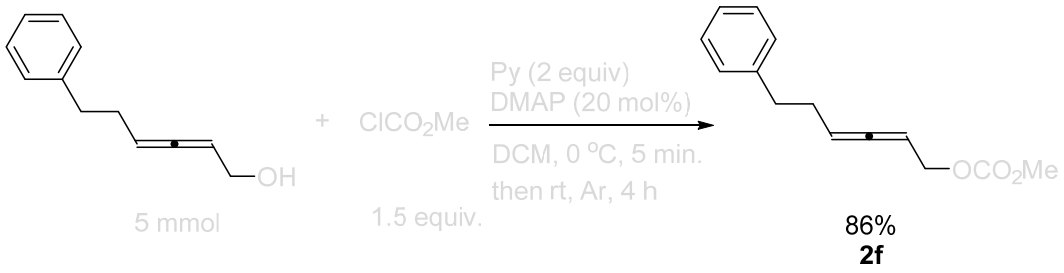
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According to **Typical Procedure I**, the reaction of pentadeca-2,3-dien-13-ynol (438.9 mg, 2 mmol), DMAP (47.1 mg, 0.4 mmol), pyridine (0.32 mL, d = 0.9819 g/cm^3, 314.2 mg, 4 mmol), DCM (8 mL), and methyl chloroformate (0.23 mL, d = 1.223 g/cm^3, 281.3 mg, 3 mmol) afforded **2e** (488.6 mg, 88%) (eluent: petroleum ether / ethyl acetate = 10/1) as a liquid: ^{1}H NMR (400 MHz, $CDCl_3$) δ 5.32-5.23 (m, 2 H, HC=C=CH), 4.63-4.57 (m, 2 H, OCH_2), 3.79 (s, 3 H, OCH_3), 2.15-2.08 (m, 2 H, CH_2), 2.06-1.98 (m, 2 H, CH_2), 1.79-1.76 (s, 3 H, CH_3), 1.51-1.24 (m, 12 H, 6×CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 205.6, 155.5, 93.0, 86.3, 79.3, 75.2, 66.3, 54.6, 29.2, 29.02, 28.97, 28.89, 28.83, 28.78, 28.1, 18.6, 3.4; MS (DART) *m/z* 296 $(M+NH_4)^+$, 279 $(M+H)^+$; IR (neat, cm^{-1}): 2925, 2854, 1965, 1748, 1443, 1367, 1252; HRMS (DART) Calcd for $C_{17}H_{27}O_3$ $(M+H)^+$: 279.1955, Found: 279.1953.

5. Synthesis of 6-phenylhexa-2,3-dienyl methyl carbonate (2f, zyz-2-112)



According to **Typical Procedure I**, the reaction of 6-phenylhexa-2,3-dienol (860.1 mg, 5 mmol), DMAP (120.1 mg, 1 mmol), pyridine (0.85 mL, d = 0.9819 g/cm^3, 834.6 mg, 10 mmol), DCM (20 mL), and methyl chloroformate (0.6 mL, d = 1.223 g/cm^3, 733.8 mg, 7.5 mmol) afforded **2f** (974.9 mg, 86%) (eluent: petroleum ether / ethyl acetate = 10/1) as a liquid: ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.31-7.24 (m, 2 H, Ar-H), 7.22-7.15 (m, 3 H, Ar-H), 5.36-5.24 (m, 2 H, HC=C=CH), 4.53 (dd, J_1 = 6.8 Hz, J_2 = 2.0 Hz, 2 H, OCH_2), 3.78 (s, 3 H, OCH_3), 2.73 (t, J = 7.6 Hz, 2 H, CH_2), 2.39-2.29 (m, 2 H, CH_2); ^{13}C

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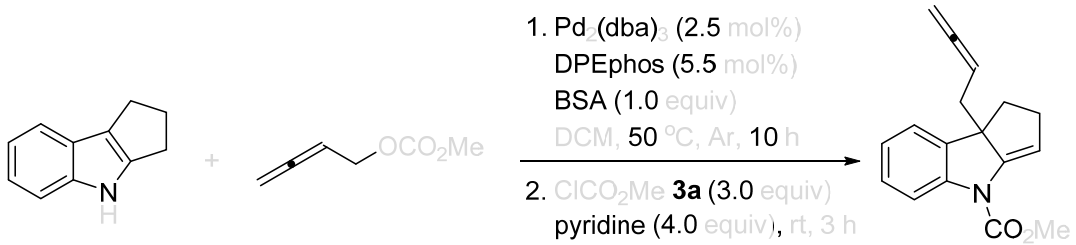
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NMR (100 MHz, $CDCl_3$) δ 205.8, 155.5, 141.3, 128.4, 128.3, 125.9, 92.3, 86.9, 66.1, 54.7, 35.1, 29.8; MS (DART) *m/z* 250 $(M+NH_4)^+$; IR (neat, cm^{-1}): 3027, 2954, 2854, 1966, 1745, 1444, 1367, 1251; HRMS (DART) Calcd for $C_{14}H_{20}O_3N$ $(M+NH_4)^+$: 250.1438, Found: 250.1436.

Palladium-catalyzed reaction of hydrocycloalk[*b*]indoles with 2,3-allenyl carbonates

1. Methyl 8b-(buta-2,3-dienyl)-2,8b-dihydrocyclopenta[*b*]indole-4(*1H*)-carbonate (4aaa, zyz-3-21)



0.2 mmol **1a** | 1.3 equiv **2a** | 80% **4aaa**

Typical Procedure II: To an oven dried Schlenk tube equipped with a polytetrafluoroethylene plug were added $Pd_2(dba)_3$ (4.6 mg, 0.005 mmol), DPEphos (5.9 mg, 0.011 mmol), and DCM (2.0 mL) sequentially under the atmosphere of Ar. The resulting mixture was stirred at room temperature for 30 min. Compound **1a** (31.3 mg, 0.2 mmol), BSA (40.6 mg, 0.2 mmol), **2a** (33.7 mg, 0.26 mmol), and DCM (2 mL) were then added sequentially. The Schlenk tube was then sealed by screwing the polytetrafluoroethylene plug tightly and stirred at 50 ºC for 10 h. After the reaction was complete as monitored by TLC, the resulting mixture was cooled to room temperature. $ClCO_2Me$ (47 μL, d = 1.223 g/cm^3, 59.9 mg, 0.6 mmol) and pyridine (63 μL, d = 0.9819 g/cm^3, 61.9 mg, 0.8 mmol) were added and the resulting mixture was stirred at room temperature for 3 h as monitored by TLC. Then a saturated aqueous solution of $NaHCO_3$ (5 mL) was added and the resulting mixture was extracted with DCM (3×10 mL). The combined organic layer was washed with brine (10 mL) and dried over anhydrous Na_2SO_4. After filtration and evaporation, the residue was purified by silica gel column chromatography to afford **4aaa** (42.5 mg, 80%) (eluent: petroleum ether / ethyl acetate = 80/1) as an oil: ^1H NMR (400 MHz, $CDCl_3$) δ 7.76 (brs, 1 H, Ar-H), 7.20-7.13 (m, 1 H, Ar-H), 7.08 (d, *J* = 7.6 Hz, 1 H, Ar-H), 6.96 (t, *J* = 7.4 Hz, 1 H, Ar-H), 5.40 (brs, 1 H,

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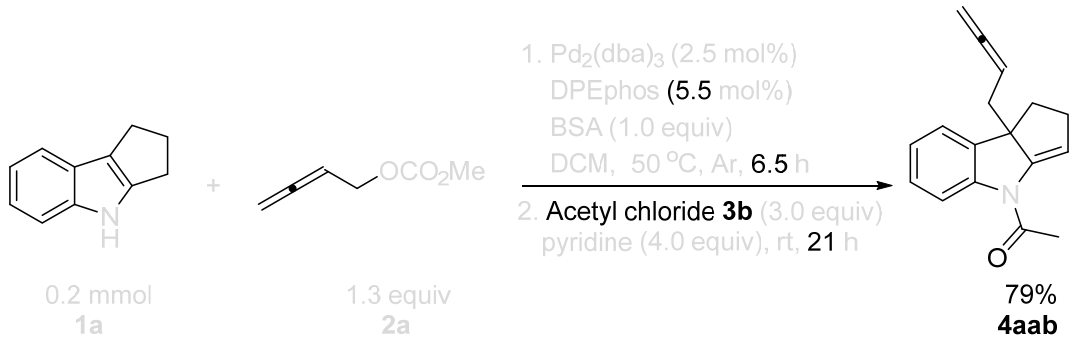
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=CH), 4.92-4.80 (m, 1 H, =CH), 4.46 (dt, J_1 = 6.4 Hz, J_2 = 2.2 Hz, 2 H, =CH_2), 3.85 (s, 3 H, OCH_3), 2.81-2.70 (m, 1 H, one proton of CH_2), 2.46-2.35 (m, 1 H, one proton of CH_2), 2.25-2.13 (m, 2 H, CH_2), 2.11-2.03 (m, 1 H, one proton of CH_2), 1.99-1.89 (m, 1 H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 210.1, 152.9, 149.8, 144.3, 135.4, 127.7, 123.7, 123.2, 115.0, 106.8, 85.3, 73.9, 58.5, 52.9, 37.1, 35.0, 32.4; MS (ESI) *m/z* 290 $(M+Na)^+$, 268 $(M+H)^+$; IR (neat, cm^{-1}): 2952, 2932, 2849, 1953, 1718, 1668, 1605, 1475, 1458, 1439, 1367, 1302, 1265, 1222, 1095. HRMS (ESI) Calcd for $C_{17}H_{18}NO_2$ $[(M+H)^+]$: 268.1331, Found: 268.1330.

2. 4-Acetyl-8b-(2,3-butadienyl)-2,8b-dihydrocyclopenta[*b*]indole (4aab, grd-17-84)



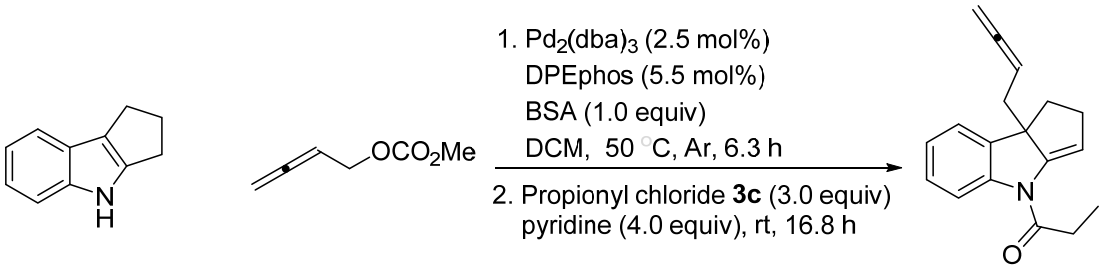
According to **Typical Procedure II**, the reaction of $Pd_2(dba)_3$ (4.6 mg, 0.005 mmol), DPEphos (6.0 mg, 0.011 mmol), DCM (2 mL), **1a** (31.5 mg, 0.2 mmol), **2a** (33.4 mg, 0.26 mmol), BSA (40.7 mg, 0.2 mmol), DCM (2 mL), pyridine (65 μL, d = 0.9819 g/cm^3, 63.8 mg, 0.8 mmol), and **3b** (43 μL, d = 1.104 g/cm^3, 47.5 mg, 0.6 mmol) afforded **4aab** (39.4 mg, 79%) (eluent: petroleum ether / ethyl acetate = 60/1) as an oil: ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.16 (brs, 1 H, Ar-H), 7.27-7.20 (m, 1 H, Ar-H), 7.16 (dd, J_1 = 7.4 Hz , J_2 = 1.0 Hz, 1 H, Ar-H), 7.07 (td, 1 H, J_1 = 7.4 Hz, J_2 = 1.1 Hz, Ar-H), 5.30 (brs, 1 H, =CH), 4.93-4.83 (m, 1 H, =CH), 4.56 (dt, J_1 = 6.8 Hz, J_2 = 2.3 Hz, 2 H, =CH_2), 2.90-2.79 (m, 1 H, one proton of CH_2), 2.55-2.47 (m, 1 H, one proton of CH_2), 2.43 (s, 3 H, CH_3), 2.32-2.26 (m, 1 H, one proton of CH_2), 2.25-2.19 (m, 2 H, CH_2), 2.09-1.99 (m, 1 H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) 209.9, 168.6, 151.3, 144.8, 135.9, 127.7, 124.0, 123.1, 117.1, 109.2, 85.2, 74.1, 59.1, 37.2, 35.0, 32.4, 24.6; IR (neat, cm^{-1}): 2931, 1953, 1675, 1602, 1455, 1376, 1350, 1284, 1205, 1106, 1014; HRMS (ESI) Calcd for $C_{17}H_{18}NO$ $([M+H]^+)$: 252.1383, Found: 252.1384.

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3. 8b-(2,3-Butadienyl)-4-propionyl-2,8b-dihydrocyclopenta[*b*]indole (4aac, grd-17-83-2)



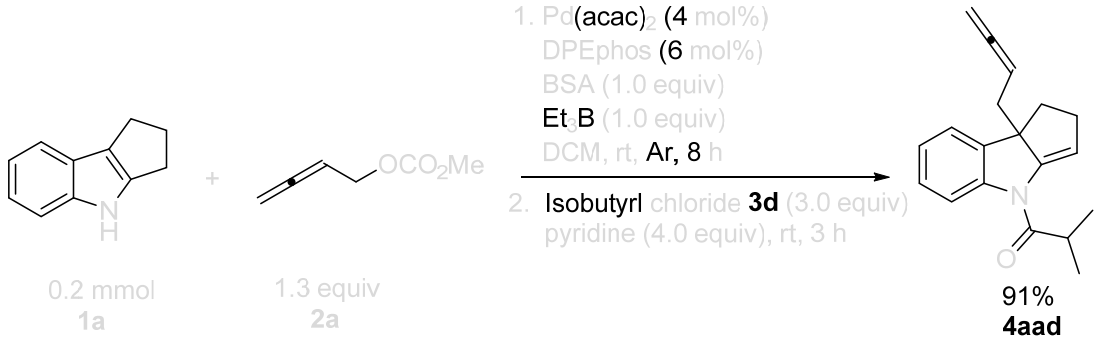
0.2 mmol **1a** | 1.3 equiv **2a** | 86% **4aac**

According to **Typical Procedure II**, the reaction of $Pd_2(dba)_3$ (4.6 mg, 0.005 mmol), DPEphos (6.0 mg, 0.011 mmol), DCM (2 mL), **1a** (31.5 mg, 0.2 mmol), **2a** (33.4 mg, 0.26 mmol), BSA (40.7 mg, 0.2 mmol), DCM (2 mL), pyridine (65 μL, d = 0.9819 g/cm^3, 63.8 mg, 0.8 mmol), and **3c** (53 μL, d = 1.065 g/cm^3, 56.4 mg, 0.6 mmol) afforded **4aac** (45.7 mg, 86%) (eluent: petroleum ether/ethyl acetate = 40/1): ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.18 (d, *J* = 7.6 Hz, 1 H, Ar-H), 7.24 (t, *J* = 7.4 Hz, 1 H, Ar-H), 7.16 (d, *J* = 7.4 Hz, 1 H, Ar-H), 7.06 (t, *J* = 7.4 Hz, 1 H, Ar-H), 5.31 (d, *J* = 2.4 Hz, 1 H, =CH), 4.93-4.84 (m, 1 H, =CH), 4.60-4.50 (m, 2 H, =CH_2), 2.90-2.61 (m, 3 H, one proton of CH_2 and another CH_2), 2.56-2.44 (m, 1 H, one proton of CH_2), 2.33-2.18 (m, 3 H, CH_2 and another one proton of CH_2), 2.08-1.98 (m, 1 H, one proton of CH_2), 1.25 (t, *J* = 7.2 Hz, 3 H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ 209.9, 172.3, 150.9, 145.0, 135.9, 127.7, 123.9, 123.1, 117.1, 109.1, 85.2, 74.0, 59.2, 37.1, 34.8, 32.4, 29.5, 8.8; IR (neat, cm^{-1}): 2984, 2942, 2851, 1954, 1675, 1600, 1453, 1376, 1325, 1258, 1197, 1169, 1104, 1072, 1020; HRMS (ESI) Calcd. for $C_{18}H_{20}NO$ ($[M+H]^+$): 266.1539, Found: 266.1541.

4. 8b-(2,3-Butadienyl)-4-(2-methyl propionyl)-2,8b-dihydrocyclopenta[*b*]indole (4aad, zyz-2-158)

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Typical Procedure III: To an oven dried Schlenk tube were added $Pd(acac)_2$ (2.4 mg, 0.008 mmol), DPEphos (5.9 mg, 0.012 mmol), and DCM (2 mL) sequentially under the atmosphere of Ar. The resulting mixture was stirred at room temperature 30 min. Compound **1a** (31.7 mg, 0.2 mmol), BSA (41.7 mg, 0.2 mmol)/DCM (1 mL), **2a** (33.7 mg, 0.26 mmol)/DCM (1 mL), and Et_3B (1M in THF, 0.2 mL, 0.2 mmol) were then added sequentially and the resulting mixture stirred at room temperature for 8 h. After the reaction was complete as monitored by TLC, pyridine (64 μL, d = 0.9819 g/cm³, 62.8 mg, 0.8 mmol) and **3d** (62 μL, d = 1.107 g/cm^3, 68.6 mg, 0.6 mmol) were added. The resulting mixture was stirred at room temperature for 3 h as monitored by TLC. A saturated aqueous solution of $NaHCO_3$ (15 mL) was added and the resulting mixture was extracted with DCM (15 mL×3). The combined organic layer was washed with brine (10 mL) and dried over anhydrous Na_2SO_4. After filtration and evaporation, the residue was purified by silica gel column chromatography to afford **4aad** (51.5 mg, 91%) (eluent: petroleum ether / ethyl acetate = 30/1) as an oil: ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.20 (d, *J* = 7.6 Hz, 1 H, Ar-H), 7.24 (t, *J* = 8.0 Hz, 1 H, Ar-H), 7.16 (d, *J* = 7.6 Hz, 1 H, Ar-H), 7.07 (t, *J* = 7.4 Hz, 1 H, Ar-H), 5.36 (brs, 1 H, =CH), 4.90 (quint, *J* = 7.3 Hz, 1 H, =CH), 4.59-4.51 (m, 2 H, $=CH_2$), 3.28 (sep, *J* = 6.7 Hz, 1 H, CH), 2.91-2.79 (m, 1 H, one proton of CH_2), 2.57-2.46 (m, 1 H, one proton of CH_2), 2.34-2.16 (m, 3 H, CH_2 and one proton of another CH_2), 2.09-1.99 (m, 1 H, one proton of CH_2), 1.28 (d, *J* = 6.8 Hz, 3 H, CH_3), 1.23 (d, *J* = 6.8 Hz, 3 H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ 210.0, 175.9, 150.8, 145.0, 136.3, 127.7, 124.0, 123.0, 117.5, 109.1, 85.3, 74.1, 59.2, 37.1, 34.8, 33.0, 32.2, 19.5, 19.2; MS (ESI) *m/z* 581 $(2M+Na)^+$, 302 $(M+Na)^+$, 280 $(M+H)^+$; IR (neat, cm^{-1}): 3070, 2966, 2933, 2852, 1954, 1672, 1602, 1472, 1456, 1389, 1362, 1259, 1193, 1108; HRMS (ESI) Calcd. for $C_{19}H_{22}NO$ ($[M+H]^+$): 280.1696, Found: 280.1693.

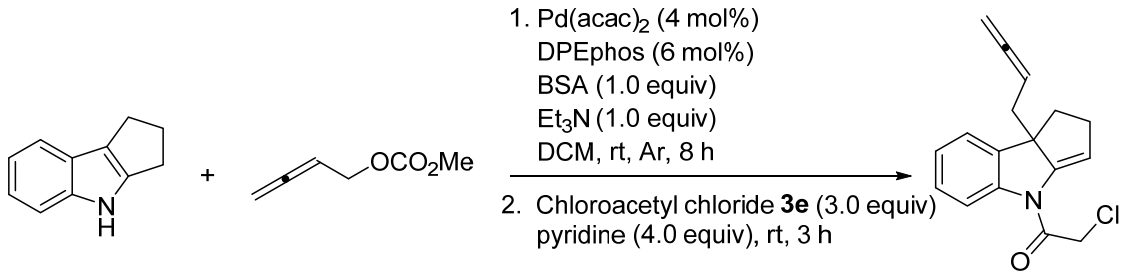
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5. 8b-(2,3-Butadienyl)-4-chloroacetyl-2,8b-dihydrocyclopenta[*b*]indole (4aae, zyz-2-155C)



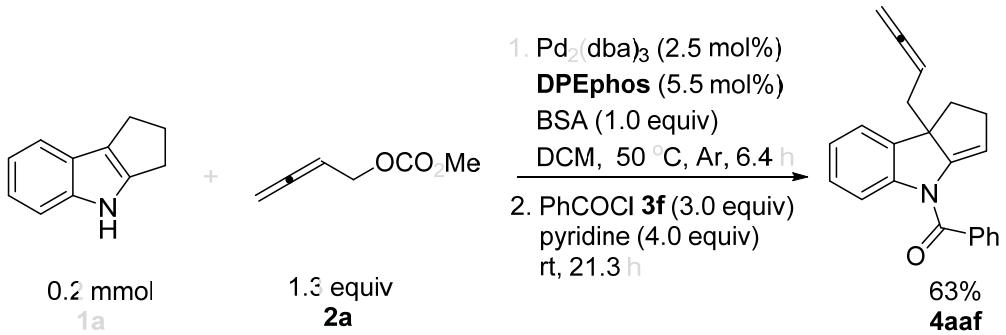
0.2 mmol **1a** + 1.3 equiv **2a** → 88% **4aae**

According to **Typical Procedure III**, the reaction of $Pd(acac)_2$ (4.6 mg, 0.008 mmol), DPEphos (5.9 mg, 0.012 mmol), DCM (2 mL), **1a** (30.9 mg, 0.2 mmol), BSA (40.7 mg, 0.2 mmol)/DCM (1 mL), **2a** (33.1 mg, 0.26 mmol)/DCM (1 mL), Et_3B (1M in THF, 0.2 mL, 0.2 mmol), pyridine (64 μL, d = 0.9819 g/cm^3, 62.8 mg, 0.8 mmol), and **3e** (48 μL, d = 1.419 g/cm^3, 68.1 mg, 0.6 mmol) afforded **4aae** (49.3 mg, 88%) (eluent: petroleum ether/ethyl acetate = 30/1) as an oil: ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.16 (d, *J* = 7.2 Hz, 1 H, Ar-H), 7.27 (t, *J* = 7.8 Hz, 1 H, Ar-H), 7.18 (d, *J* = 7.2 Hz, 1 H, Ar-H), 7.12 (t, *J* = 7.4 Hz, 1 H, Ar-H), 5.50 (brs, 1 H, =CH), 4.88 (quint, *J* = 7.2 Hz, 1 H, =CH), 4.57 (d, *J* = 6.4 Hz, 2 H, $=CH_2$), 4.45 (d, *J* = 13.6 Hz, 1 H, one proton of CH_2Cl), 4.41 (d, *J* = 13.2 Hz, 1 H, one proton of CH_2Cl), 2.94-2.80 (m, 1 H, one proton of CH_2), 2.60-2.48 (m, 1 H, one proton of CH_2), 2.31 (dd, J_1 = 12.0, J_2 = 6.0 Hz, 1 H, one proton of CH_2), 2.25-2.23 (m, 2 H, CH_2), 2.07 (dd, J_1 = 21.2 Hz, J_2 = 9.6 Hz, 1 H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 209.9, 163.8, 150.0, 144.2, 136.2, 127.9, 124.9, 123.2, 117.4, 110.7, 85.0, 74.2, 59.2, 42.8, 37.2, 35.0, 32.4; MS (ESI) *m/z* 308 $(M(^{35}Cl)+Na)^+$, 288 $(M(^{37}Cl)+H)^+$, 286 $(M(^{35}Cl)+H)^+$, 288 $(M(^{37}Cl)+H)^+$; IR (neat, cm^{-1}): 2936, 2852, 1953, 1672, 1602, 1473, 1456, 1379, 1314, 1260, 1195, 1171, 1229, 1107, 1080, 1019; HRMS (ESI) Calcd for $C_{17}H_{17}{}^{35}ClNO$ $([M+H]^+)$: 286.0993, Found: 286.0993.

6. 8b-(2,3-Butadienyl)-4-benzoyl-2,8b-dihydrocyclopenta[*b*]indole (4aaf, grd-17-79-2)

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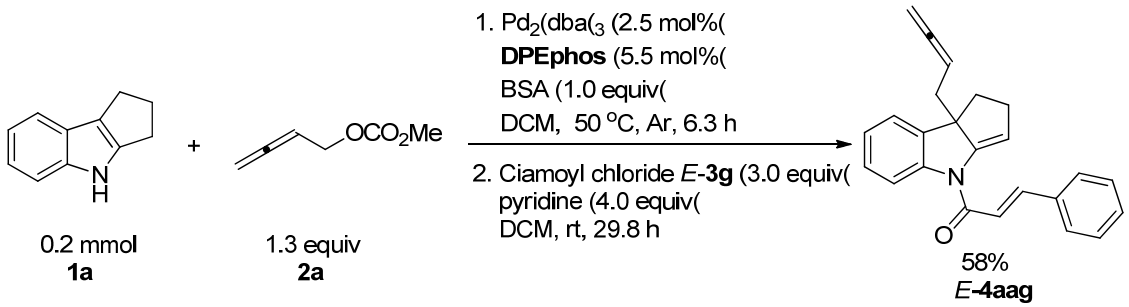
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0.2 mmol **1a** | 1.3 equiv **2a** | 63% **4aaf**

According to **Typical Procedure II**, the reaction of $Pd_2(dba)_3$ (4.6 mg, 0.005 mmol), DPEphos (6.0 mg, 0.011 mmol), DCM (2 mL), **1a** (31.5 mg, 0.2 mmol), BSA (40.7 mg, 0.2 mmol), **2a** (33.4 mg, 0.26 mmol), DCM (2 mL), pyridine (65 μL, d = 0.9819 g/cm^3, 63.8 mg, 0.8 mmol), and **3f** (70 μL, d = 1.211 g/cm^3, 84.8 mg, 0.6 mmol) afforded **4aaf** (39.2 mg, 63%) (eluent: petroleum ether/ethyl acetate = 50/1): ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.06 (d, J = 4.8 Hz, 1 H, Ar-H), 7.71-7.64 (m, 2 H, Ar-H), 7.54-7.46 (m, 1 H, Ar-H), 7.46-7.39 (m, 2 H, Ar-H), 7.28-7.22 (m, 1 H, Ar-H), 7.22-7.18 (m, 1 H, Ar-H), 7.13-7.08 (m, 1 H, Ar-H), 5.06-4.95 (m, 1 H, =CH), 4.68-4.58 (m, 2 H, =CH_2), 4.45 (brs, 1 H, =CH), 2.72-2.60 (m, 1 H, one proton of CH_2), 2.38 (dt, J_1 = 8.0, J_2 = 2.4 Hz, 2 H, CH_2), 2.35-2.23 (m, 2 H, one proton of CH_2 and one proton of another CH_2), 2.06-1.96 (m, 1 H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 210.0, 168.6, 151.3, 145.2, 136.3, 136.0, 131.0, 128.25, 128.16, 127.7, 124.3, 123.4, 116.8, 109.8, 85.7, 74.3, 58.8, 37.5, 35.2, 32.2; IR (neat, cm^{-1}): 3062, 2931, 2851, 1953, 1659, 1600, 1454, 1362, 1306, 1174, 1106, 1053, 1024; HRMS (ESI) Calcd for $C_{22}H_{20}NO$ ([M+H]$^+$): 314.1539, Found: 314.1540.

7. 8b-(2,3-Butadienyl)-4-cinnamoyl-2,8b-dihydrocyclopenta[*b*]indole (4aag, grd-17-83-3)



0.2 mmol **1a** | 1.3 equiv **2a** | 58% ***E*-4aag**

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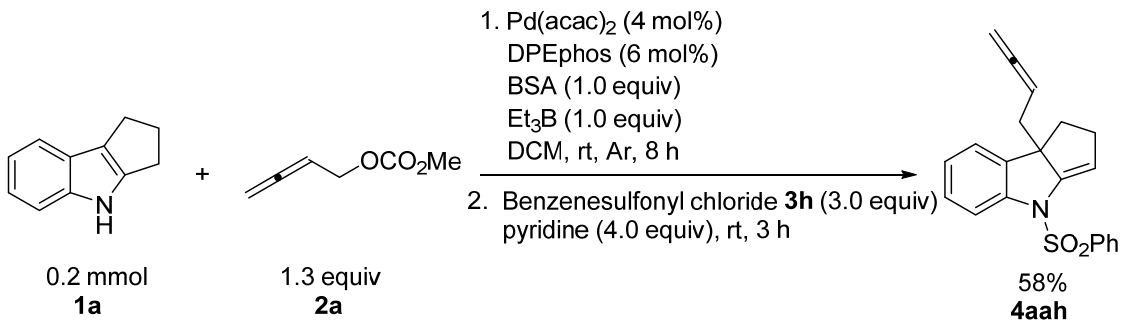
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According to **Typical Procedure II**, the reaction of $Pd_2(dba)_3$ (4.6 mg, 0.005 mmol), DPEphos (6.0 mg, 0.011 mmol), DCM (2 mL), **1a** (31.5 mg, 0.2 mmol), **2a** (33.4 mg, 0.26 mmol), BSA (40.7 mg, 0.2 mmol), DCM (2 mL), pyridine (65 μL, d = 0.9819 g/cm^3, 63.8 mg, 0.8 mmol), and *E*-**3g** (100 mg, 0.6 mmol) afforded *E*-**4aag** (39.6 mg, 58%) (eluent: petroleum ether/ethyl acetate = 50/1) as an oil: ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.32 (d, *J* = 7.6 Hz, 1 H, Ar-H), 7.83 (d, *J* = 15.6 Hz, 1 H, =CH), 7.62-7.56 (m, 2 H, Ar-H), 7.46-7.38 (m, 3 H, Ar-H), 7.32-7.22 (m, 2 H, Ar-H and =CH), 7.20 (d, *J* = 6.8 Hz, 1 H, =CH), 7.11 (t, *J* = 7.2 Hz, 1 H, Ar-H), 5.41 (d, *J* = 2.4 Hz, 1 H, =CH), 4.98-4.88 (m, 1 H, =CH), 4.62-4.50 (m, 2 H, =CH_2), 2.97-2.84 (m, 1 H, one proton of CH_2), 2.60-2.48 (m, 1 H, one proton of CH_2), 2.39-2.27 (m, 3 H, CH_2 and one pronton of another CH_2), 2.16-2.03 (m, 1 H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 210.0, 164.3, 151.2, 144.9, 143.2, 136.3, 134.9, 130.0, 128.9, 128.1, 127.9, 124.3, 123.2, 119.6, 117.3, 111.0, 85.2, 74.2, 58.8, 37.2, 35.3, 32.2; IR (neat, cm^{-1}): 3063, 2929, 1953, 1661, 1619, 1453, 1367, 1284, 1257, 1198, 1167, 1107, 1078; HRMS (ESI) Calcd for $C_{24}H_{22}NO$ ($[M+H]^+$): 340.1696, Found: 340.1700.

8. 8b-(2,3-Butadienyl)-4-(phenylsulfonyl)-2,8b-dihydrocyclopenta[*b*]indole (4aah, zyz-2-155B)



0.2 mmol **1a** + 1.3 equiv **2a** → 58% **4aah**

According to **Typical Procedure III**, the reaction of $Pd(acac)_2$ (2.4 mg, 0.008 mmol), DPEphos (6.0 mg, 0.012 mmol), DCM (2 mL), **1a** (30.7 mg, 0.2 mmol), BSA (40.9 mg, 0.2 mmol)/DCM (1 mL), **2a** (33.9 mg, 0.26 mmol)/DCM (1 mL), Et_3B (1M in THF, 0.2 mL, 0.2 mmol), pyridine (64 μL, d = 0.9819 g/cm^3, 62.8 mg, 0.8 mmol), and **3h** (77 μL, d = 1.384 g/cm^3, 106.6 mg, 0.6 mmol) afforded **4aah** (40.2 mg, 58%) (eluent: petroleum ether/ethyl acetate = 40/1) as an oil: ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.89 (d, *J* = 7.6 Hz, 2 H, Ar-H), 7.68 (d, *J* = 8.0 Hz, 1 H, Ar-H), 7.55 (t, *J* = 7.4 Hz, 1 H, Ar-H), 7.44 (t, *J* = 7.6

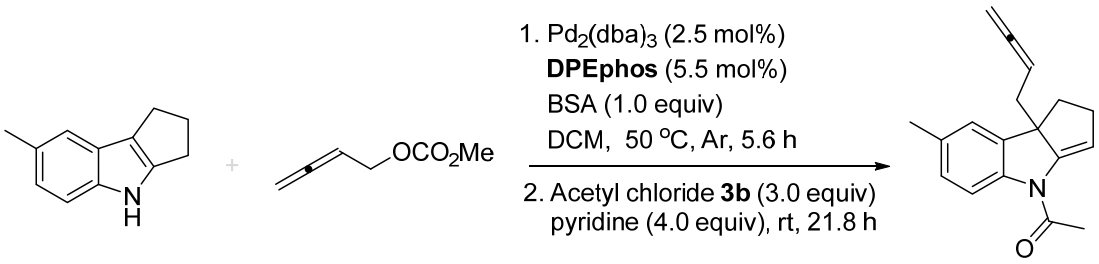
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Hz, 2 H, Ar-H), 7.25-7.20 (m, 1 H, Ar-H), 7.08 (d, J = 6.8 Hz, 1 H, Ar-H), 7.01 (t, J = 7.4 Hz, 1 H, Ar-H), 5.60 (d, J = 2.4 Hz, 1 H, =CH), 4.82 (quint, J = 7.3 Hz, 1 H, =CH), 4.56-4.47 (m, 2 H, $=CH_2$), 2.83-2.70 (m, 1 H, one proton of CH_2), 2.52-2.39 (m, 1 H, one proton of CH_2), 2.23 (dd, J_1 = 11.8 Hz, J_2 = 5.8 Hz, 1 H, one proton of CH_2), 2.02-1.84 (m, 2 H, one proton of CH_2 and one proton of another CH_2), 1.67-1.59 (m, 1 H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 210.1, 149.7, 144.0, 137.7, 135.6, 133.5, 128.9, 127.9, 127.1, 124.2, 123.8, 114.2, 107.8, 85.1, 74.0, 59.1, 37.0, 34.5, 32.3; MS (ESI) *m/z* 372 $(M+Na)^+$, 350 $(M+H)^+$; IR (neat, cm^{-1}): 3066, 2934, 2852, 1953, 1664, 1603, 1450, 1364, 1169, 1091, 1022; HRMS (ESI) Calcd for $C_{21}H_{20}NO_2S$ ($[M+H]^+$): 350.1209, Found: 350.1205.

9. 4-Acetyl-8b-(2,3-butadienyl)-7-methyl-2,8b-dihydrocyclopenta[*b*]indole (4bab, grd-17-102)



0.2 mmol **1b** | 1.3 equiv **2a** | 83% **4bab**

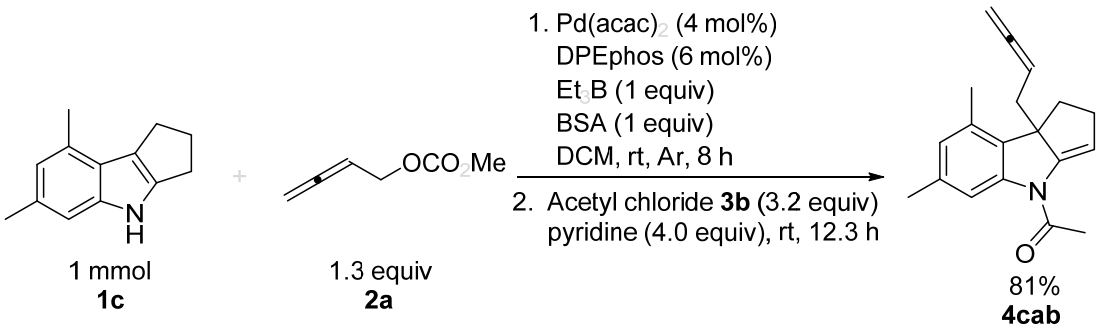
According to **Typical Procedure II**, the reaction of $Pd_2(dba)_3$ (4.6 mg, 0.005 mmol), DPEphos (6.0 mg, 0.011 mmol), DCM (2 mL), **1b** (34.3 mg, 0.2 mmol), **2a** (37.1 mg, 2.9 mmol), BSA (40.7 mg, 0.2 mmol), DCM (2 mL), pyridine (65 μL, d = 0.9819 g/cm^3, 63.8 mg, 0.8 mmol), and **3b** (43 μL, d = 1.104 g/cm^3, 47.5 mg, 0.6 mmol) afforded **4bab** (43.8 mg, 83%) (eluent: petroleum ether/ ethyl acetate = 50:1) as an oil: ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.03 (brs, 1 H, Ar-H), 7.03 (d, J = 6.8 Hz, 1 H, Ar-H), 6.97 (s, 1 H, Ar-H), 5.35-5.22 (m, 1 H, =CH), 4.95-4.83 (m, 1 H, =CH), 4.62-4.53 (m, 2 H, $=CH_2$), 2.89-2.78 (m, 1 H, one proton of CH_2), 2.54-2.46 (m, 1 H, one proton of CH_2), 2.41 (s, 3 H, CH_3), 2.33 (s, 3 H, CH_3), 2.30-2.19 (m, 3 H, one proton of CH_2 and another CH_2), 2.08-1.97 (m, 1 H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 209.9, 168.3, 151.6, 142.5, 136.0, 133.6, 128.1, 123.8, 116.8, 109.0, 85.3, 74.1, 59.0, 37.2, 34.9, 32.4, 24.5, 21.1; IR

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(neat, cm^{-1}): 2929, 2852, 1954, 1673, 1473, 1372, 1346, 1303, 1195, 1013; HRMS (ESI) Calcd for $C_{18}H_{20}NO$ ($[M+H]^+$): 266.1539, Found: 266.1541.

10. 4-Acetyl-8b-(2,3-butadienyl)-6,8-dimethyl-2,8b-dihydrocyclopenta[*b*]indole (4cab, zyz-2-168)



Typical Procedure IV: To a 50 mL oven dried pear-shaped Schlenk flask were added $Pd(acac)_2$ (12.3mg, 0.04 mmol), DPEphos (32.7 mg, 0.06 mmol), and DCM (10 mL) under the atmosphere of Ar. The resulting mixture was stirred at room temperature for 30 min. Then compound **1c** (185.9 mg, 1.0 mmol), BSA (203.9 mg, 1.0 mmol)/DCM (5 mL), **2a** (167.7 mg, 1.3 mmol)/DCM (5 mL), and Et_3B (1 M in THF, 1.0 mL, 1.0 mmol) were added sequentially. The resulting mixture was stirred at room temperature for 8 h as monitored by TLC. Then **3b** (0.23 mL, d = 1.104 g/cm^3, 253.9 mg, 3.2 mmol) and pyridine (0.32 mL, d = 0.9819 g/cm^3, 314.2 mg, 4.0 mmol) were added. The resulting mixture was stirred at room temperature for 12.5 h as monitored by TLC. A saturated aqueous solution of $NaHCO_3$ (20 mL) was added to quench the reaction. The organic layer was separated and the aqueous layer was extracted with DCM (20 mL × 2). The combined organic layer was washed with brine (20 mL) and dried over anhydrous Na_2SO_4. After filtration and evaporation, the residue was purified by silica gel column chromatography to afford **4cab** (227.3 mg, 81%) (eluent: petroleum ether/ethyl acetate = 60/1) as an oil: ^1H NMR (400 MHz, $CDCl_3$) δ 7.86 (brs, 1 H, Ar-H), 6.70 (s, 1 H, Ar-H), 5.26 (brs, 1 H, =CH), 4.83-4.71 (m, 1 H, =CH), 4.58-4.45 (m, 2 H, $=CH_2$), 2.89-2.76 (m, 1 H, one proton of CH_2), 2.53-2.43 (m, 1 H, one proton of CH_2), 2.40 (s, 3 H, CH_3), 2.40 (s, 3 H, CH_3), 2.38-2.32 (m, 1 H, one proton of CH_2), 2.31 (s, 3 H, CH_3), 2.28-2.14 (m, 6 H, CH_3, CH_2 and one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 209.5, 168.3, 151.4, 144.9, 137.5, 132.6, 130.5, 126.6, 115.2, 108.9, 85.2, 73.8, 58.6, 36.3, 36.0, 32.5, 24.5,

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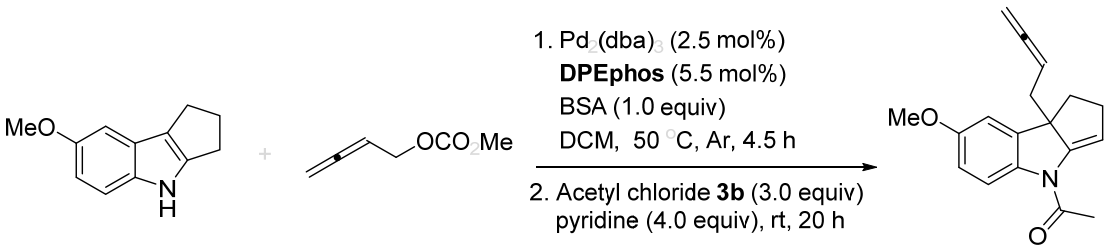
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21.5, 17.9; MS (EI, 70 eV) *m/z* (%) 279 (M^+, 8.08), 184 (100); IR (neat, cm^{-1}): 2928, 2853, 1954, 1673, 1594, 1411, 1372, 1351, 1303, 1266, 1252, 1232, 1094, 1070, 1032; HRMS (ESI) Calcd for $C_{19}H_{22}NO$ ($[M+H]^+$): 280.1696, Found: 280.1692.

11. 4-Acetyl-8b-(2,3-butadienyl)-7-methoxy-2,8b-dihydrocyclopenta[*b*]indole (4dab, grd-17-94)



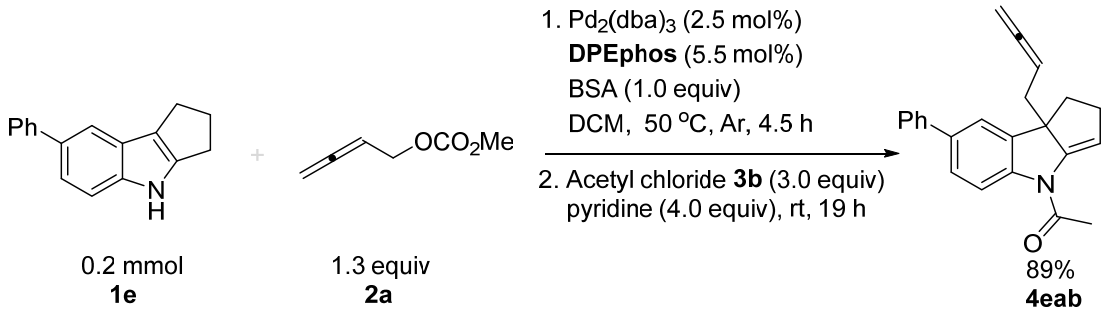
0.2 mmol **1d** | 1.3 equiv **2a** | 88% **4dab**

According to **Typical Procedure II**, the reaction of $Pd_2(dba)_3$ (4.6 mg, 0.005 mmol), DPEphos (6.0 mg, 0.011 mmol), DCM (2 mL), **1d** (37.5 mg, 0.2 mmol), **2a** (33.4 mg, 0.26 mmol), BSA (40.7 mg, 0.2 mmol), DCM (2 mL), pyridine (65 μL, d = 0.9819 g/cm^3, 63.8 mg, 0.8 mmol), and **3b** (43 μL, d = 1.104 g/cm^3, 47.5 mg, 0.6 mmol) afforded **4dab** (49.5 mg, 88%) (eluent: petroleum ether/ethyl acetate = 40/1) as an oil: ^1H NMR (400 MHz, $CDCl_3$) δ 8.09 (d, J = 6.0 Hz, 1 H, Ar-H), 6.77-6.71 (m, 2 H, Ar-H), 5.27 (brs, 1 H, =CH), 4.94-4.85 (m, 1 H, =CH), 4.58 (dt, J_1 = 6.8 Hz, J_2 = 2.2 Hz, 2 H, $=CH_2$), 3.79 (s, 3 H, OCH_3), 2.89-2.78 (m, 1 H, one proton of CH_2), 2.56-2.46 (m, 1 H, one proton of CH_2), 2.40 (s, 3 H, CH_3), 2.30-2.19 (m, 3 H, CH_2 and one proton of another CH_2), 2.09-1.98 (m, 1 H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 210.0, 168.0, 156.4, 151.5, 138.5, 137.4, 117.8, 111.7, 109.7, 109.0, 85.2, 74.1, 59.1, 55.5, 37.1, 34.8, 32.3, 24.4; IR (neat, cm^{-1}): 2934, 2846, 1956, 1664, 1610, 1475, 1449, 1379, 1344, 1293; HRMS (ESI) Calcd for $C_{18}H_{20}NO_2$ ($[M+H]^+$): 282.1489, Found: 282.1487.

12. 4-Acetyl-8b-(2,3-butadienyl)-7-phenyl-2,8b-dihydrocyclopenta[*b*]indole (4eab, grd-17-97)

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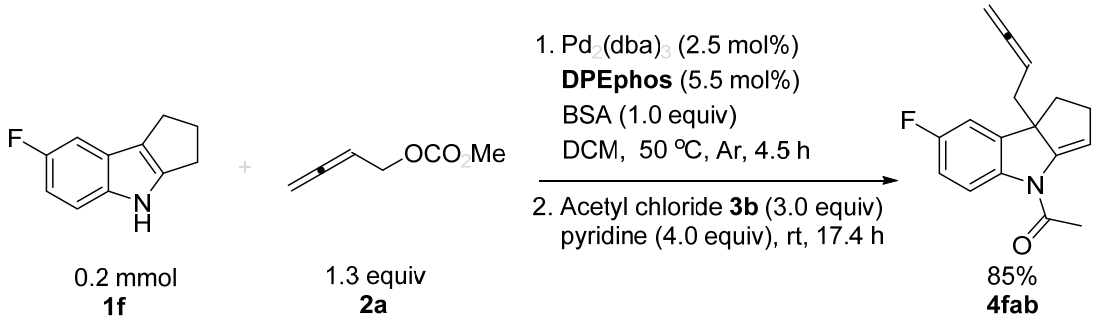
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0.2 mmol **1e** | 1.3 equiv **2a** | 89% **4eab**

According to **Typical Procedure II**, the reaction of $Pd_2(dba)_3$ (4.6 mg, 0.005 mmol), DPEphos (6.0 mg, 0.011 mmol), DCM (2 mL), **1e** (46.7 mg, 0.2 mmol), **2a** (33.4 mg, 0.26 mmol), BSA (40.7 mg, 0.2 mmol), DCM (2 mL), pyridine (65 μL, d = 0.9819 g/cm^3, 63.8 mg, 0.8 mmol), and **3b** (43 μL, d = 1.104 g/cm^3, 47.5 mg, 0.6 mmol) afforded **4eab** (58.2 mg, 89%) (eluent: petroleum ether/ethyl acetate = 40/1) as an oil: ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.21 (brs, 1 H, Ar-H), 7.57 (d, J = 8.0 Hz, 2 H, Ar-H), 7.48 (dd, J_1 = 8.4, J_2 =1.6 Hz, 1 H, Ar-H), 7.44-7.37 (m, 3 H, Ar-H), 7.31 (t, J = 7.4 Hz, 1 H, Ar-H), 5.31 (brs, 1 H, =CH), 4.94 (quint, J = 7.3 Hz, 1 H, =CH), 4.61-4.49 (m, 2 H, =CH_2), 2.92-2.80 (m, 1 H, one proton of CH_2), 2.57-2.48 (m, 1 H, one proton of CH_2), 2.44 (s, 3 H, CH_3), 2.36-2.22 (m, 3 H, CH_2 and one proton of another CH_2), 2.14-2.03 (m, 1 H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 210.0, 168.5, 151.4, 144.2, 140.8, 137.1, 136.5, 128.7, 127.0, 126.8, 126.5, 121.8, 117.2, 109.2, 85.2, 74.2, 59.2, 37.2, 34.9, 32.4, 24.6; IR (neat, cm^{-1}): 3031, 2930, 1953, 1675, 1601, 1466, 1371, 1344, 1300, 1204, 1124, 1011; HRMS (ESI) Calcd for $C_{23}H_{22}NO$ ([M+H]$^+$): 328.1696, Found: 328.1696.

13. 4-Acetyl-8b-(2,3-butadienyl)-7-fluoro-2,8b-dihydrocyclopenta[*b*]indole (4fab, grd-18-6)



0.2 mmol **1f** | 1.3 equiv **2a** | 85% **4fab**

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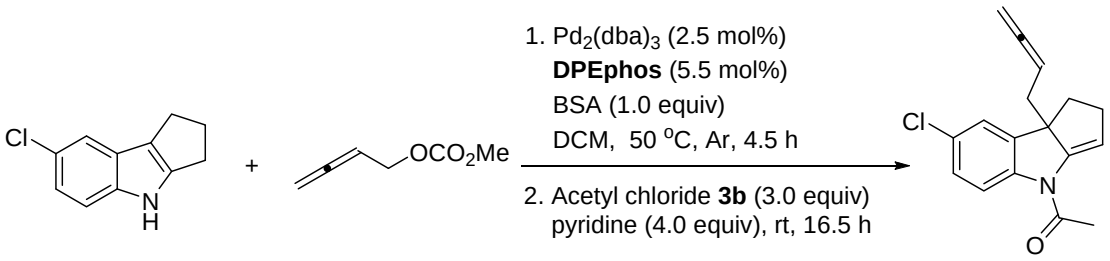
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According to **Typical Procedure II**, the reaction of $Pd_2(dba)_3$ (4.6 mg, 0.005 mmol), DPEphos (6.0 mg, 0.011 mmol), DCM (2 mL), **1f** (35.1 mg, 0.2 mmol), **2a** (33.3 mg, 0.26 mmol), BSA (40.7 mg, 0.2 mmol), DCM (2 mL), pyridine (65 μL, d = 0.9819 g/cm^3, 63.8 mg, 0.8 mmol), and **3b** (49 μL, d = 1.104 g/cm^3, 54.1 mg, 0.7 mmol) afforded **4fab** (45.7 mg, 85%) (eluent: petroleum ether/ethyl acetate = 50/1) as an oil: ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.13 (brs, 1 H, Ar-H), 6.92 (td, J_1 = 8.8 Hz, J_2 = 2.8 Hz, 1 H, Ar-H), 6.87 (dd, J_1 = 8.0, J_2 = 2.8 Hz, 1 H, Ar-H), 5.31 (brs, 1 H, =CH), 4.93-4.84 (m, 1 H, =CH), 4.58 (dt, J_1 = 6.6, J_2 = 2.3 Hz, 2 H, =CH_2), 2.90-2.79 (m, 1 H, one proton of CH_2), 2.57-2.47 (m, 1 H, one proton of CH_2), 2.42 (s, 3 H, CH_3), 2.30-2.18 (m, 3 H, CH_2 and one proton of another CH_2), 2.09-1.98 (m, 1 H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 210.0, 168.3, 159.5 (d, J = 242.1 Hz), 151.1, 140.81 (d, J = 2.4 Hz), 137.86 (d, J = 7.6 Hz), 118.08 (d, J = 7.2 Hz), 113.8 (d, J = 22.6 Hz), 110.7 (d, J = 23.8 Hz), 109.5, 84.8, 74.3, 59.1, 37.0, 34.8, 32.2, 24.4; ^{19}F NMR (376 MHz, $CDCl_3$) δ -118.1 (s); IR (neat, cm^{-1}): 2934, 2852, 1954, 1675, 1599, 1465, 1374, 1345, 1305, 1256, 1013; HRMS (ESI) Calcd for $C_{17}H_{17}FNO$ ($[M+H]^+$): 270.1289, Found: 270.1291.

14. 4-Acetyl-8b-(2,3-butadienyl)-7-chloro-2,8b-dihydrocyclopenta[*b*]indole (4gab, grd-18-7)



0.2 mmol **1g** | 1.3 equiv **2a** | 85% **4gab**

According to **Typical Procedure II**, the reaction of $Pd_2(dba)_3$ (4.6 mg, 0.005 mmol), DPEphos (6.0 mg, 0.011 mmol), DCM (2 mL), **1g** (38.4 mg, 0.2 mmol), **2a** (33.4 mg, 0.26 mmol), BSA (40.7 mg, 0.2 mmol), DCM (2 mL), pyridine (65 μL, d = 0.9819 g/cm^3, 63.8 mg, 0.8 mmol), and **3b** (43 μL, d = 1.104 g/cm^3, 47.5 mg, 0.6 mmol) afforded **4gad** (49.0 mg, 85%) (eluent: petroleum ether/ethyl acetate = 50/1) as an oil: ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.10 (d, J = 7.2 Hz, 1 H, Ar-H), 7.20 (dd, J_1 = 8.4 Hz, J_2 = 1.8 Hz, 1 H, Ar-H), 7.13 (d, J = 1.2 Hz, 1 H, Ar-H), 5.31 (brs, 1 H, =CH), 4.88 (quint, 1 H, =CH),

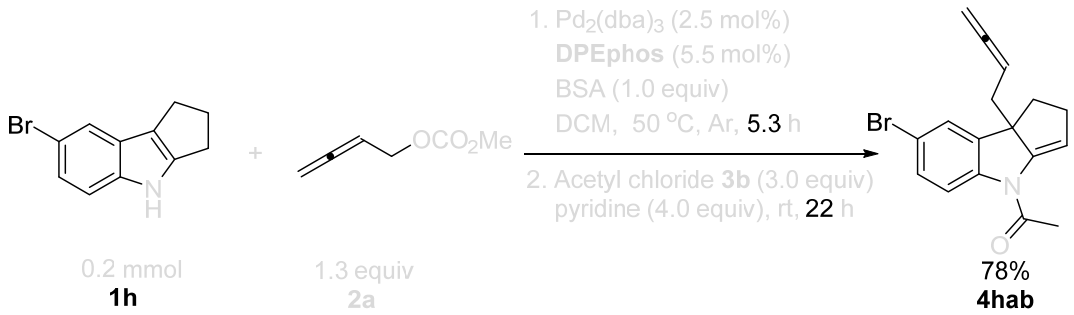
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4.62-4.53 (m, 2 H, =CH_2), 2.90-2.78 (m, 1 H, one proton of CH_2), 2.57-2.47 (m, 1 H, one proton of CH_2), 2.42 (s, 3 H, CH_3), 2.32-2.15 (m, 3 H, CH_2 and one proton of another CH_2), 2.03 (dd, 1 H, J_1 = 20.6 Hz, J_2 = 9.8 Hz, one proton of CH_2); ^{13}C NMR (75 MHz, $CDCl_3$) δ 210.1, 168.5, 150.8, 143.3, 137.6, 129.1, 127.5, 123.5, 118.1, 109.6, 84.7, 74.3, 59.1, 36.9, 34.8, 32.3, 24.5; IR (neat, cm^{-1}): 2986, 2957, 2934, 2847, 1953, 1679, 1459, 1373, 1341, 1301, 1261, 1169, 1070; HRMS (ESI) Calcd for $C_{17}H_{17}{}^{35}ClNO$ ([M+H]$^+$): 286.0993, Found: 286.0995.

15. 4-Acetyl-7-bromo-8b-(2,3-butadienyl)-2,8b-dihydrocyclopenta[*b*]indole (4hab, grd-17-101)



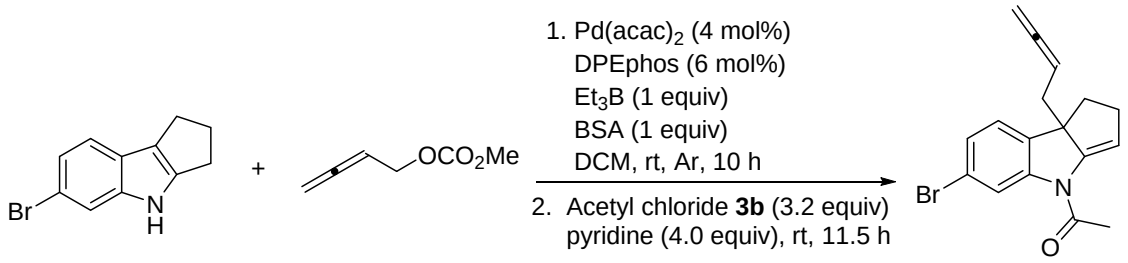
According to **Typical Procedure II**, the reaction of $Pd_2(dba)_3$ (4.6 mg, 0.005 mmol), DPEphos (6.0 mg, 0.011 mmol), DCM (2 mL), **1h** (47.3 mg, 0.2 mmol), **2a** (37.1 mg, 0.26 mmol), BSA (40.7 mg, 0.2 mmol), DCM (2 mL), pyridine (65 μL, d = 0.9819 g/cm^3, 63.8 mg, 0.8 mmol), and **3b** (43 μL, d = 1.104 g/cm^3, 47.5 mg, 0.6 mmol) afforded **4had** (51.3 mg, 78%) (eluent: petroleum ether/ethyl acetate = 50/1) as an oil: ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.05 (d, J = 8.4 Hz, 1 H, Ar-H), 7.35 (dd, J_1 = 8.6 Hz, J_2 = 1.8 Hz, 1 H, Ar-H), 7.28 (d, J = 2.0 Hz, 1 H, Ar-H), 5.31 (brs, 1 H, =CH), 4.93-4.83 (m, 1 H, =CH), 4.59 (dt, J_1 = 6.4 Hz, J_2 = 2.0 Hz, 2 H, =CH_2), 2.89-2.79 (m, 1 H, one proton of CH_2), 2.57-2.48 (m, 1 H, one proton of CH_2), 2.42 (s, 3 H, CH_3), 2.31-2.16 (m, 3 H, CH_2 and one proton of another CH_2), 2.08-1.98 (m, 1 H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 210.1, 168.5, 150.7, 143.8, 138.0, 130.5, 126.4, 118.6, 116.8, 109.6, 84.7, 74.4, 59.2, 37.0, 34.8, 32.3, 24.6; IR (neat, cm^{-1}): 2932, 2852, 1953, 1678, 1456, 1370, 1342, 1302, 1257, 1174, 1012; HRMS (ESI) Calcd for $C_{17}H_{17}{}^{79}BrNO$ ([M+H]$^+$): 330.0488, Found: 330.0489.

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16. 4-Acetyl-6-bromo-8b-(2,3-butadienyl)-2,8b-dihydrocyclopenta[*b*]indole (4iab, zyz-2-166)



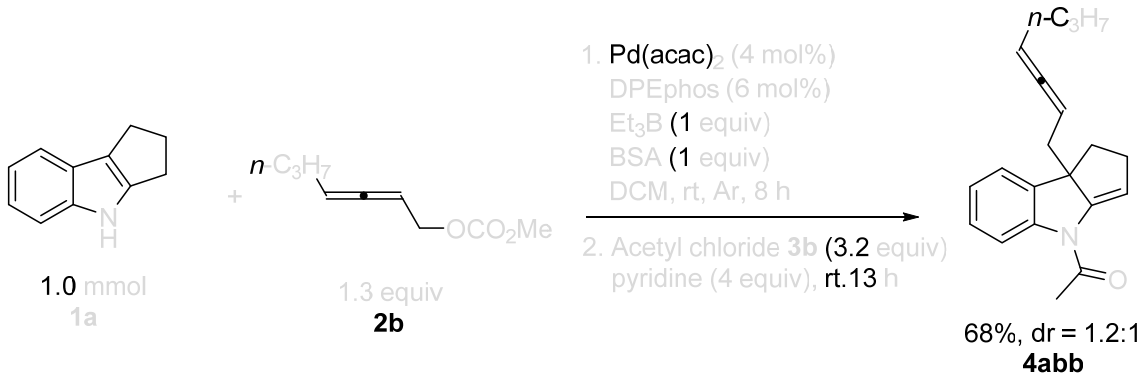
1.0 mmol **1i** | 1.3 equiv **2a** | 80% **4iab**

According to **Typical Procedure IV**, the reaction of $Pd(acac)_2$ (12.2 mg, 0.04 mmol), DPEphos (32.9 mg, 0.06 mmol), DCM (10 mL), **1i** (236.7 mg, 1.0 mmol), BSA (205.1 mg, 1.0 mmol)/DCM (5 mL), **2a** (167.9 mg, 1.3 mmol)/DCM (5 mL), Et_3B (1 M in THF, 1.0 mL, 1.0 mmol), **3b** (0.23 mL, d = 1.104 g/cm^3, 253.9 mg, 3.2 mmol), and pyridine (0.32 mL, d = 0.9819 g/cm^3, 314.2 mg, 4.0 mmol) afforded **4iab** (265.1 mg, 80%) (eluent: petroleum ether/ethyl acetate = 60/1) as an oil: ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.37 (brs, 1 H, Ar-H), 7.22 (dd, J_1 = 7.8 Hz, J_2 = 1.8 Hz, 1 H, Ar-H), 7.02 (d, J = 8.0 Hz, 1 H, Ar-H), 5.31 (d, J = 1.6 Hz, 1 H, =CH), 4.91-4.81 (m, 1 H, =CH), 4.61-4.52 (m, 2 H, =CH_2), 2.90-2.79 (m, 1 H, one proton of CH_2), 2.57-2.47 (m, 1 H, one proton of CH_2), 2.42 (s, 3 H, CH_3), 2.32-2.14 (m, 3 H, CH_2 and one proton of another CH_2), 2.06-1.97 (m, 1 H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 209.9, 168.5, 150.8, 145.7, 134.8, 126.8, 124.2, 121.0, 120.2, 109.5, 84.8, 74.2, 58.8, 36.9, 34.9, 32.3, 24.5; MS (EI, 70eV) *m/z* (%) 331 ((M(^{81}Br))$^+$, 4.98), 329 ((M(^{79}Br)$^+$, 5.25), 234 (100); IR (neat, cm^{-1}): 2933, 2851, 1953, 1678, 1596, 1574, 1408, 1373, 1302, 1254, 1203, 1172, 1117, 1089, 1058, 1031, 1014; HRMS (ESI) Calcd for $C_{17}H_{17}{}^{79}BrNO$ ([M+H]$^+$): 330.0488, Found: 330.0482.

17. 4-Acetyl-8b-(2,3-heptadienyl)-2,8b-dihydrocyclopenta[*b*]indole (4abb, zyz-2-105)

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1.0 mmol
1a

1.3 equiv
2b

68%, dr = 1.2:1
4abb

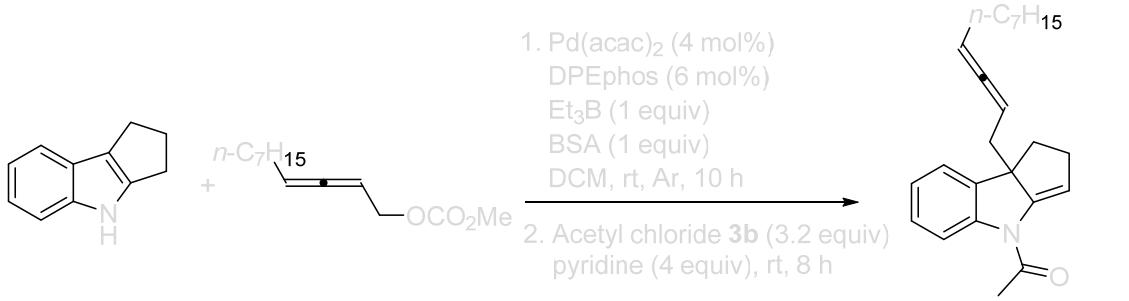
According to **Typical Procedure IV**, the reaction of $Pd(acac)_2$ (12.3 mg, 0.04 mmol), DPEphos (32.7 mg, 0.06 mmol), DCM (10 mL), **1a** (157.9 mg, 1.0 mmol), BSA (215.0 mg, 1.0 mmol)/DCM (5 mL), **2b** (221.9 mg, 1.3 mmol)/DCM (5 mL)/DCM(5 mL), Et_3B (1 M in THF, 1.0 mL, 1.0 mmol), pyridine (0.32 mL, d = 0.9819 g/cm^3, 314.2 mg, 4.0 mmol), and **3b** (0.23 mL, d = 1.104 g/cm^3, 253.9 mg, 3.2 mmol) afforded **4abb** (198.3 mg, 68%, dr =1.2:1) (eluent: petroleum ether/ethyl acetate = 60/1) as an oil: ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.16 (brs, 1 H, Ar-H), 7.25-7.20 (m, 1 H, Ar-H), 7.19-7.13 (m, 1 H, Ar-H), 7.10-7.03 (m, 1 H, Ar-H), 5.28 (brs, 1 H, =CH), 5.00-4.77 (m, 2 H, CH=C=CH), 2.90-2.78 (m, 1 H, one proton of CH_2), 2.54-2.45 (m, 1 H, one proton of CH_2), 2.44-2.40 (m, 3 H, CH_3), 2.34-2.26 (m, 1 H, one proton of CH_2), 2.25-2.10 (m, 2 H, CH_2), 2.09-1.98 (m, 1 H, one proton of CH_2), 1.91-1.77 (m, 2 H, CH_2), 1.37-1.27 (m, 2 H, CH_2), 0.90-0.83 (m, 3 H, CH_3); MS (EI, 70 eV) *m/z* (%): 293 (M^+, 6.49), 156 (100); IR (neat, cm^{-1}): 2956, 2930, 2853, 1960, 1677, 1602, 1472, 1456, 1375, 1350, 1304, 1282, 1206, 1174, 1108, 1068, 1012; Anal. calcd for $C_{20}H_{23}NO$ (%): C 81.87, H 7.90, N 4.77, Found: C 81.63, H 8.06, N 4.77.

18. 4-Acetyl-8b-(2,3-undecadienyl)-2,8b-dihydrocyclopenta[*b*]indole (4acb, zyz-2-113)

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1.0 mmol **1a** | 1.3 equiv **2c** | 60%, dr = 1.4:1 **4acb**

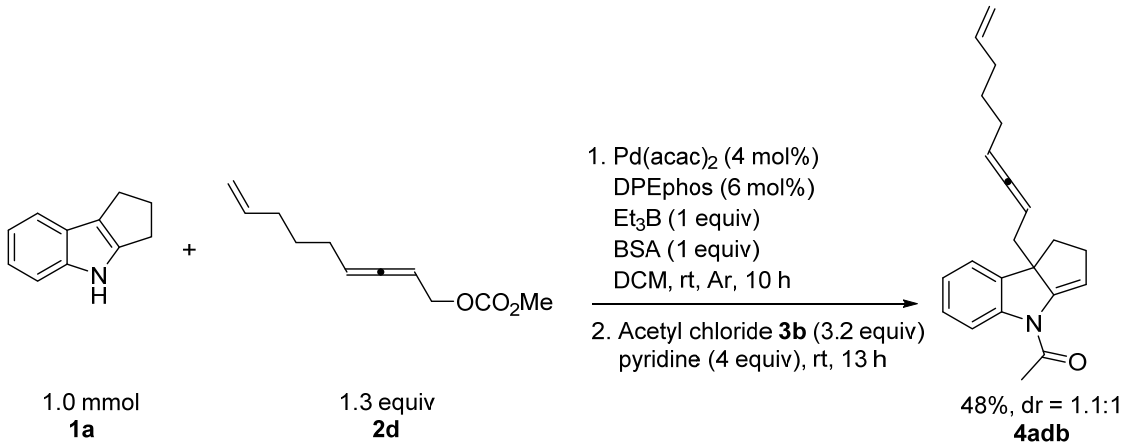
According to **Typical Procedure IV**, the reaction of $Pd(acac)_2$ (12.3 mg, 0.04 mmol), DPEphos (32.7 mg, 0.06 mmol), DCM (10 mL), **1a** (157.9 mg, 1.0 mmol), BSA (214.7 mg, 1.0 mmol)/DCM (5 mL), **2c** (294.7 mg, 1.3 mmol)/DCM (5 mL), Et_3B (1 M in THF, 1.0 mL, 1.0 mmol), pyridine (0.32 mL, d = 0.9819 g/cm^3, 314.2 mg, 4.0 mmol), and **3b** (0.23 mL, d = 1.104 g/cm^3, 253.9 mg, 3.2 mmol) afforded **4acb** (210.7 mg, 60%, dr = 1.4:1) (eluent: petroleum ether/ethyl acetate = 60/1) as an oil: ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.16 (brs, 1 H, Ar-H), 7.22 (t, *J* = 7.8 Hz, 1 H, Ar-H), 7.19-7.12 (m, 1 H, Ar-H), 7.06 (t, *J* = 7.6 Hz, 1 H, Ar-H), 5.29 (brs, 1 H, =CH), 5.01-4.78 (m, 2 H, CH=C=CH), 2.90-2.79 (m, 1 H, one proton of CH_2), 2.55-2.45 (m, 1 H, one proton of CH_2), 2.45-2.40 (m, 3 H, CH_3), 2.35-2.26 (m, 1 H, one proton of CH_2), 2.24-2.13 (m, 2 H, one proton of CH_2 and another CH_2), 2.09-1.98 (m, 1 H, one proton of CH_2), 1.93-1.76 (m, 2 H, CH_2), 1.36-1.20 (m, 10 H, 5×CH_2), 0.88 (t, *J* = 6.6 Hz, 3 H, CH_3); MS (EI, 70 eV) *m/z* (%): 349 (M^+, 8.35), 156 (100); IR (neat, cm^{-1}): 2954, 2924, 2852, 1960, 1679, 1602, 1456, 1375, 1350, 1282, 1206, 1174, 1108, 1012; HRMS (ESI) Calced for $C_{24}H_{32}NO$ $(M+H)^+$: 350.2478, Found: 350.2474.

19. 4-Acetyl-8b-(2,3,8-nonatrienyl)-2,8b-dihydrocyclopenta[*b*]indole (4adb, zyz-2-128)

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1.0 mmol **1a** | 1.3 equiv **2d** | 48%, dr = 1.1:1 **4adb**

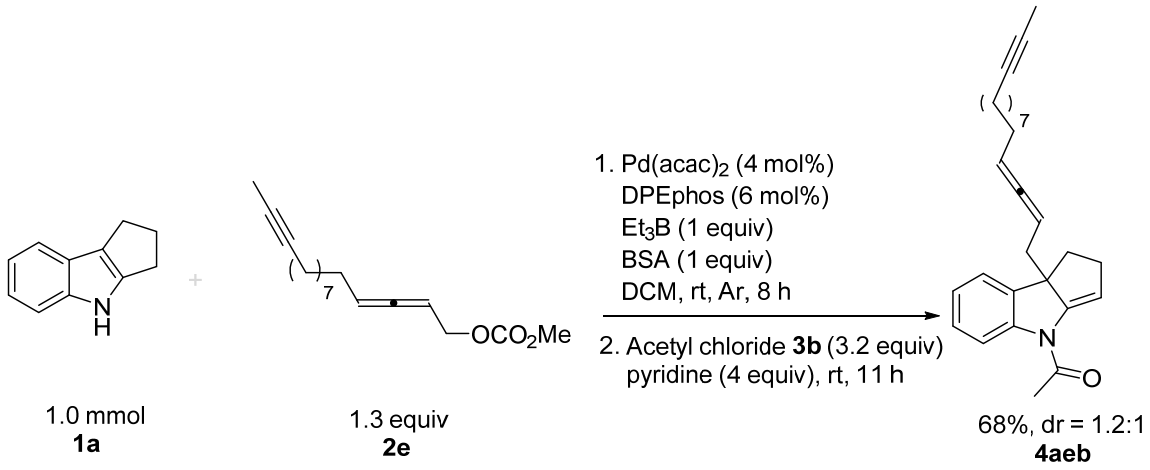
According to **Typical Procedure IV**, the reaction of $Pd(acac)_2$ (12.4 mg, 0.04 mmol), DPEphos (32.9 mg, 0.06 mmol), DCM (10 mL), **1a** (157.7 mg, 1.0 mmol), BSA (214.3 mg, 1.0 mmol)/DCM (5 mL), **2d** (255.9 mg, 1.3 mmol)/DCM (5 mL), Et_3B (1 M in THF, 1.0 mL, 1.0 mmol), pyridine (0.32 mL, d = 0.9819 g/cm^3, 314.2 mg, 4.0 mmol), and **3b** (0.23 mL, d = 1.104 g/cm^3, 253.9 mg, 3.2 mmol) afforded **4adb** (152.3 mg, 48%, dr = 1.1:1) (eluent: petroleum ether/ethyl acetate = 60/1) as an oil: ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.15 (brs, 1 H, Ar-H), 7.26-7.19 (m, 1 H, Ar-H), 7.19-7.13 (m, 1 H, Ar-H), 7.10-7.03 (m, 1 H, Ar-H), 5.85-5.71 (m, 1 H, =CH), 5.29 (brs, 1 H, =CH), 5.04-4.80 (m, 4 H, CH=C=CH and $=CH_2$), 2.90-2.79 (m, 1 H, one proton of CH_2), 2.55-2.46 (m, 1 H, one proton of CH_2), 2.42 (s, 3 H, CH_3), 2.36-2.26 (m, 1 H, one proton of CH_2), 2.26-2.12 (m, 2 H, CH_2), 2.08-1.98 (m, 3 H, CH_2 and one proton of another CH_2), 1.95-1.83 (m, 2 H, CH_2), 1.40 (quint, J = 7.6 Hz, 2 H, CH_2); MS (EI, 70 eV) *m/z* (%): 319 (M^+, 3.87), 156 (100); IR (neat, cm^{-1}): 3073, 2929, 2851, 1960, 1677, 1602, 1456, 1376, 1303, 1282, 1206, 1174, 1108, 1012; Anal. calcd for $C_{22}H_{25}NO$ (%): C 82.72, H 7.89, N 4.38, Found: C 82.50, H 8.12, N 4.35.

20. 4-Acetyl-8b-(2,3-pentadecadiene-13-yneyl)-2,8b-dihydrocyclopenta[*b*]indole (4aeb, zyz-2-137)

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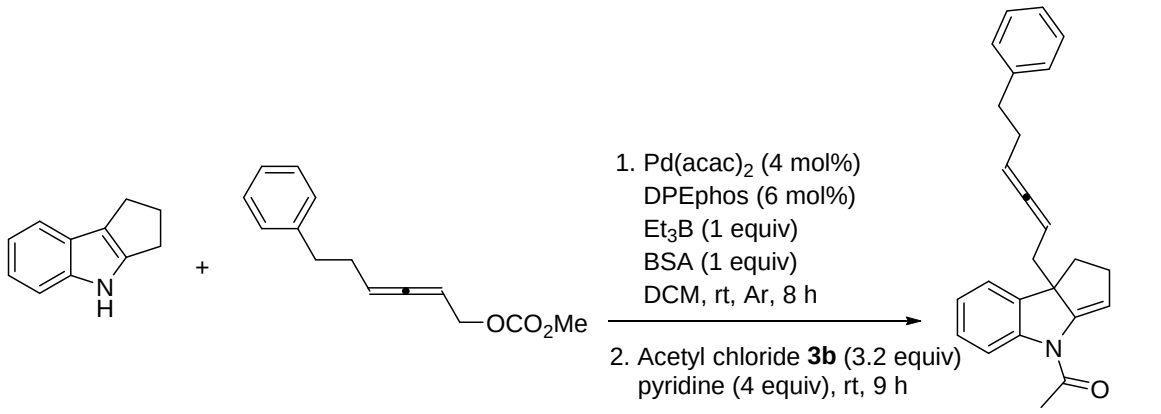
According to **Typical Procedure IV**, the reaction of $Pd(acac)_2$ (12.3 mg, 0.04 mmol), DPEphos (32.8 mg, 0.06 mmol), DCM (10 mL), **1a** (157.9 mg, 1.0 mmol), BSA (214.9 mg, 1.0 mmol)/DCM (5 mL), **2e** (360.7 mg, 1.3 mmol)/DCM (5 mL), Et_3B (1 M in THF, 1.0 mL, 1.0 mmol), pyridine (0.32 mL, d = 0.9819 g/cm^3, 314.2 mg, 4.0 mmol), and **3b** (0.23 mL, d = 1.104 g/cm^3, 253.9 mg, 3.2 mmol) afforded **4aeb** (268.1 mg, 68%, dr = 1.2:1) (eluent: petroleum ether/ethyl acetate = 60/1) as an oil: ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.15 (brs, 1 H, Ar-H), 7.22 (t, *J* = 7.8 Hz, 1 H, Ar-H), 7.18-7.13 (m, 1 H, Ar-H), 7.06 (t, *J* = 7.4 Hz, 1 H, Ar-H), 5.28 (brs, 1 H, =CH), 5.01-4.78 (m, 2 H, HC=C=CH), 2.90-2.78 (m, 1 H, one proton of CH_2), 2.55-2.45 (m, 1 H, one proton of CH_2), 2.42 (s, 3 H, CH_3), 2.35-2.00 (m, 6 H, 3×CH_2), 1.92-1.80 (m, 2 H, CH_2), 1.77 (s, 3 H, CH_3), 1.52-1.40 (m, 2 H, CH_2), 1.39-1.21 (m, 10 H, 5×CH_2); MS (EI, 70 eV) *m/z* (%): 401 (M^+, 11.99), 156 (100); IR (neat, cm^{-1}): 2925, 2852, 1960, 1679, 1602, 1472, 1456, 1376, 1350, 1304, 1282, 1206, 1174, 1108, 1012; Anal. calcd for $C_{28}H_{35}NO$ (%): C 83.74, H 8.78, N 3.49, Found: C 83.68, H 8.78, N 3.47.

21. 4-Acetyl-8b-6-phenyl-2,3-hexadienyl-2,8b-dihydrocyclopenta[*b*]indole (4afb, zyz-2-139)

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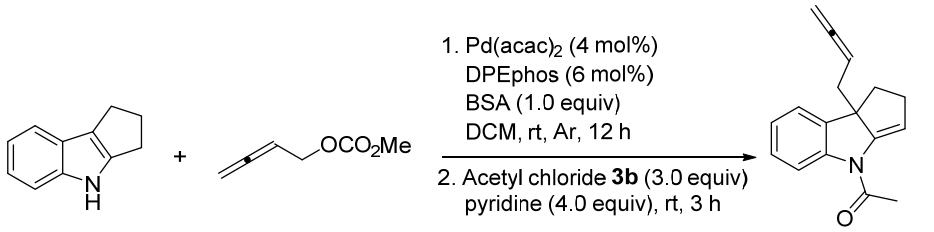
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1.0 mmol **1a** | 1.3 equiv **2f** | 78%, dr = 1.4:1 **4afb**

According to **Typical Procedure IV**, the reaction of $Pd(acac)_2$ (12.3 mg, 0.04 mmol), DPEphos (32.8 mg, 0.06 mmol), DCM (10 mL), **1a** (157.9 mg, 1.0 mmol), BSA (202.9 mg, 1.0 mmol)/DCM (5 mL), **2f** (302.5 mg, 1.3 mmol)/DCM (5 mL), Et_3B (1 M in THF, 1.0 mL, 1.0 mmol), **3b** (0.23 mL, d = 1.104 g/cm^3, 253.9 mg, 3.2 mmol), and pyridine (0.32 mL, d = 0.9819 g/cm^3, 314.2 mg, 4.0 mmol) afforded **4afb** (277.7 mg, 78%, dr = 1.4:1) (eluent: petroleum ether/ethyl acetate = 60/1) as an oil: ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.16 (brs, 1 H, Ar-H), 7.29-7.10 (m, 7 H, Ar-H), 7.04 (t, *J* = 7.4 Hz, 1 H, Ar-H), 5.26 (brs, 1 H, =CH), 5.12-4.77 (m, 2 H, HC=C=CH), 2.87-2.74 (m, 1 H, one proton of CH_2), 2.64-2.52 (m, 2 H, CH_2), 2.51-2.43 (m, 1 H, one proton of CH_2), 2.40 (s, 3 H, CH_3), 2.29-1.95 (m, 6 H, 3×CH_2); MS (EI, 70 eV) *m/z* (%): 355 (M^+, 9.14), 156 (100); IR (neat, cm^{-1}): 3062, 3025, 2930, 2851, 1960, 1676, 1602, 1473, 1376, 1282, 1205, 1174, 1107, 1078, 1012; Anal. calcd for $C_{25}H_{25}NO$ (%): C 84.47, H 7.09, N 3.94, Found: C 84.51, H 7.15, N 3.71.

22. Gram scale of 4-acetyl-8b-(2,3-butadienyl)-2,8b-dihydrocyclopenta[*b*]indole (4aab, zyz-3-12)



5 mmol **1a** | 1.3 equiv **2a** | 1.0410 g (83%) **4aab**

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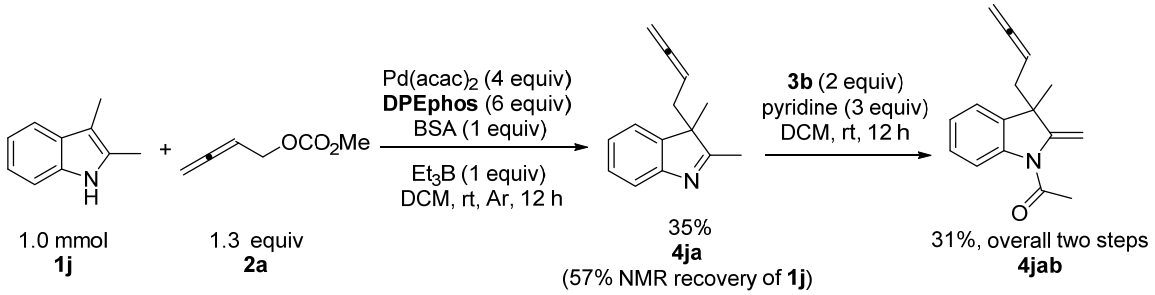
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According to **Typical Procedure IV**, the reaction of $Pd(acac)_2$ (61.0 mg, 0.2 mmol), DPEphos (161.3 mg, 0.3 mmol), DCM (50 mL), **1a** (786.7 mg, 5.0 mmol), BSA (1.0179 g, 1.0 mmol)/DCM (25 mL), **2a** (833.9 mg, 6.5 mmol)/DCM (25 mL), Et_3B (1 M in THF, 5.0 mL, 5.0 mmol), pyridine (1.6 mL, d = 0.9819 g/cm^3, 1.5710 g, 20 mmol) and **3b** (1.1 mL, d = 1.104 g/cm^3, 1.2144 g, 15 mmol) afforded **4aab** (1.0410 g, 83%) (eluent: petroleum ether/ethyl acetate = 60/1) as an oil: ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.16 (brs, 1 H, Ar-H), 7.24 (t, *J* = 7.8 Hz, 1 H, Ar-H), 7.16 (d, J_1 = 7.2 Hz, 1 H, Ar-H), 7.07 (t, 1 H, J_1 = 7.4 Hz, Ar-H), 5.30 (brs, 1 H, =CH), 4.89 (quint, *J* = 7.6 Hz, 1 H, =CH), 4.59-4.52 (m, 2 H, $=CH_2$), 2.90-2.79 (m, 1 H, one proton of CH_2), 2.55-2.46 (m, 1 H, one proton of CH_2), 2.43 (s, 3 H, CH_3), 2.32-2.19 (m, 3 H, CH_2 and one proton of another CH_2), 2.10-1.98 (m, 1 H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 209.9, 168.6, 151.3, 144.8, 135.9, 127.7, 124.0, 123.1, 117.1, 109.2, 85.2, 74.1, 59.1, 37.1, 34.9, 32.3, 24.6.

23. 3-(Buta-2,3-dienyl)-2,3-dimethyl-*3H*-indole (4ja, zyz-2-189) and 1-acetyl-3-(buta-2,3-dienyl)-3-methyl-2-methylene indoline (4jab, zyz-2-190)



To a oven dried Schlenk tube were added $Pd(acac)_2$ (12.2 mg, 0.04 mmol), DPEphos (32.4 mg, 0.06 mmol), and DCM (10 mL) under the atmosphere of Ar. The resulting mixture was stirred at room temperature for 30 min. Compound **1j** (146.7 mg, 1.0 mmol), **2a** (167.7 mg, 1.3 mmol)/DCM (5 mL), BSA (205.1 mg, 1.0 mmol)/DCM (5 mL), and Et_3B (1 M in THF, 1.0 mL, 1.0 mmol) were added sequentially. The resulting mixture was stirred at room temperature for 12 h as monitored by TLC. A saturated aqueous solution of $NaHCO_3$ (20 mL) was added to quench the reaction. The organic layer was separated and the aqueous layer was extracted with DCM (20 mL × 2). The combined organic layer was washed with brine (20 mL) and dried over anhydrous Na_2SO_4, After filtration and evaporation, the residue was purified by silica gel column chromatography to afford **4ja** (69.3 mg, 35%) (eluent: petroleum ether/ethyl acetate = 10/1 (600 mL) to

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1/1 (400 mL)) as an oil: ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.53 (d, *J* = 8.0 Hz, 1 H, and Ar-H), 7.34-7.25 (m, 2 H, Ar-H), 7.19 (t, *J* = 7.2 Hz, 1 H, Ar-H), 4.61-4.38 (m, 3 H, =CH and $=CH_2$), 2.62-2.54 (m, 1 H, one proton of CH_2), 2.44-2.36 (m, 1 H, one proton of CH_2), 2.27 (s, 3 H, CH_3), 1.32 (s, 3 H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ 209.3, 186.4, 154.2, 142.9, 127.7, 125.0, 121.7, 119.8, 84.2, 74.6, 57.6, 36.1, 21.7, 15.8; MS (ESI) *m/z* 198 $(M+H)^+$; IR (neat, cm^{-1}): 3049, 2962, 2928, 2867, 1954, 1578, 1451, 1427, 1376, 1309, 1254, 1206, 1185, 1093, 1014; HRMS (ESI) Calcd for $C_{14}H_{16}N$ $(M+H)^+$: 198.1277, Found: 198.1275.

Compound **4ja** (69.3 mg, 0.35 mmol) was dissolved in DCM (10 mL). Pyridine (85 μL, d = 0.9817 g/cm^3, 83.4 mg, 1.05 mmol) and **3b** (50 μL, d = 1.104 g/cm^3, 55.2 mg, 0.7 mmol) were added sequentially and the resulting mixture was stirred in room temperature for 12 h as monitored by TLC. A saturated aqueous solution of $NaHCO_3$ (10 mL) was added to quench the reaction. The organic layer was separated and the aqueous layer was extracted with DCM (10 mL × 2). The combined organic layer was washed with brine (10 mL) and dried over anhydrous Na_2SO_4. After filtration and evaporation, the residue was purified by silica gel column chromatography to afford **4jab** (74.2 mg, 31%, overall two steps) (eluent: petroleum ether/ethyl acetate = 20/1) as an oil: ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.98 (d, *J* = 8.0 Hz, 1 H, Ar-H), 7.23 (t, *J* = 7.8 Hz, 1 H, Ar-H), 7.16 (d, *J* = 7.2 Hz, 1 H, Ar-H), 7.10 (t, *J* = 7.2 Hz, 1 H, Ar-H), 5.15 (s, 1 H, one proton of $=CH_2$), 4.84-4.71 (m, 2 H, one proton of $=CH_2$ and =CH), 4.58-4.43 (m, 2 H, $=CH_2$), 2.50 (s, 3 H, CH_3), 2.41-2.27 (m, 2 H, CH_2), 1.44 (s, 3 H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ 209.9, 169.6, 154.5, 141.8, 136.3, 127.7, 124.2, 122.3, 116.5, 95.8, 84.7, 74.0, 48.0, 42.2, 25.9, 25.7; MS (ESI) *m/z* 262 $(M+Na)^+$, 240 $(M+H)^+$; IR (neat, cm^{-1}): 3051, 2965, 2932, 1954, 1669, 1599, 1477, 1462, 1367, 1341, 1321, 1283, 1226, 1206, 1176, 1108, 1072, 1051, 1031, 1013; HRMS (ESI) Calcd for $C_{16}H_{18}NO$ $([M+H]^+)$: 240.1383, Found: 240.1382.

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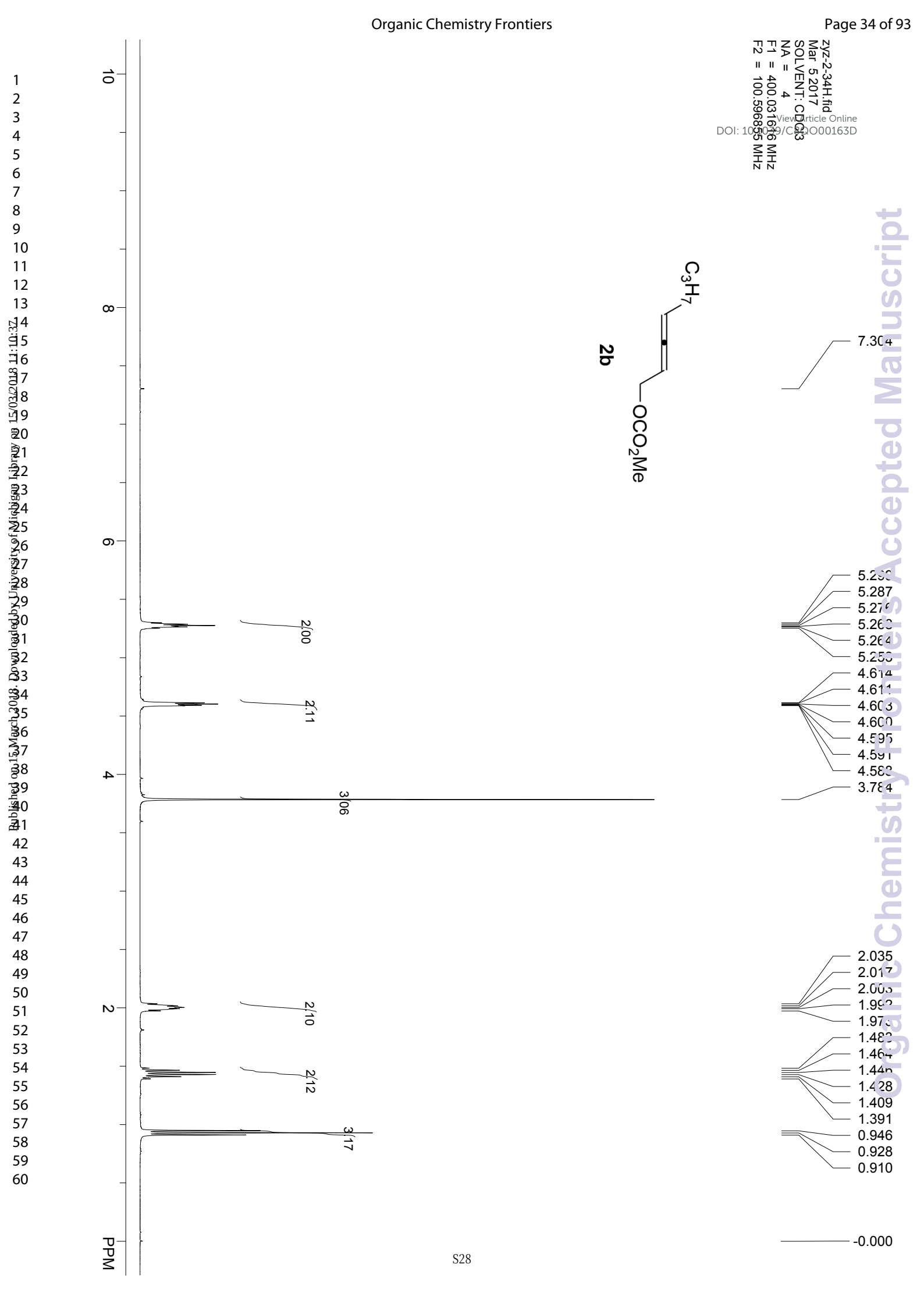
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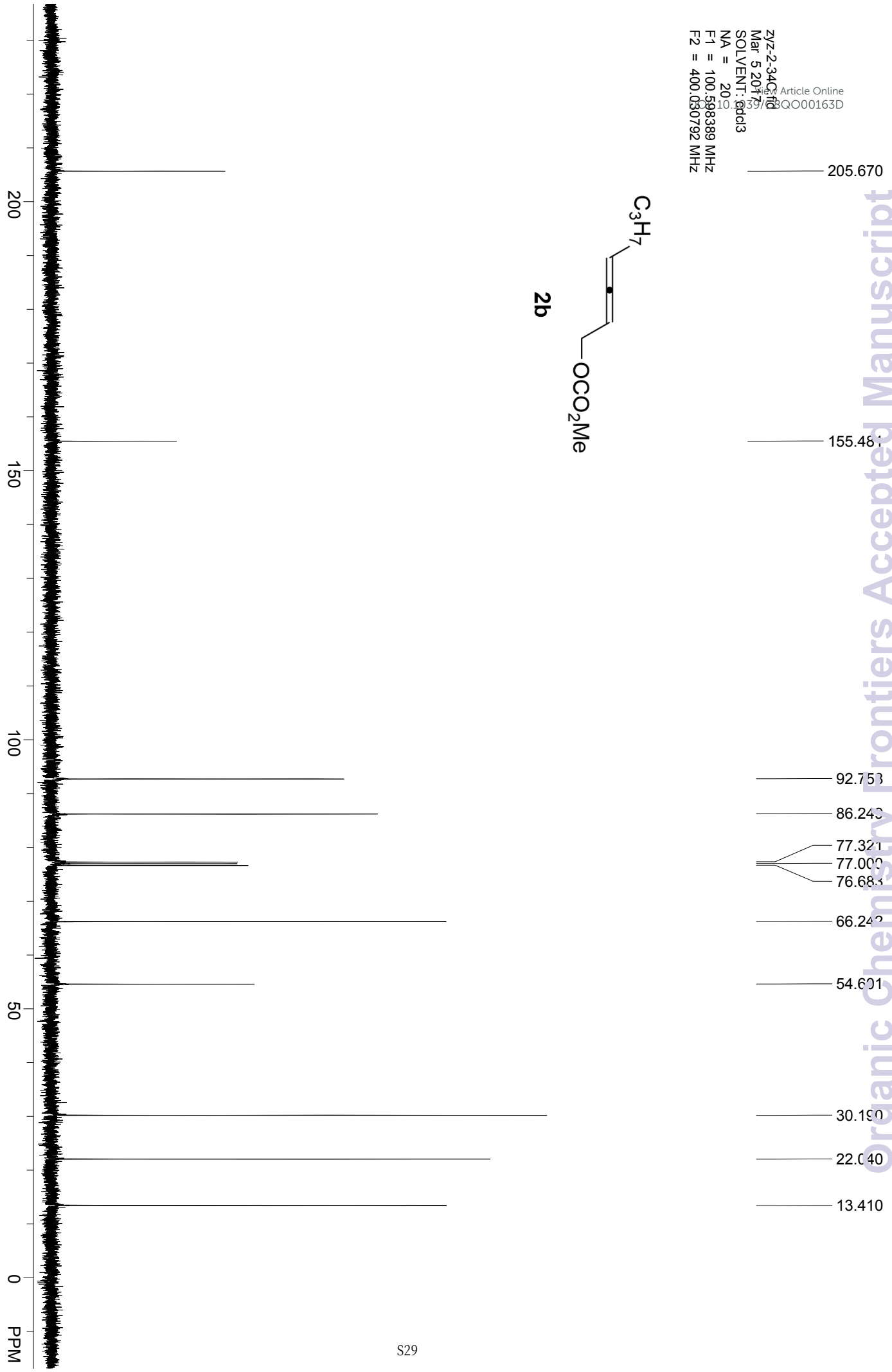
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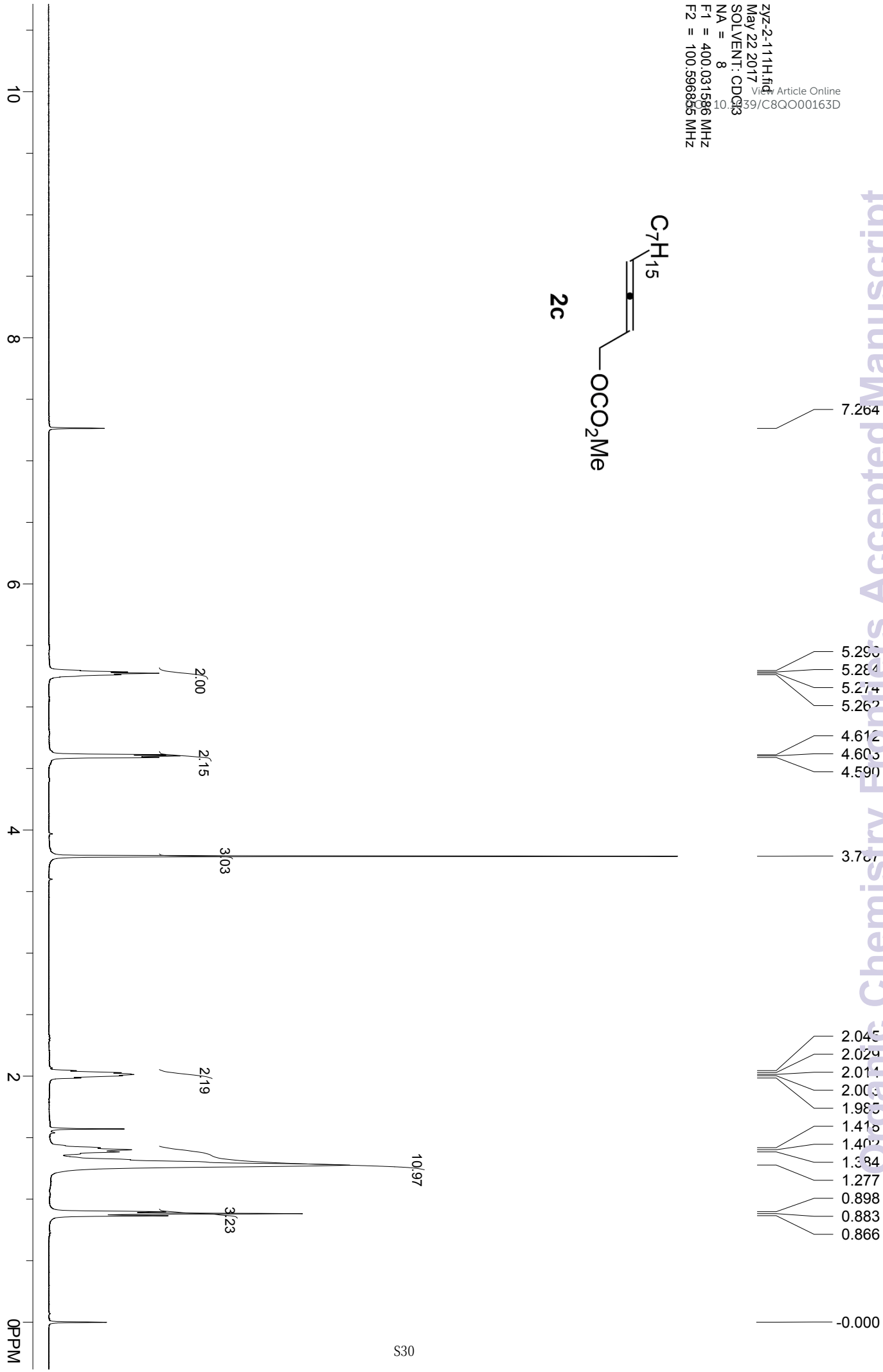
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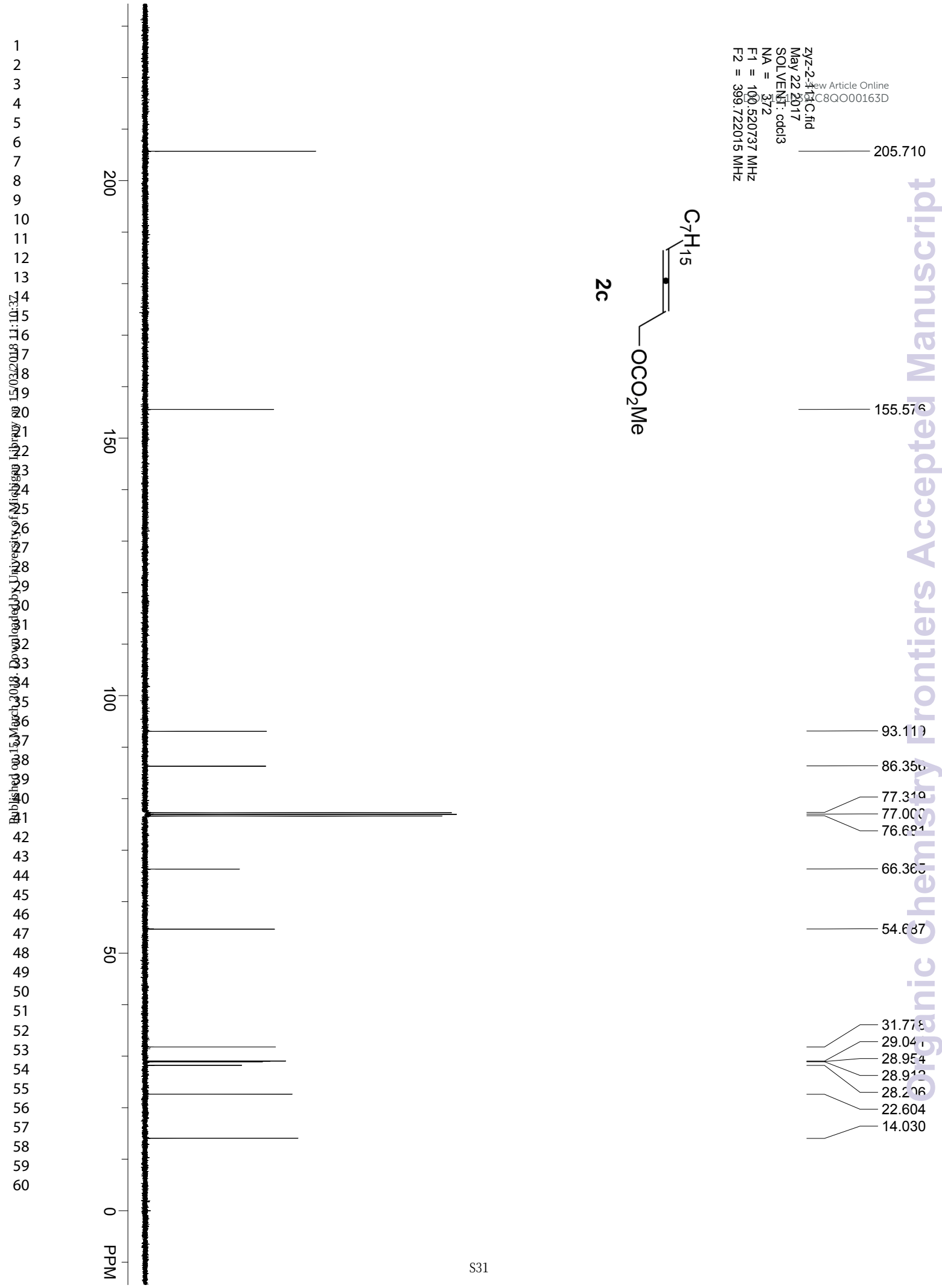
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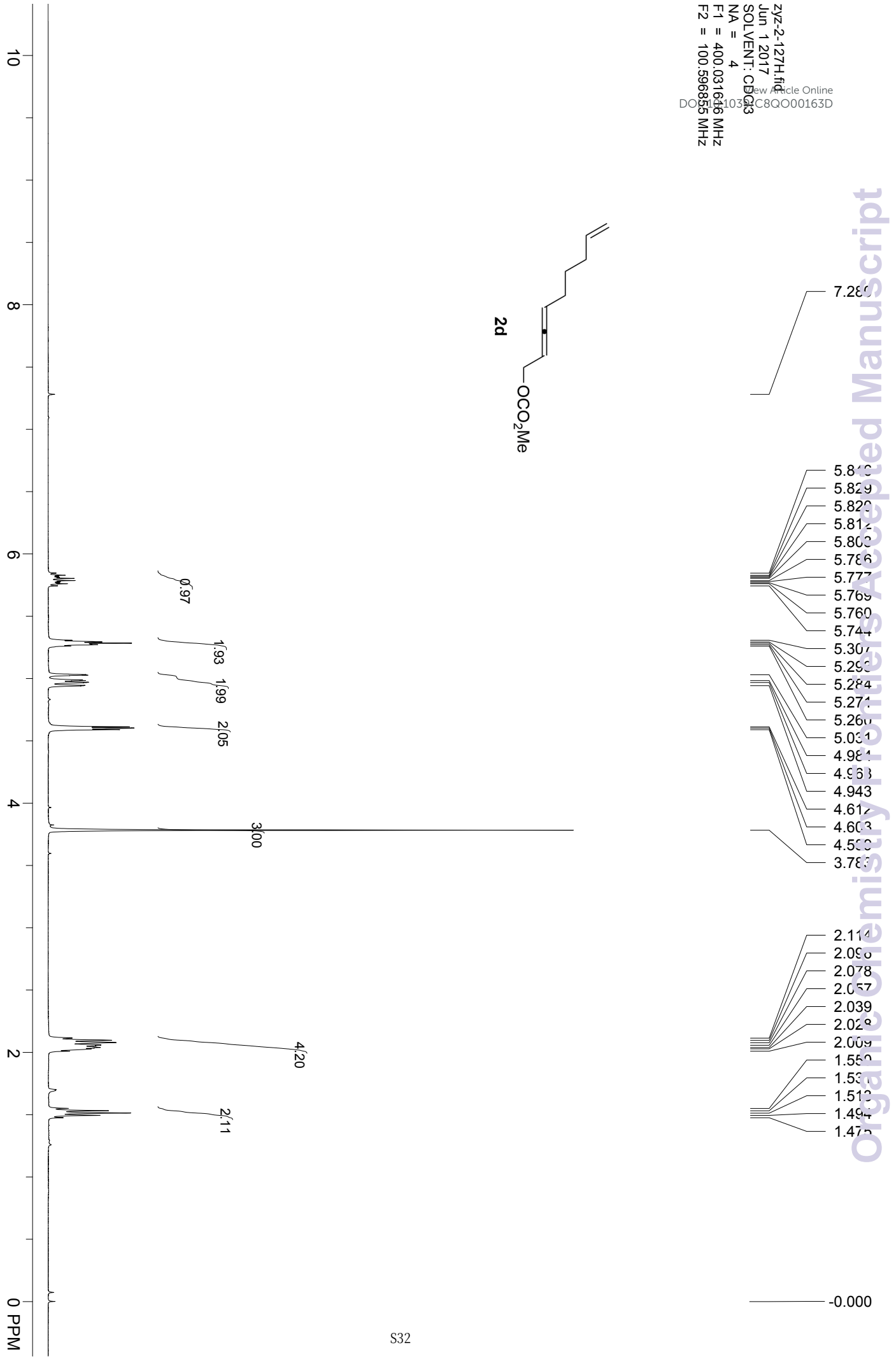


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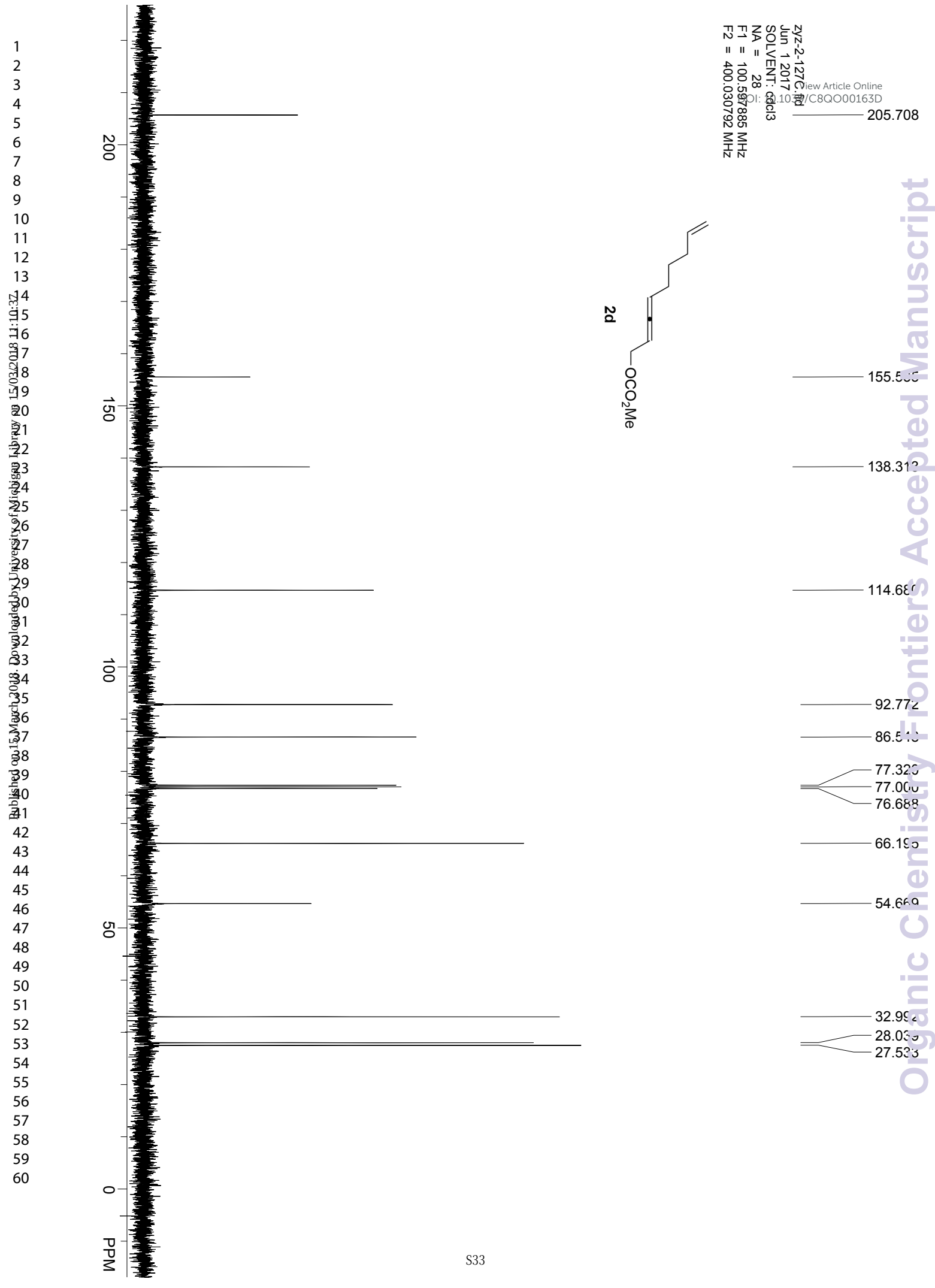
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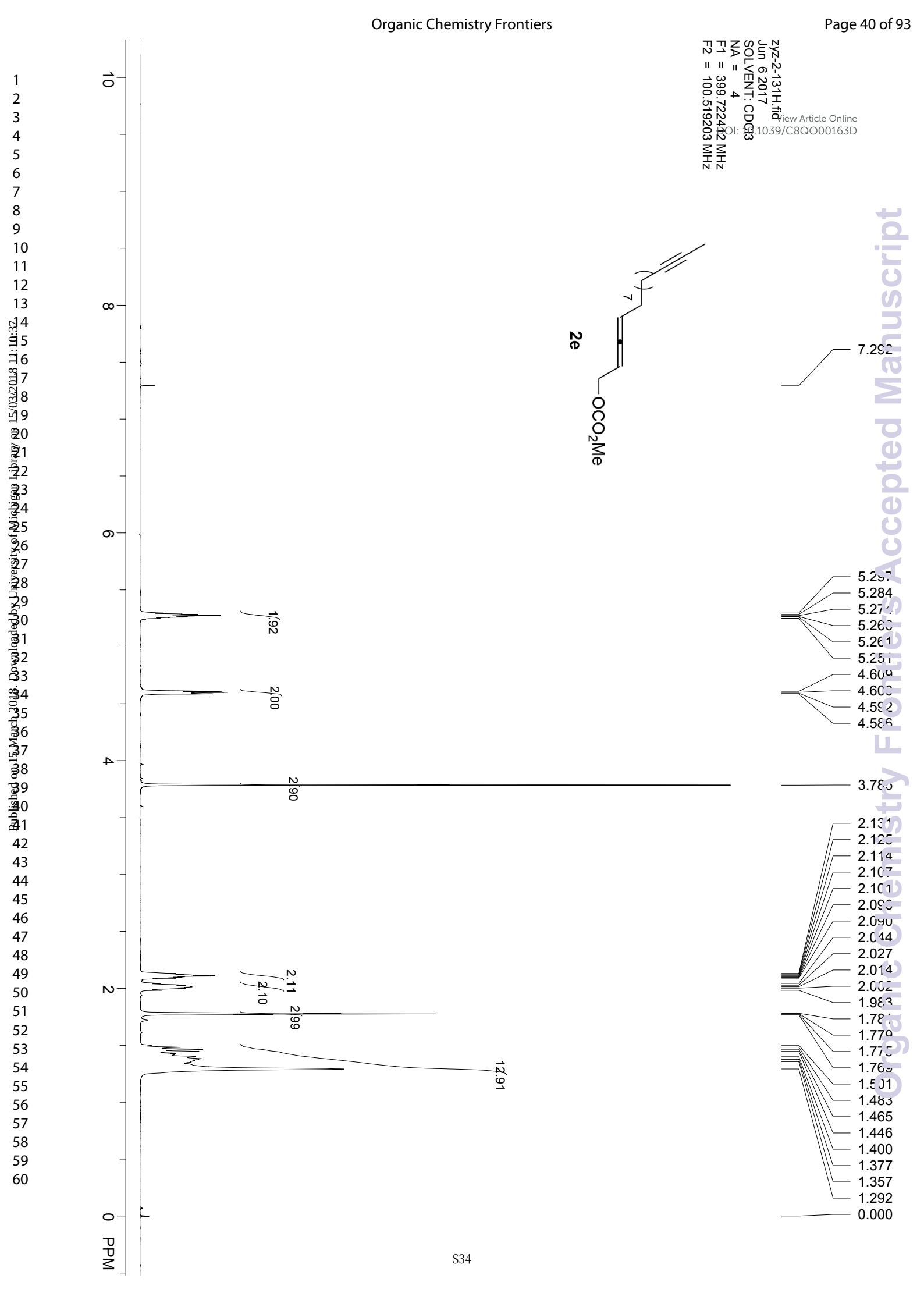


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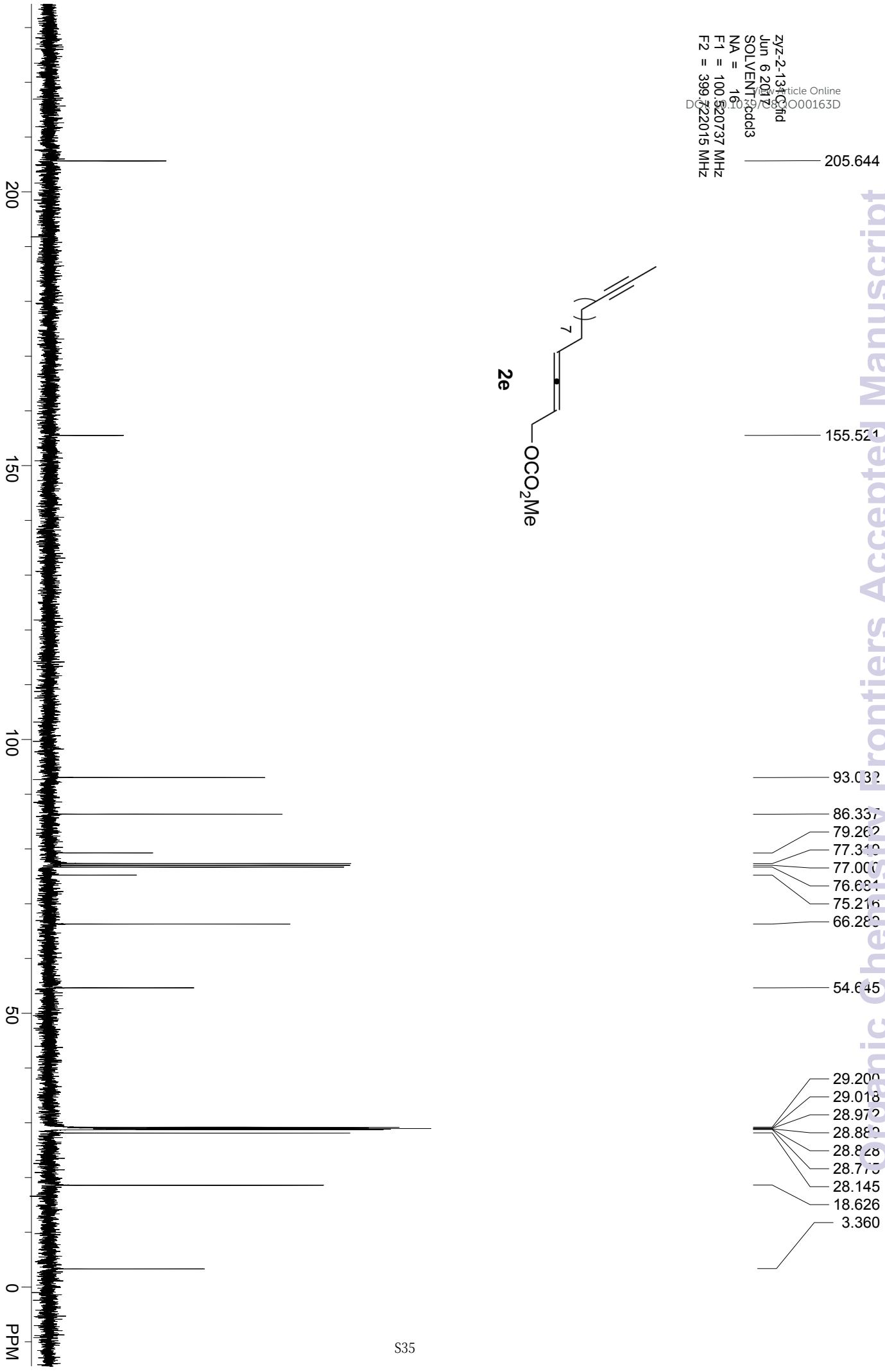
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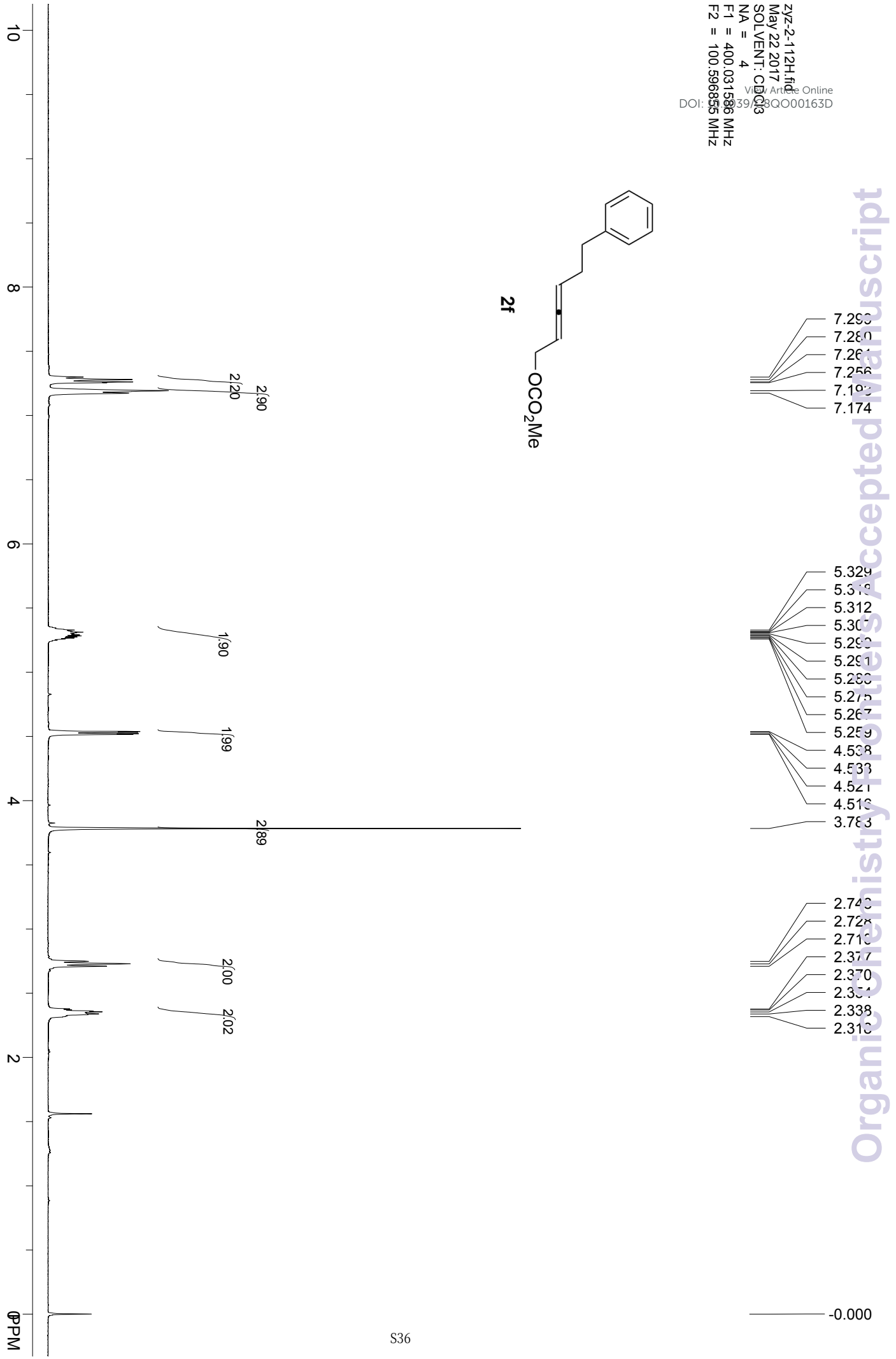
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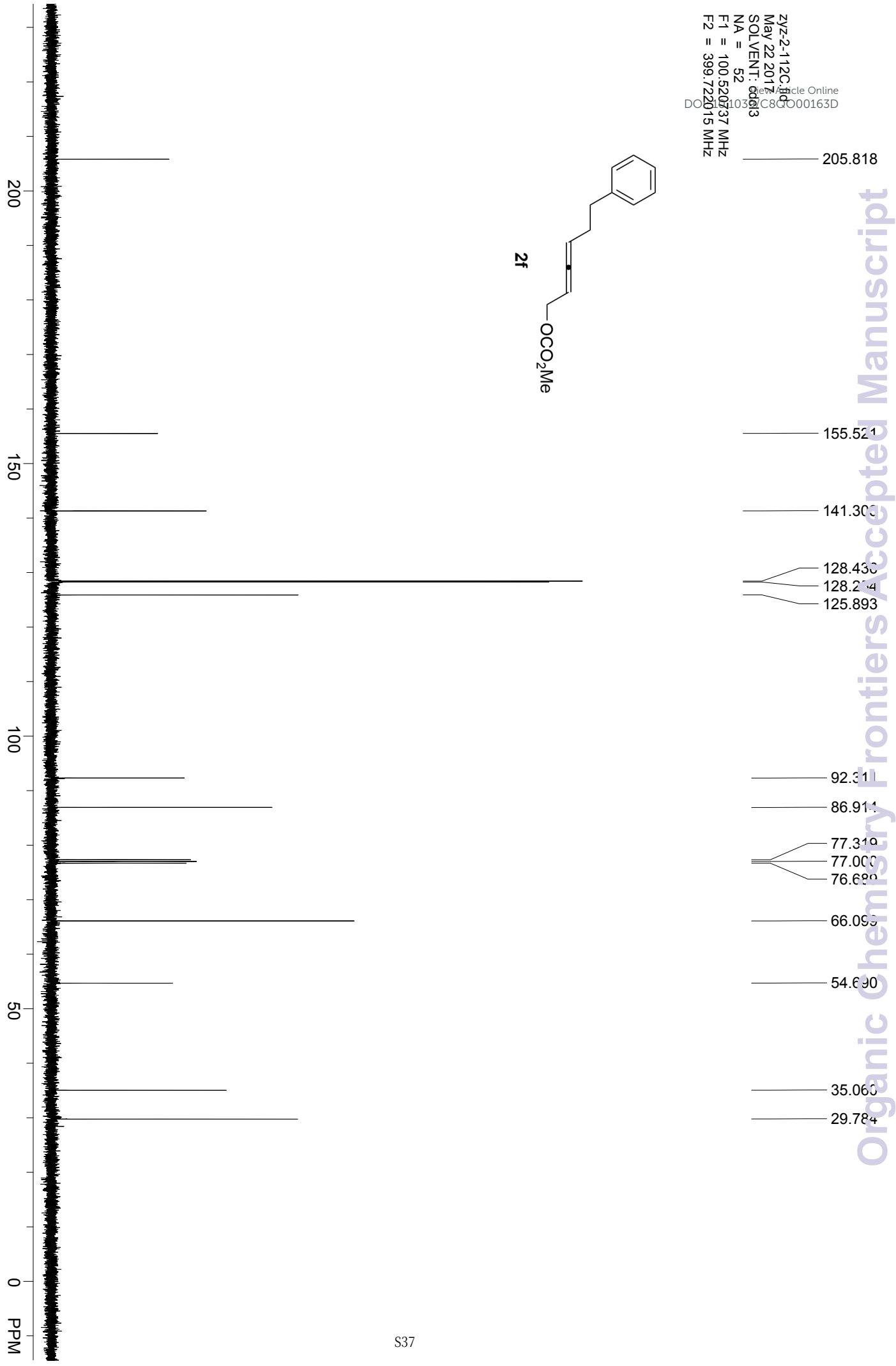
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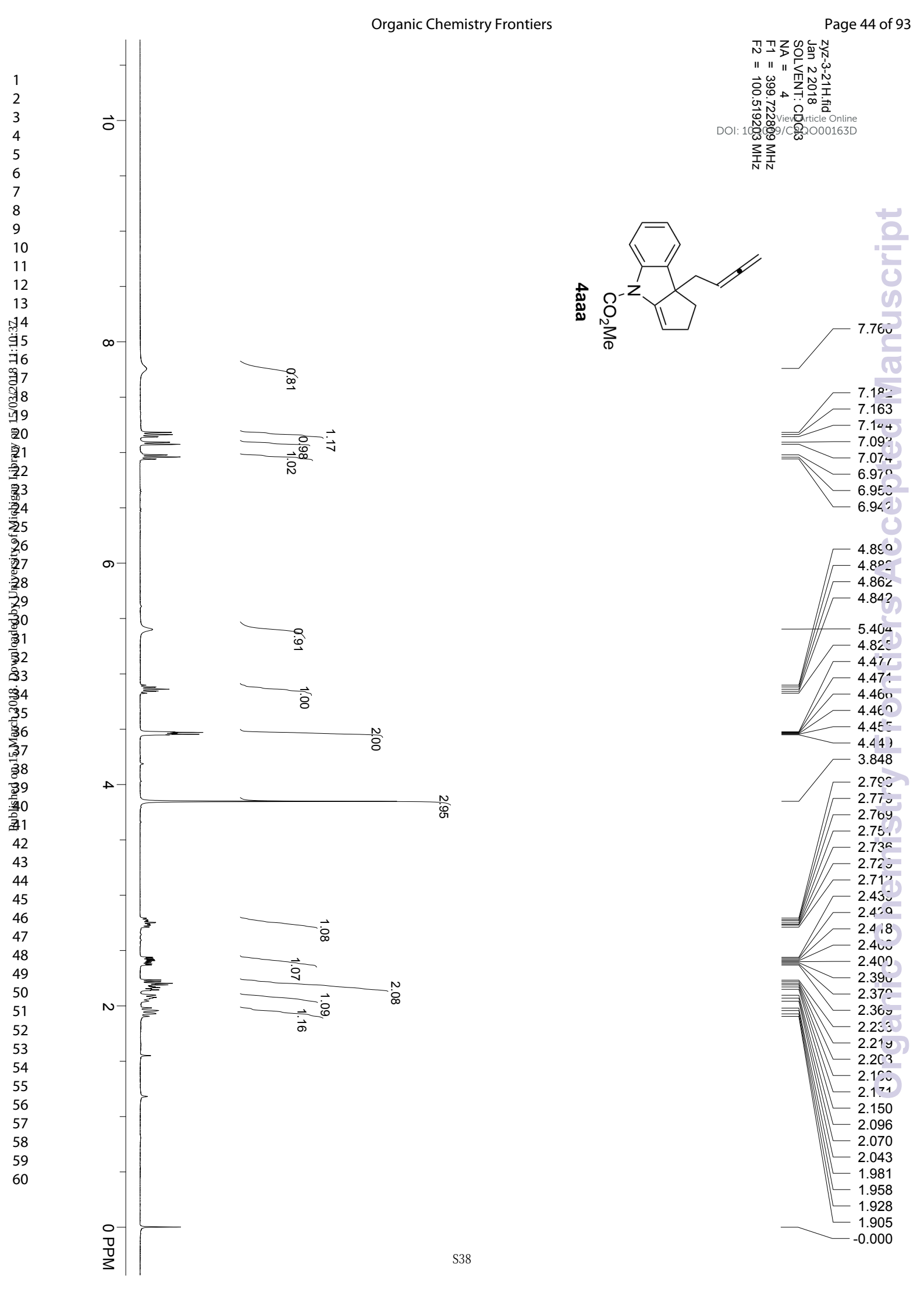


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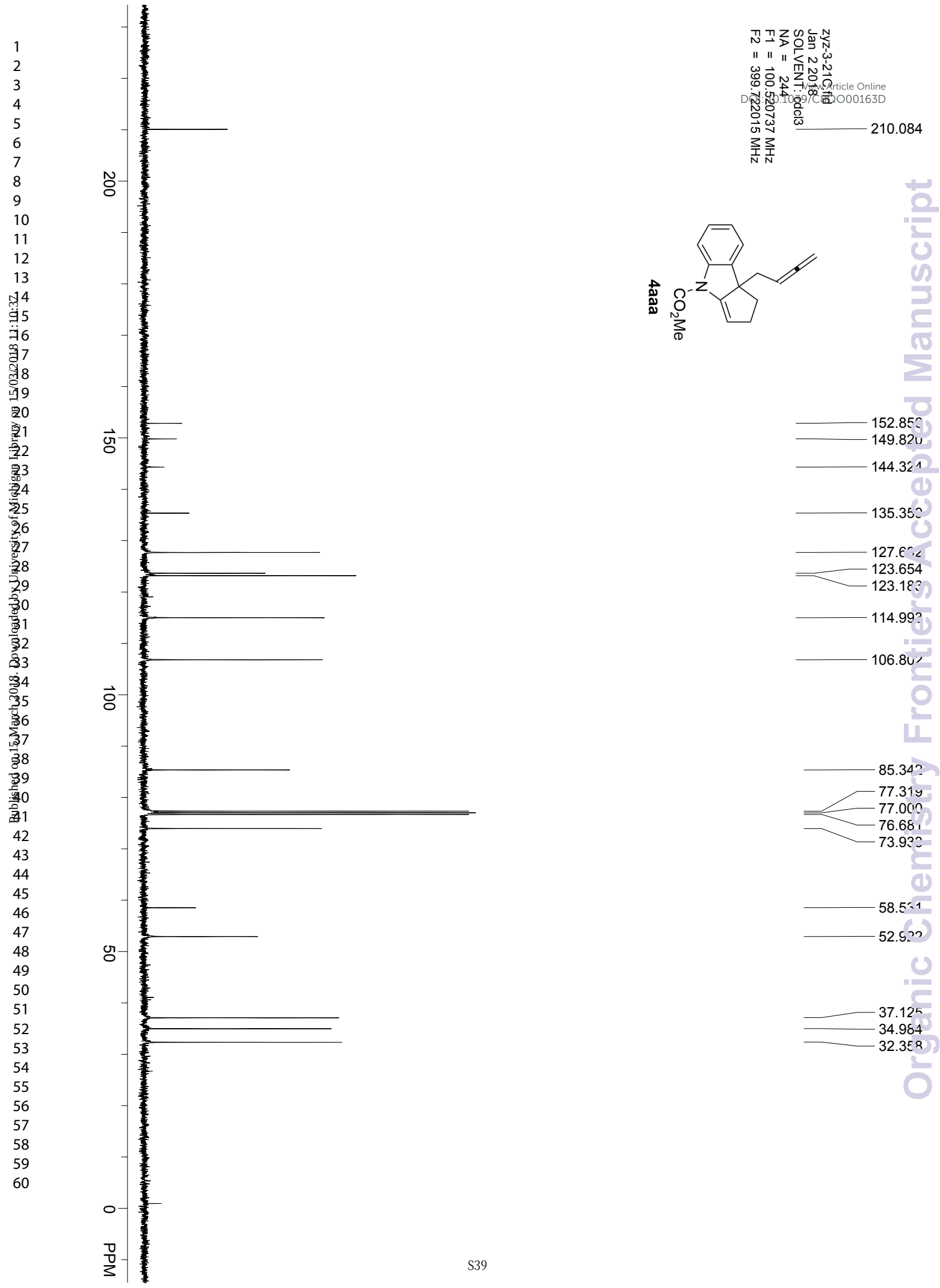
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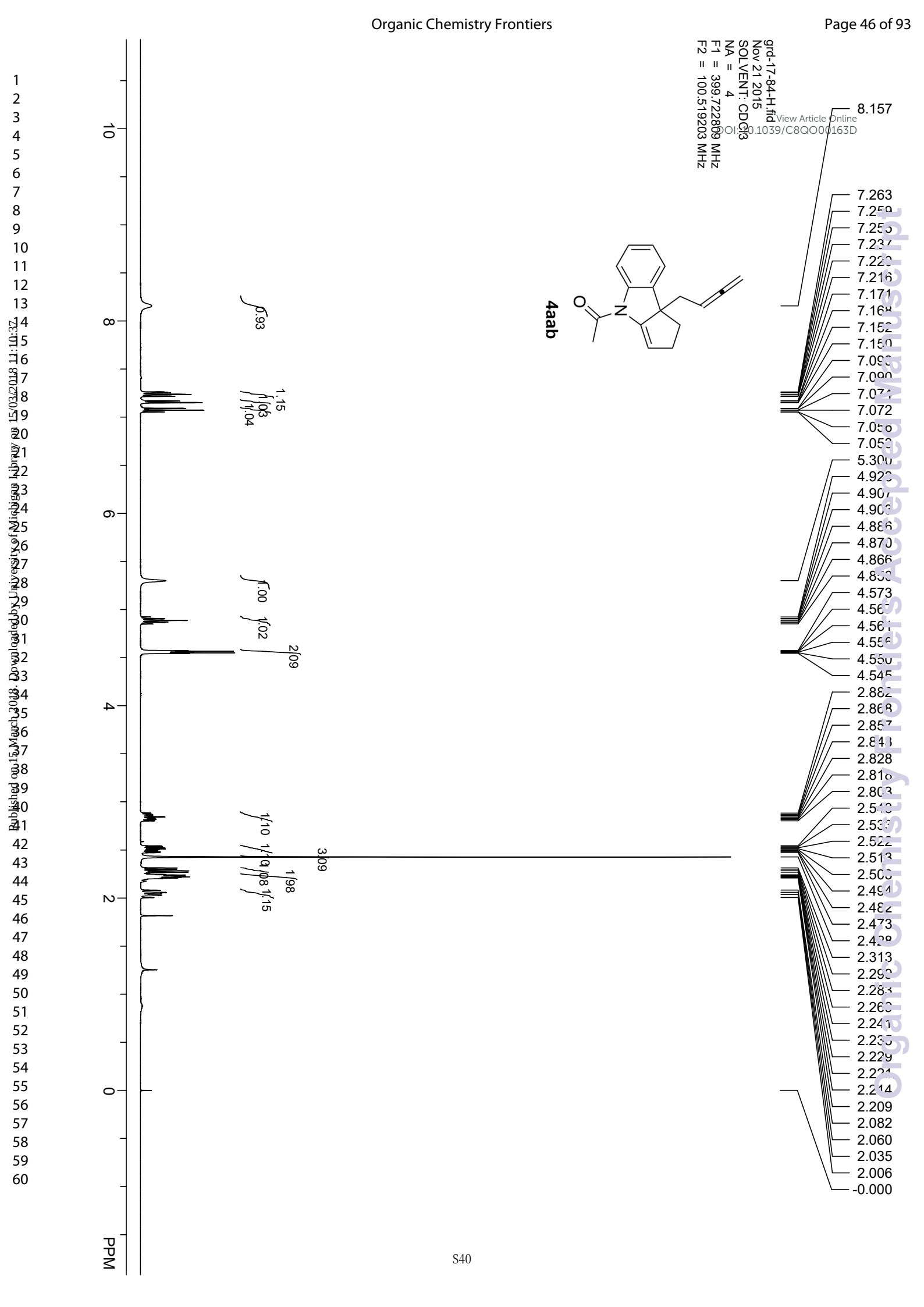
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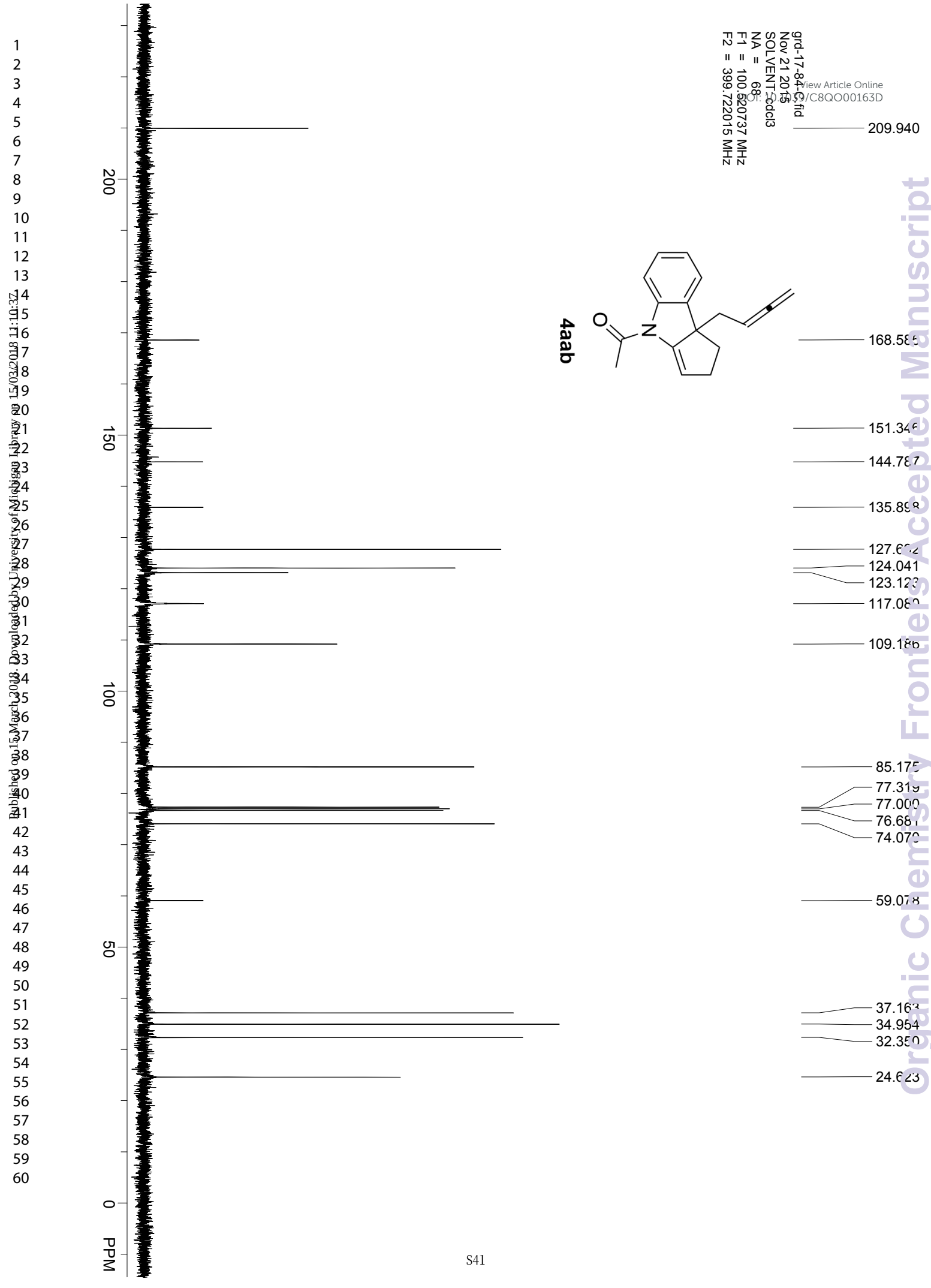
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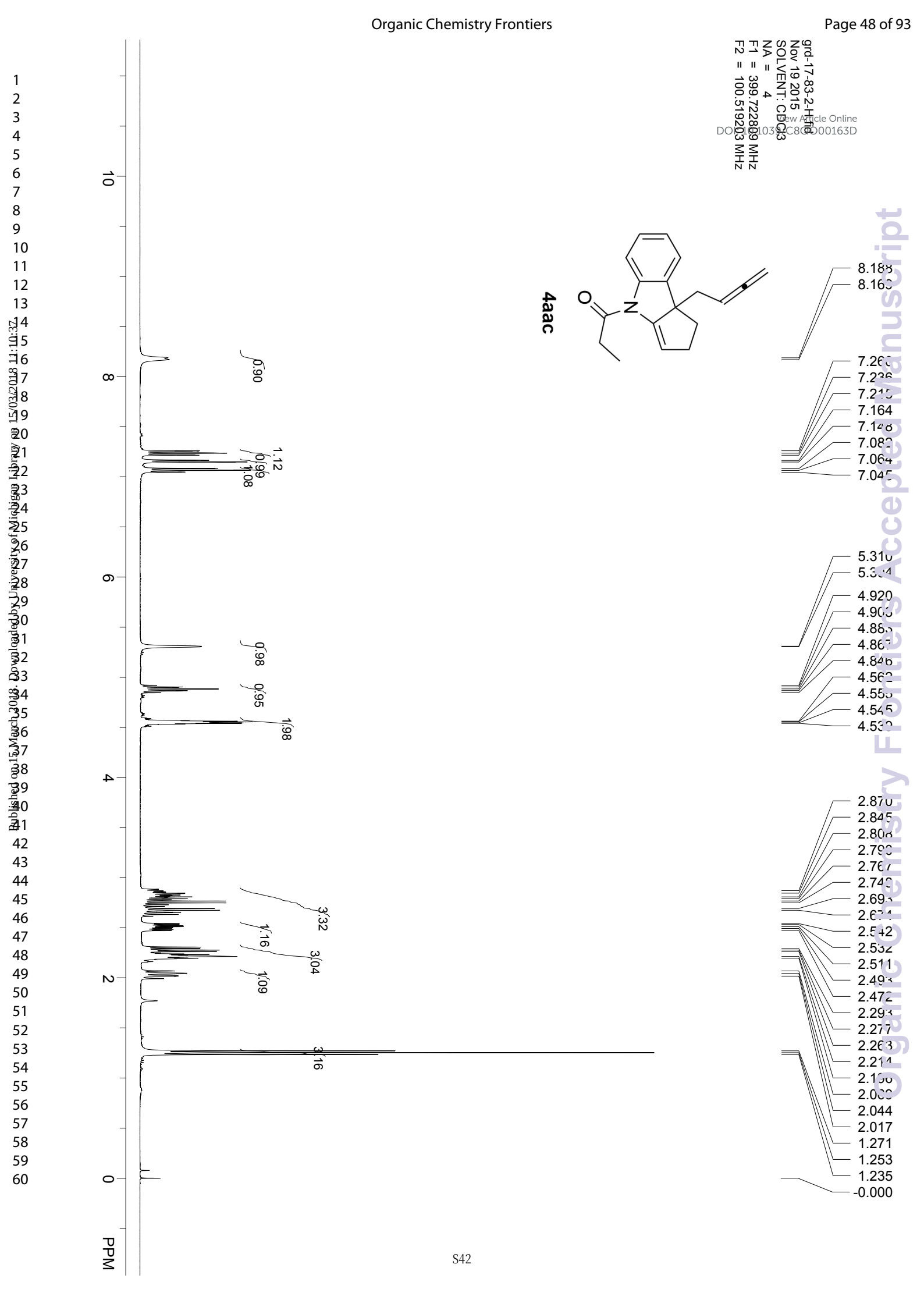
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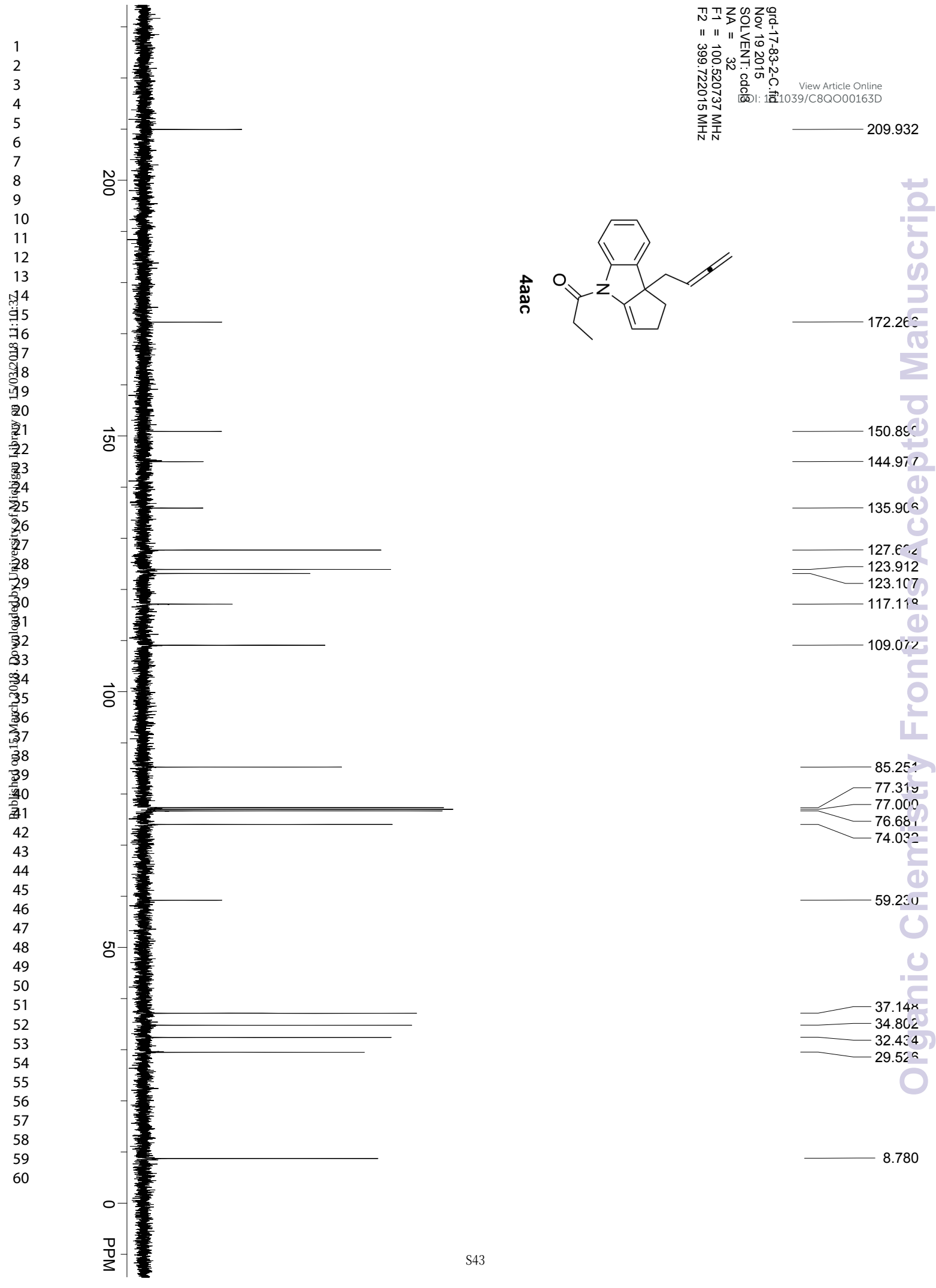
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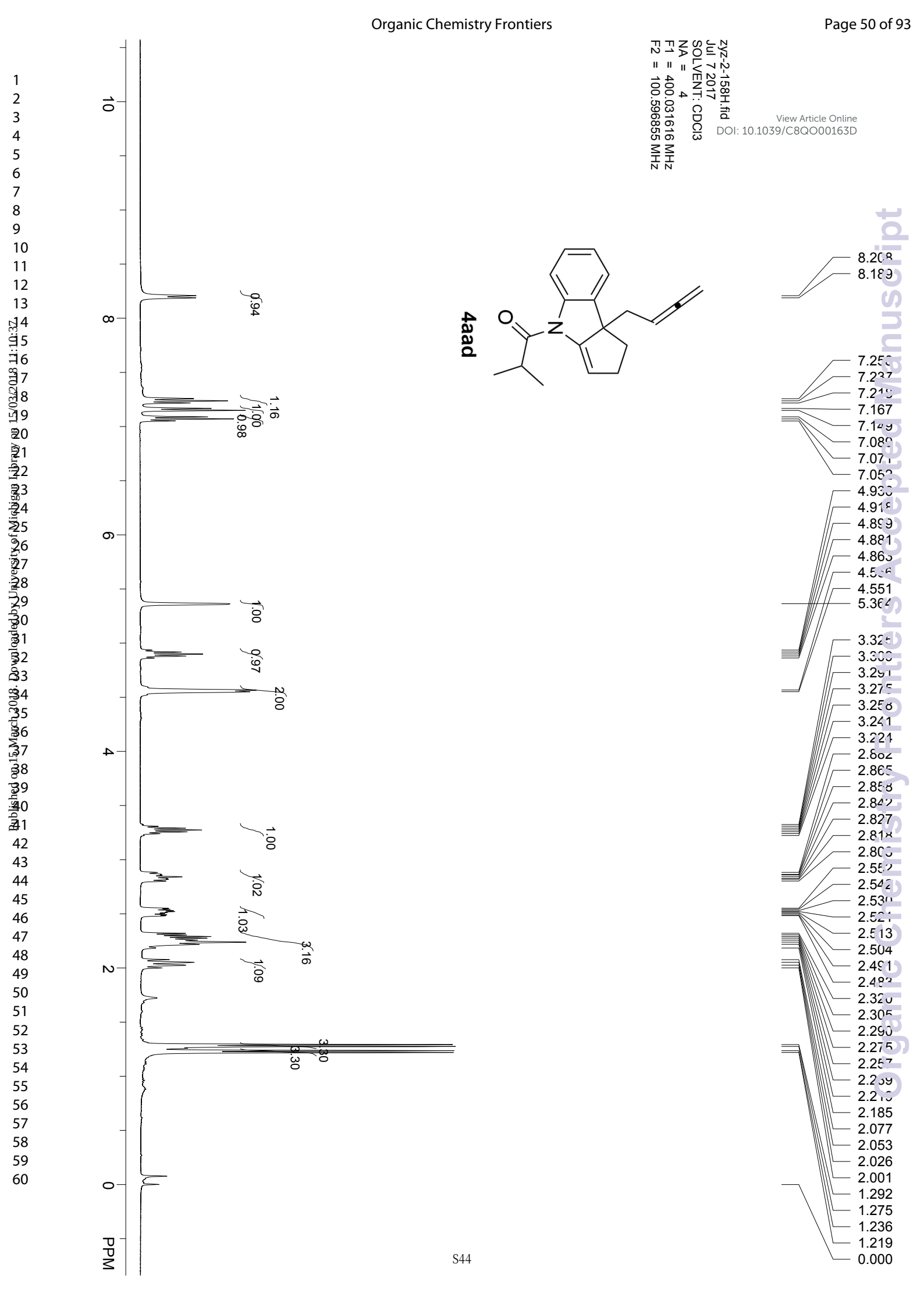
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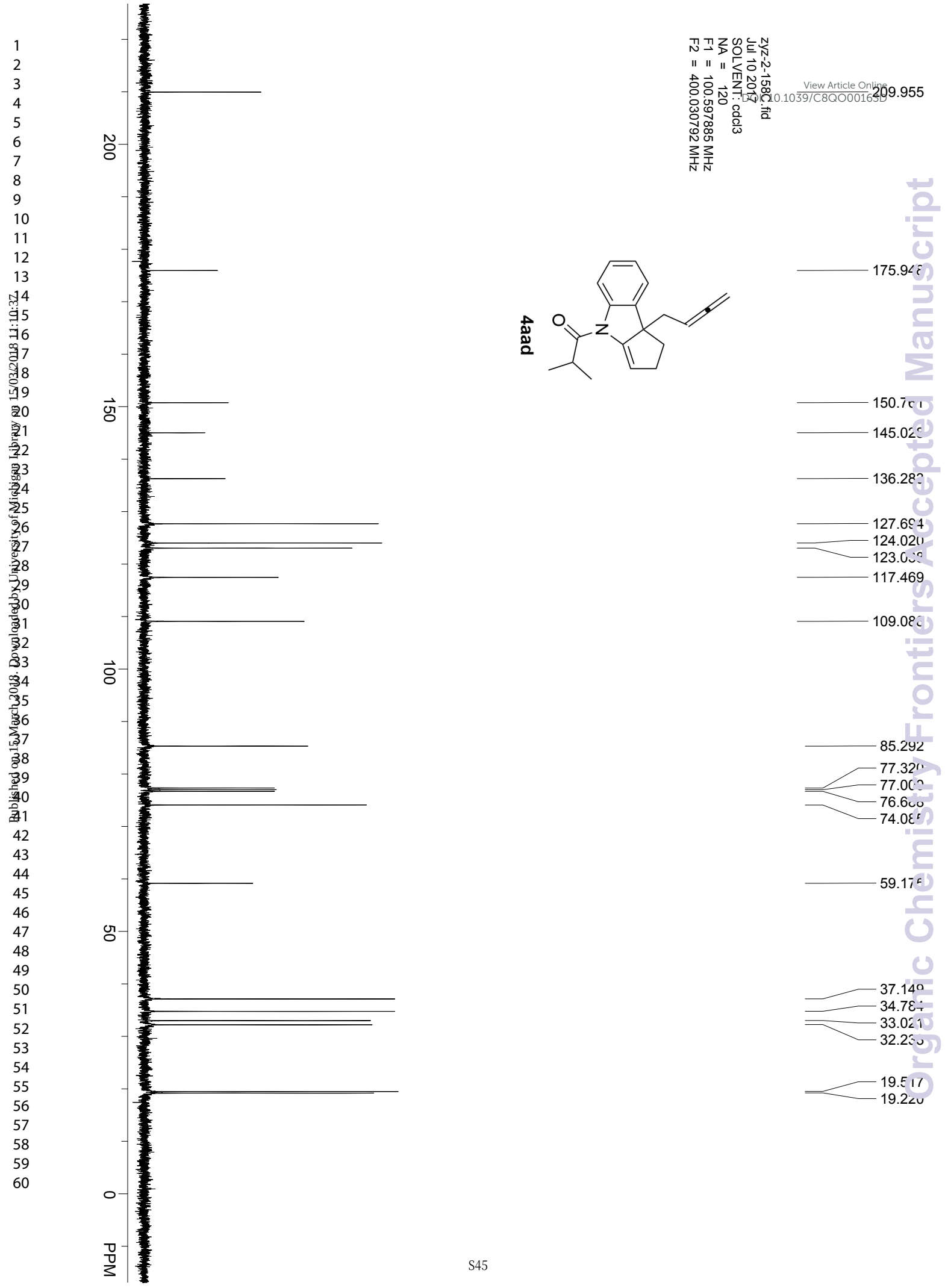
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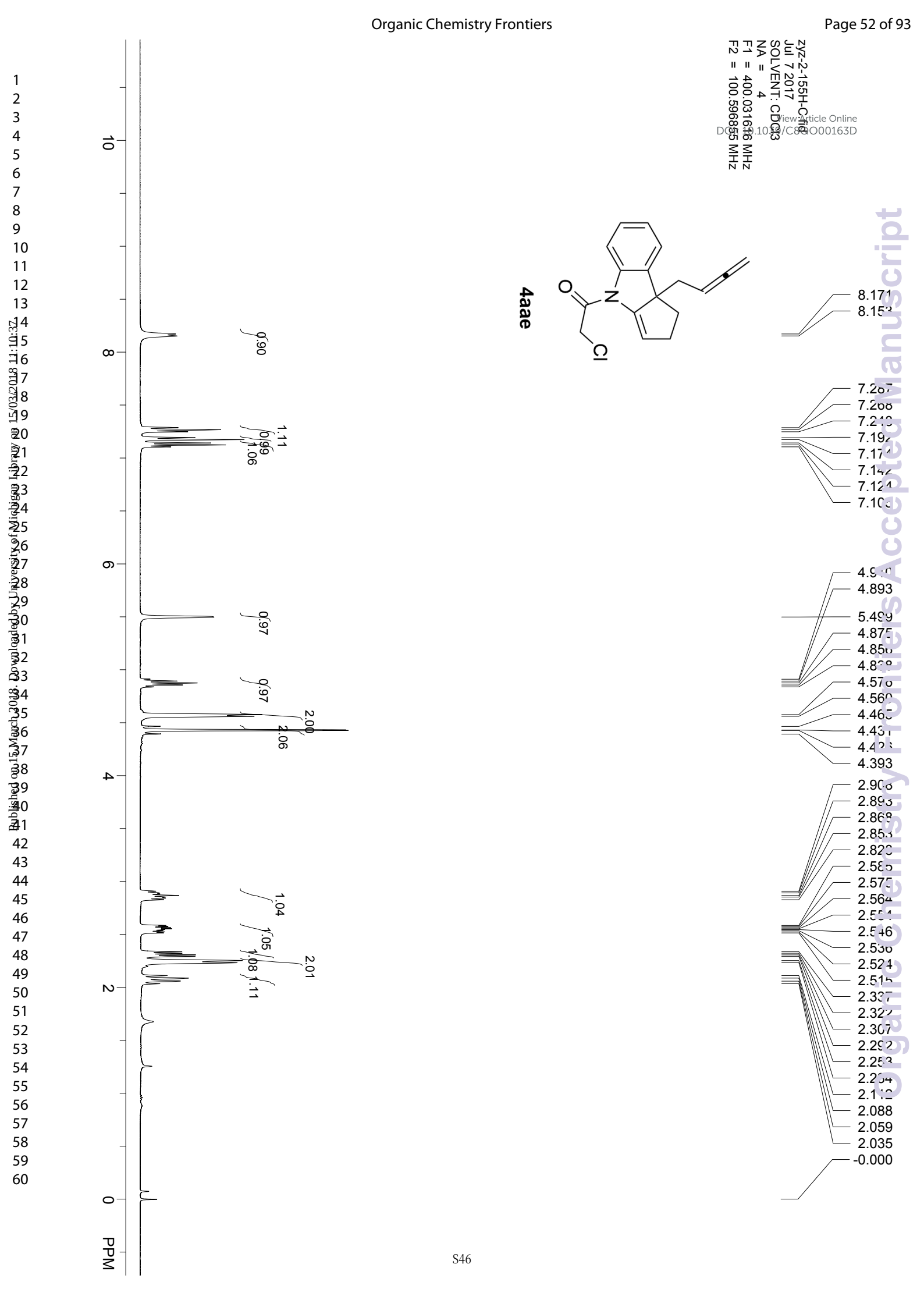
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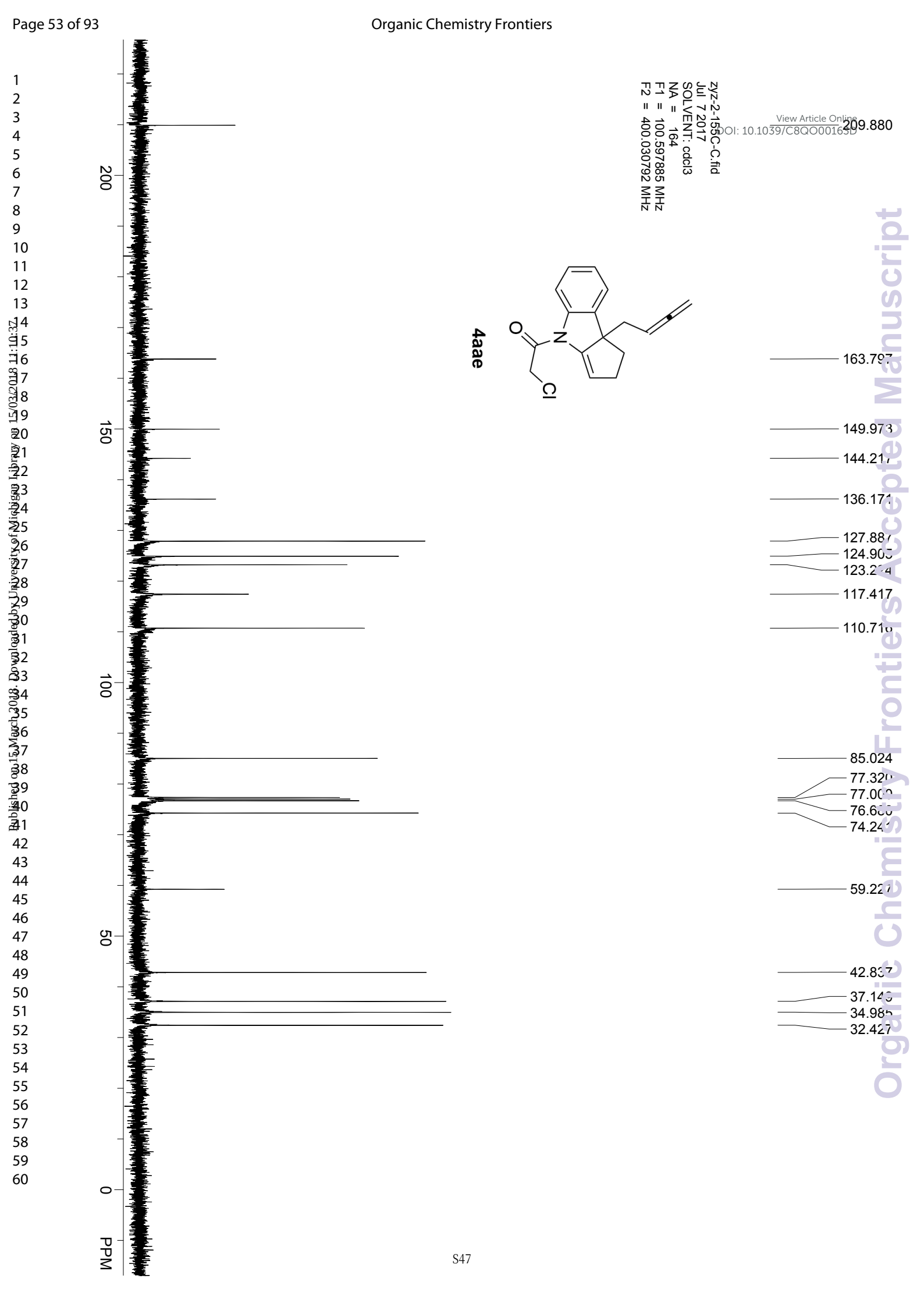
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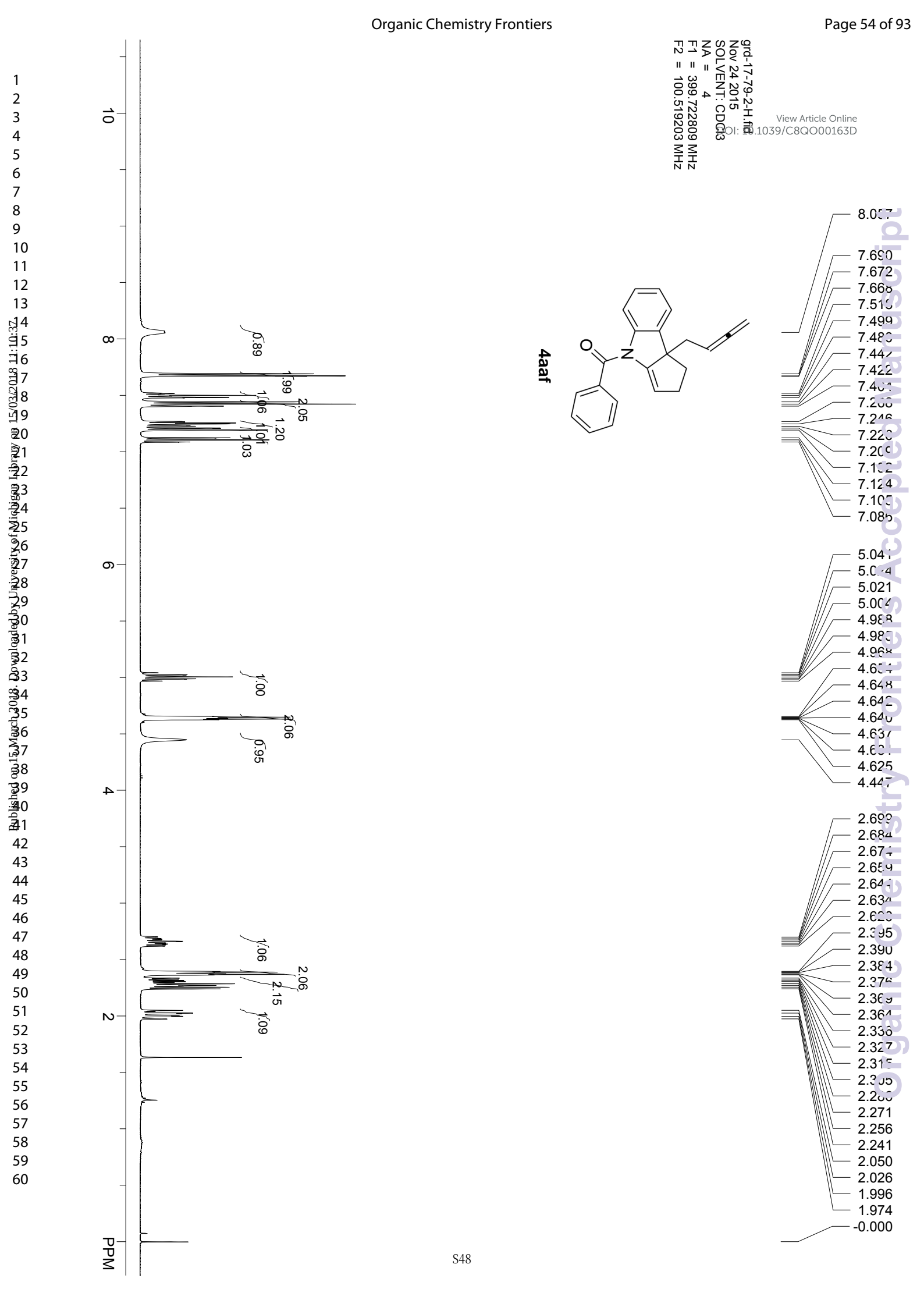
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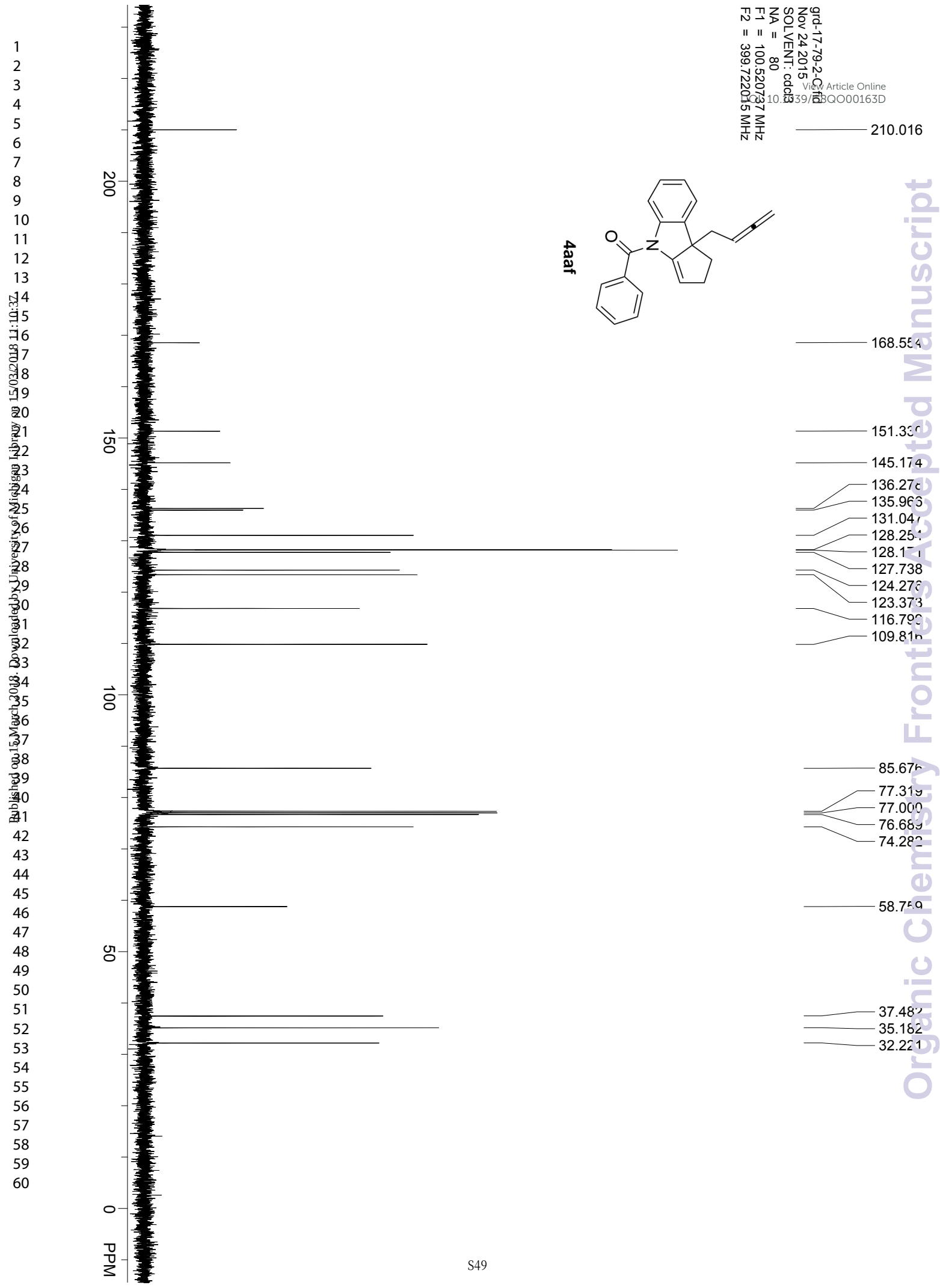


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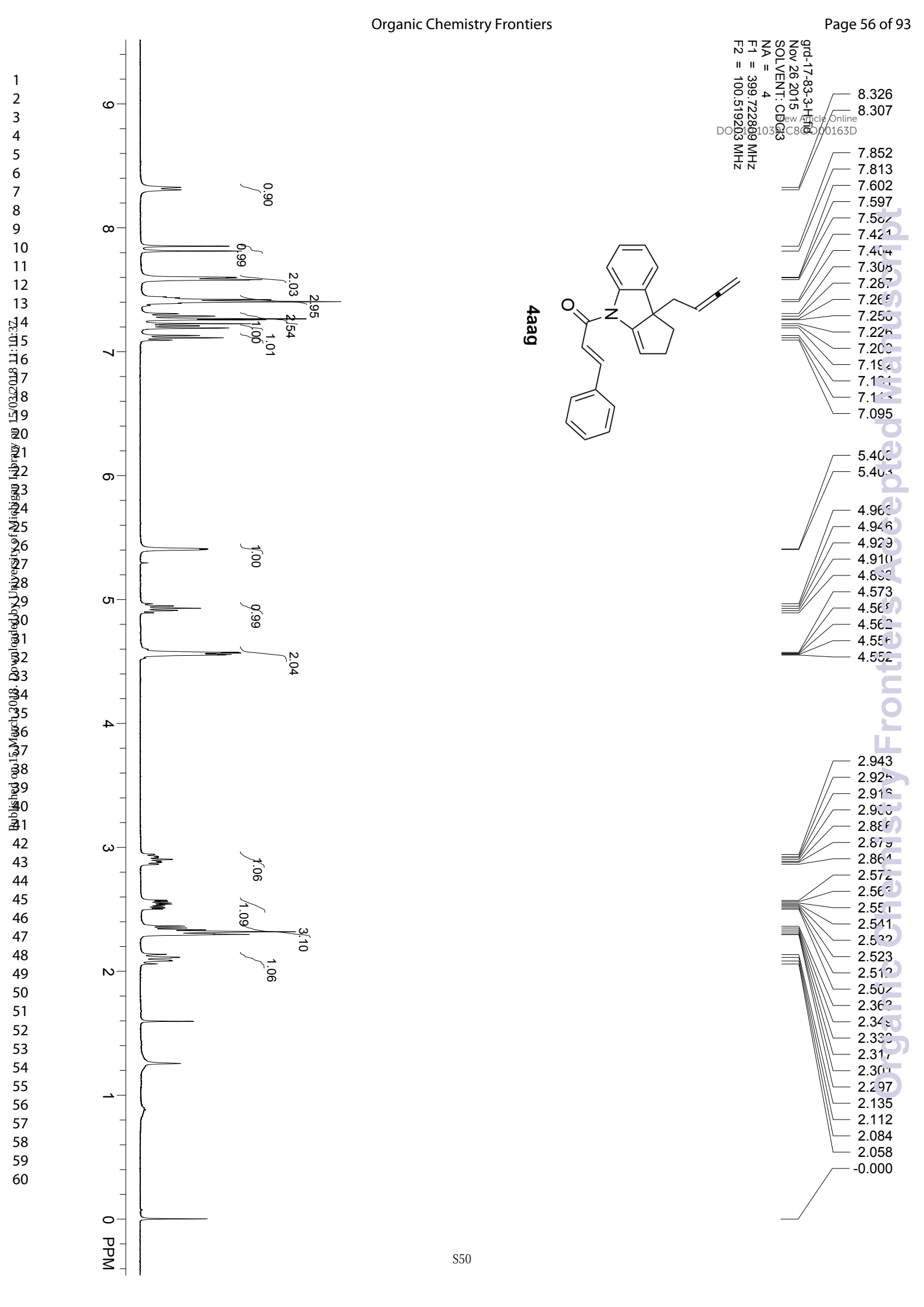
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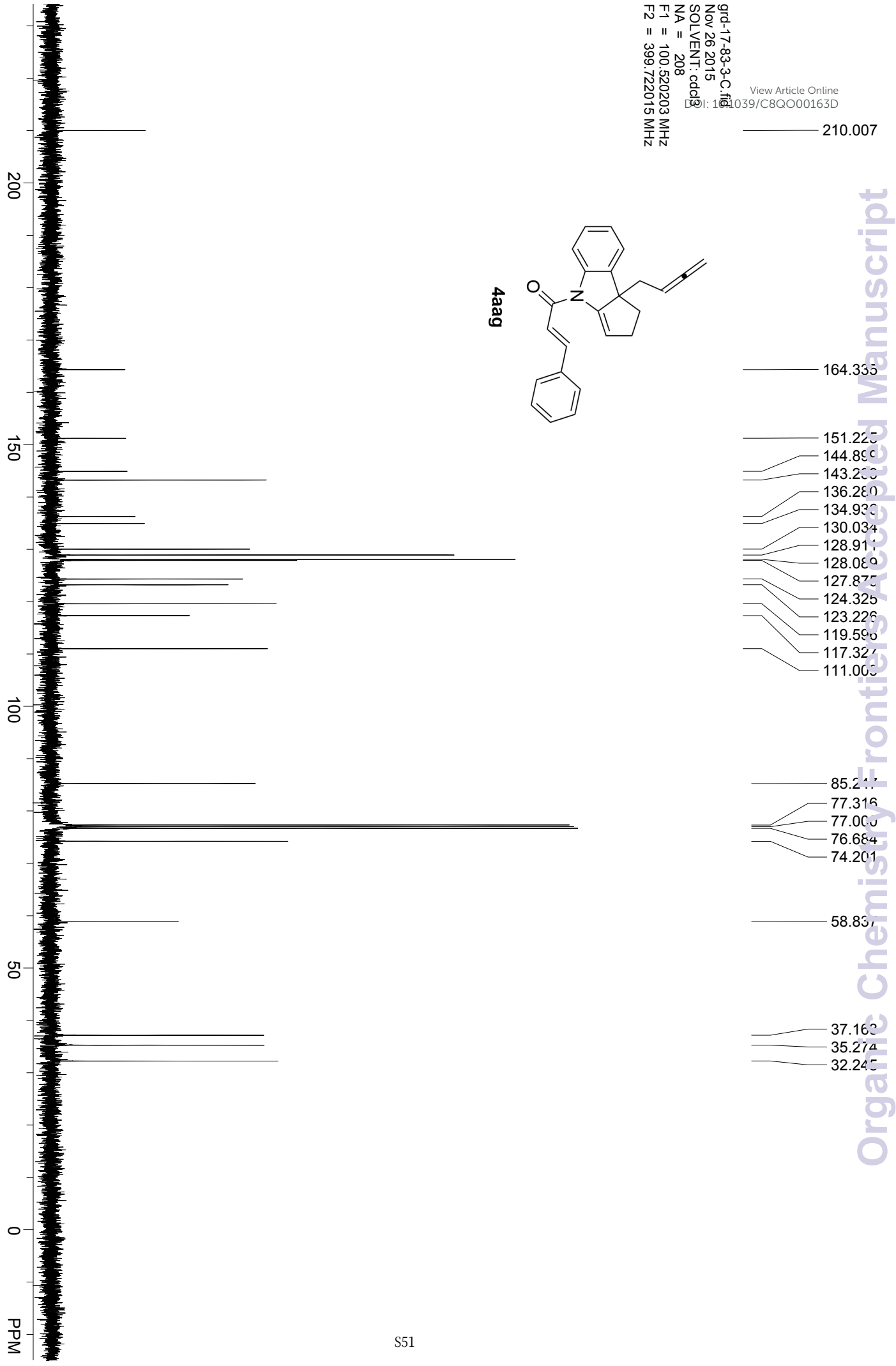
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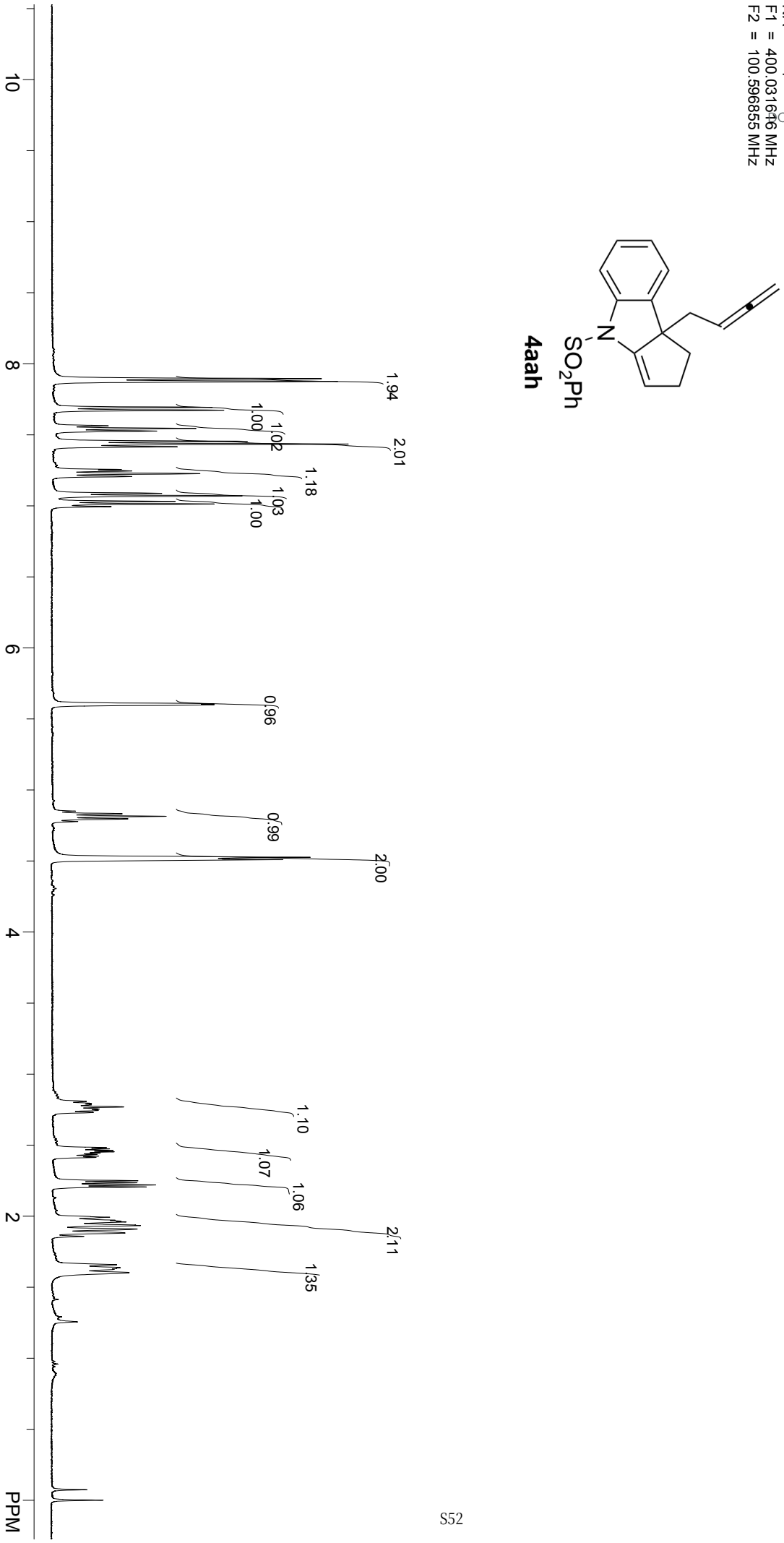


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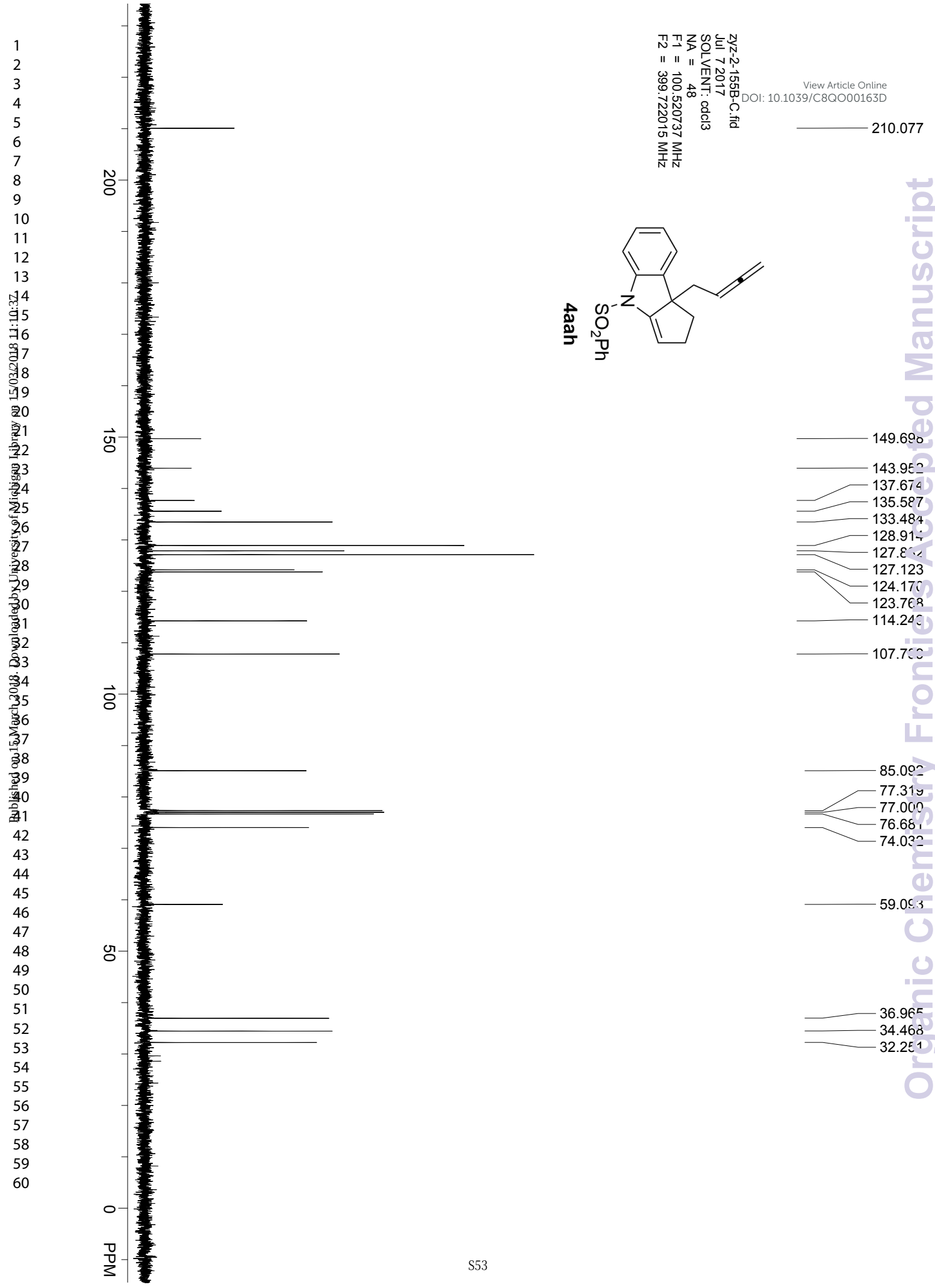
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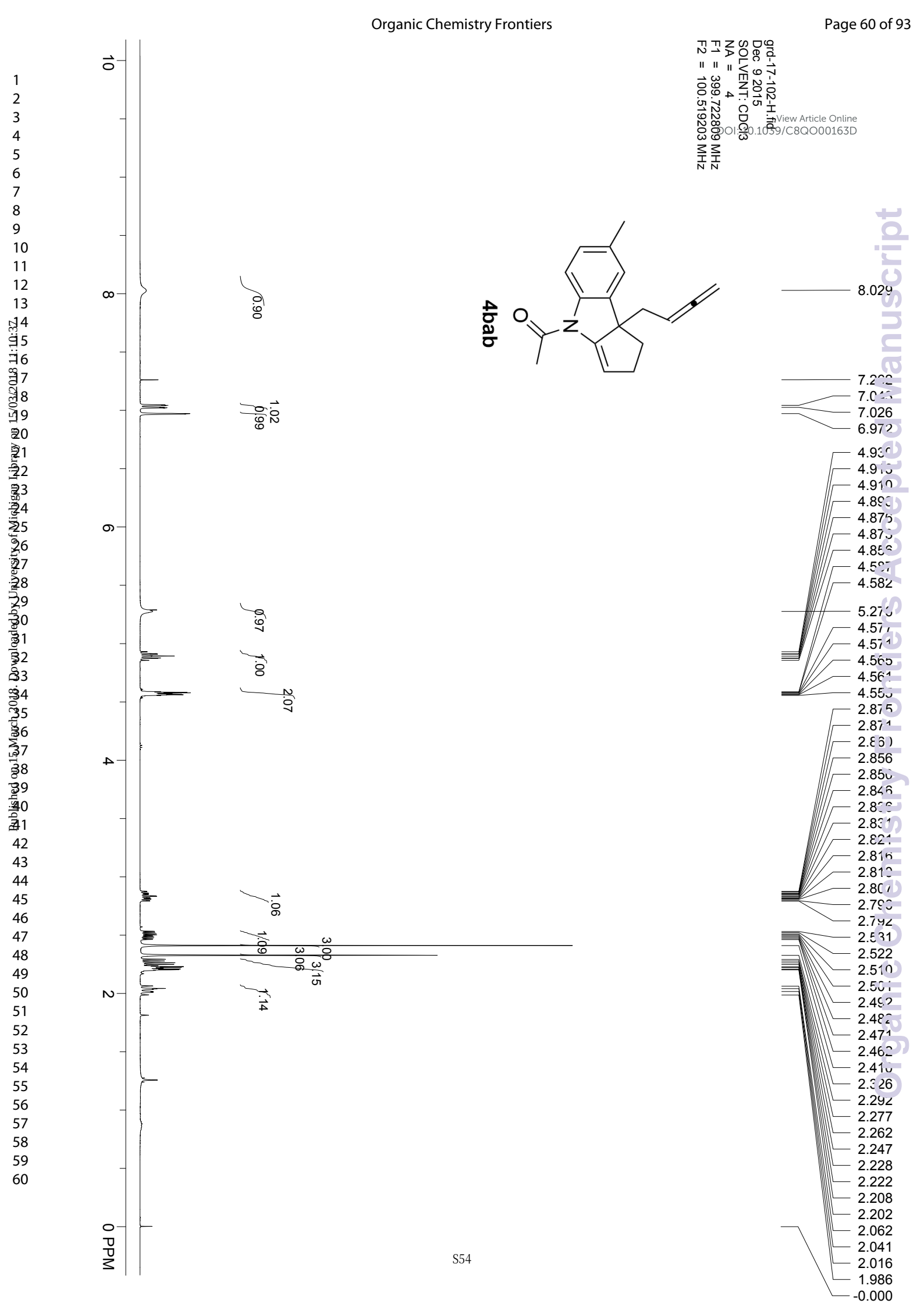
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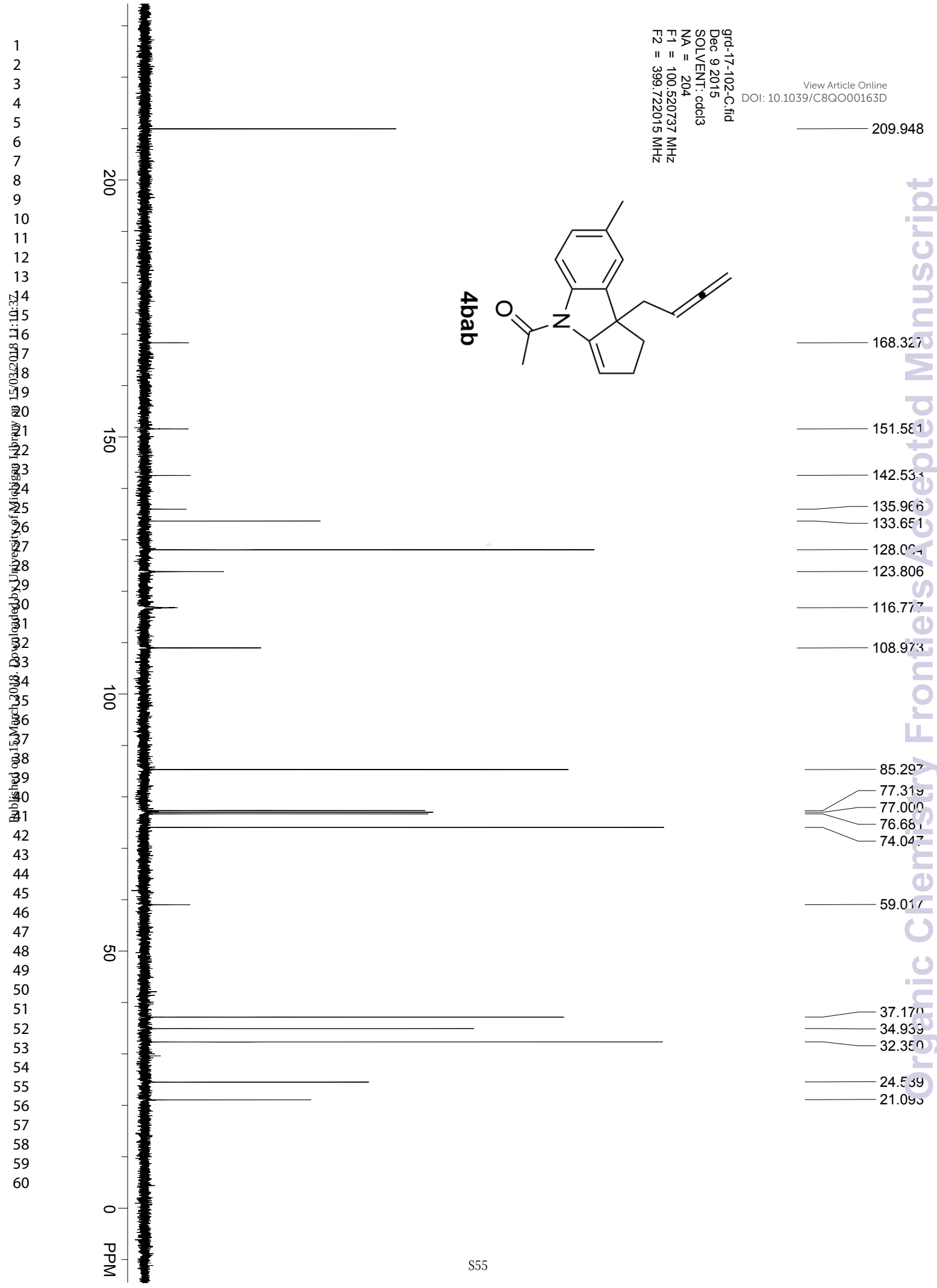
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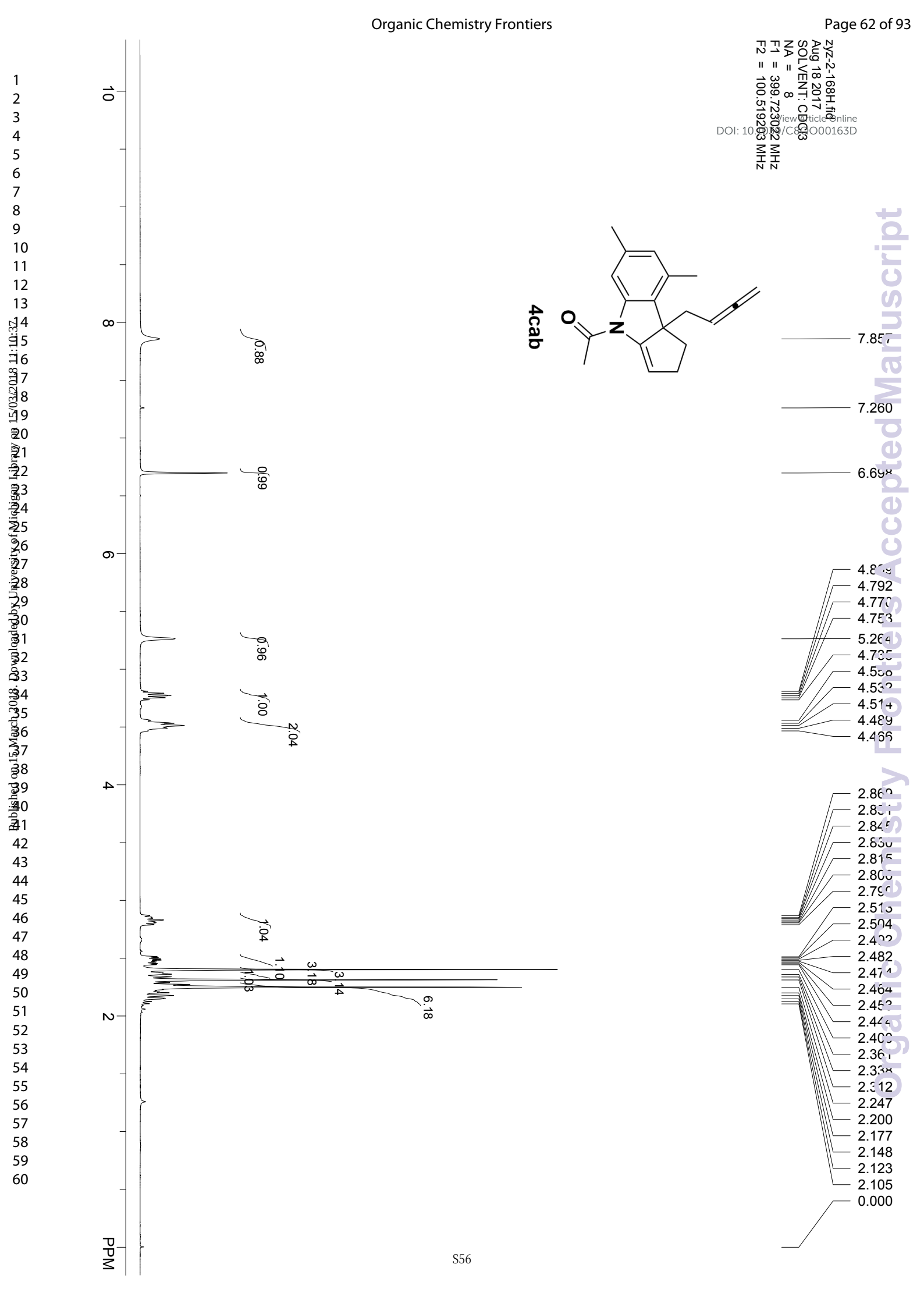
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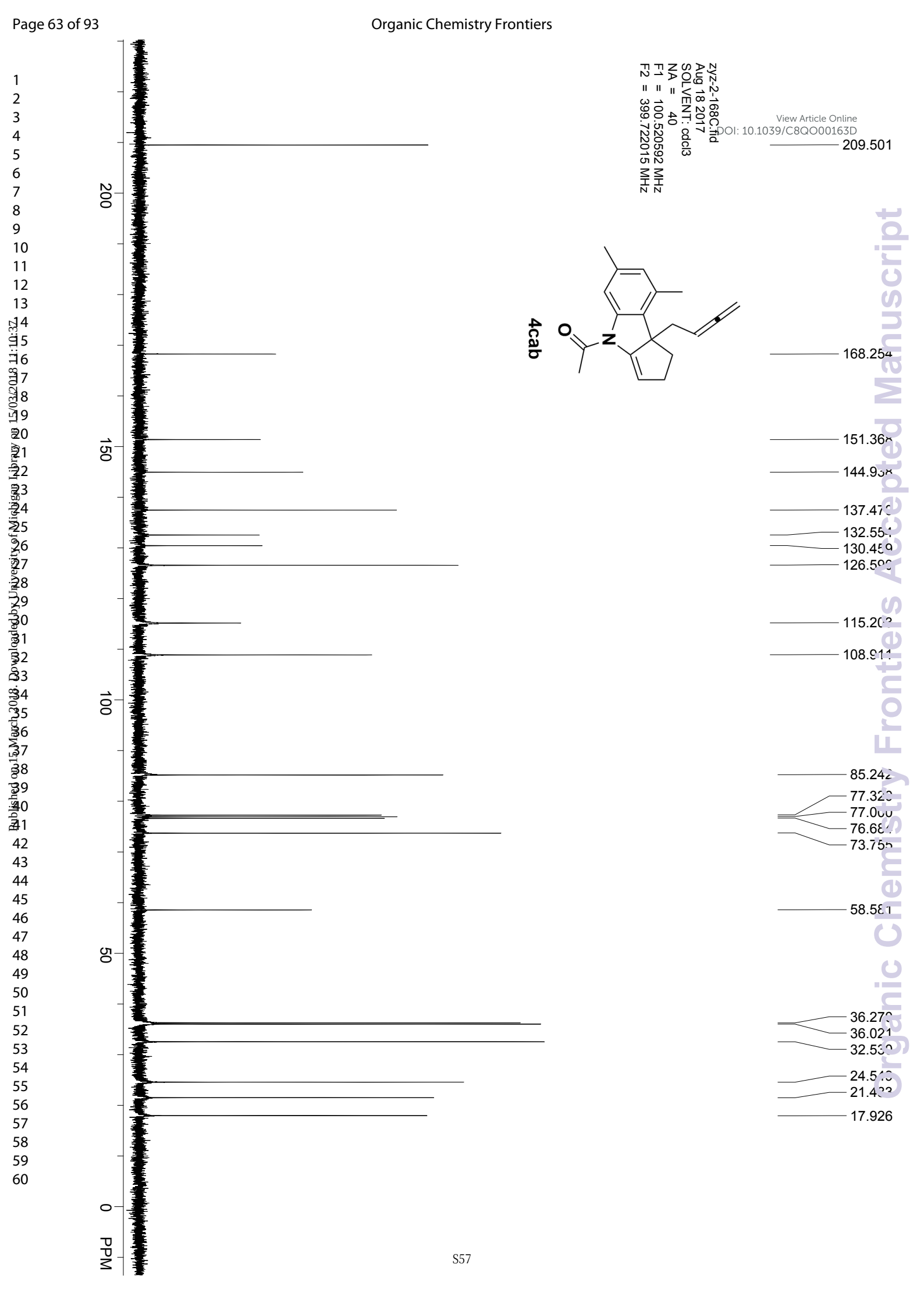
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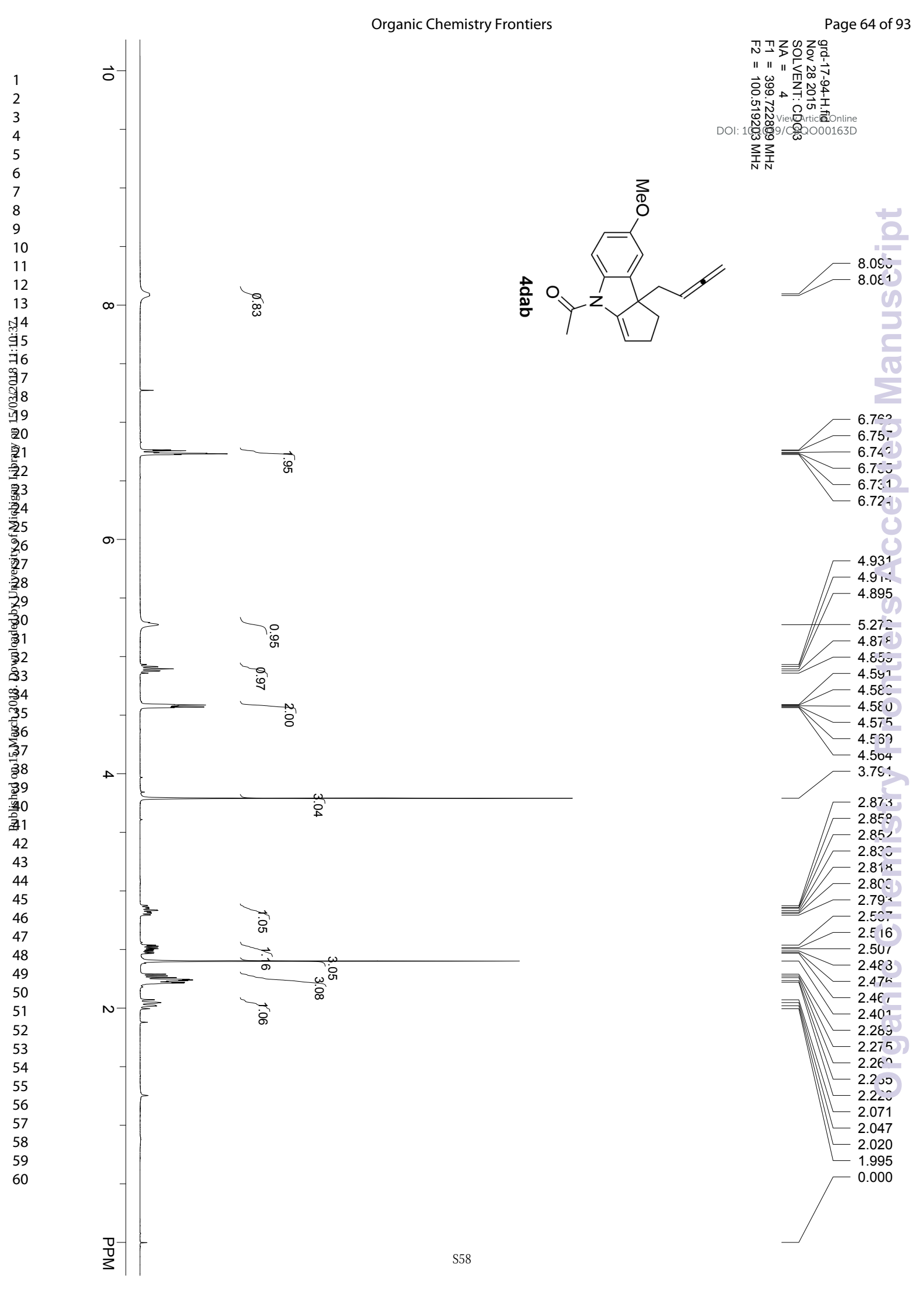
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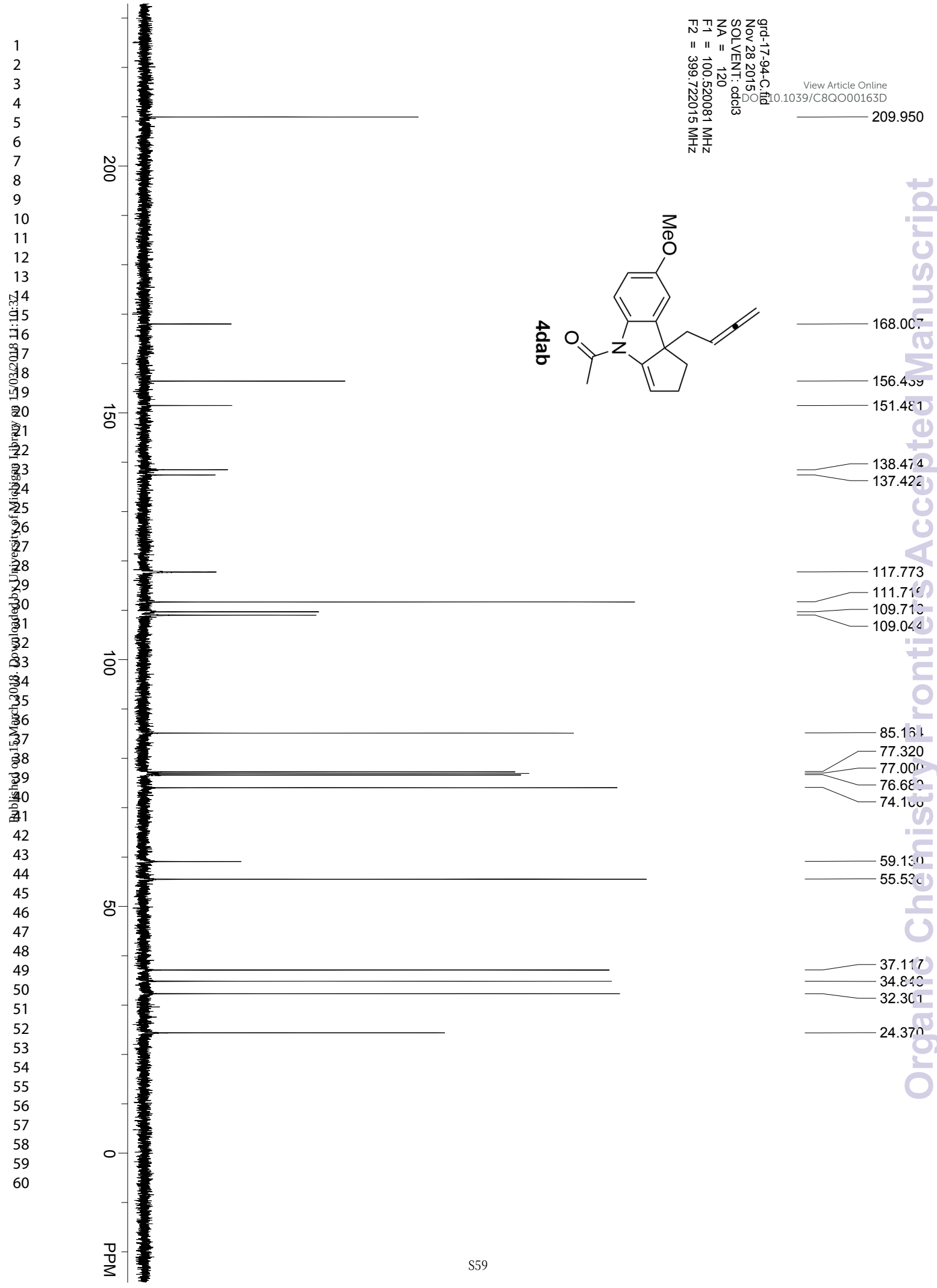
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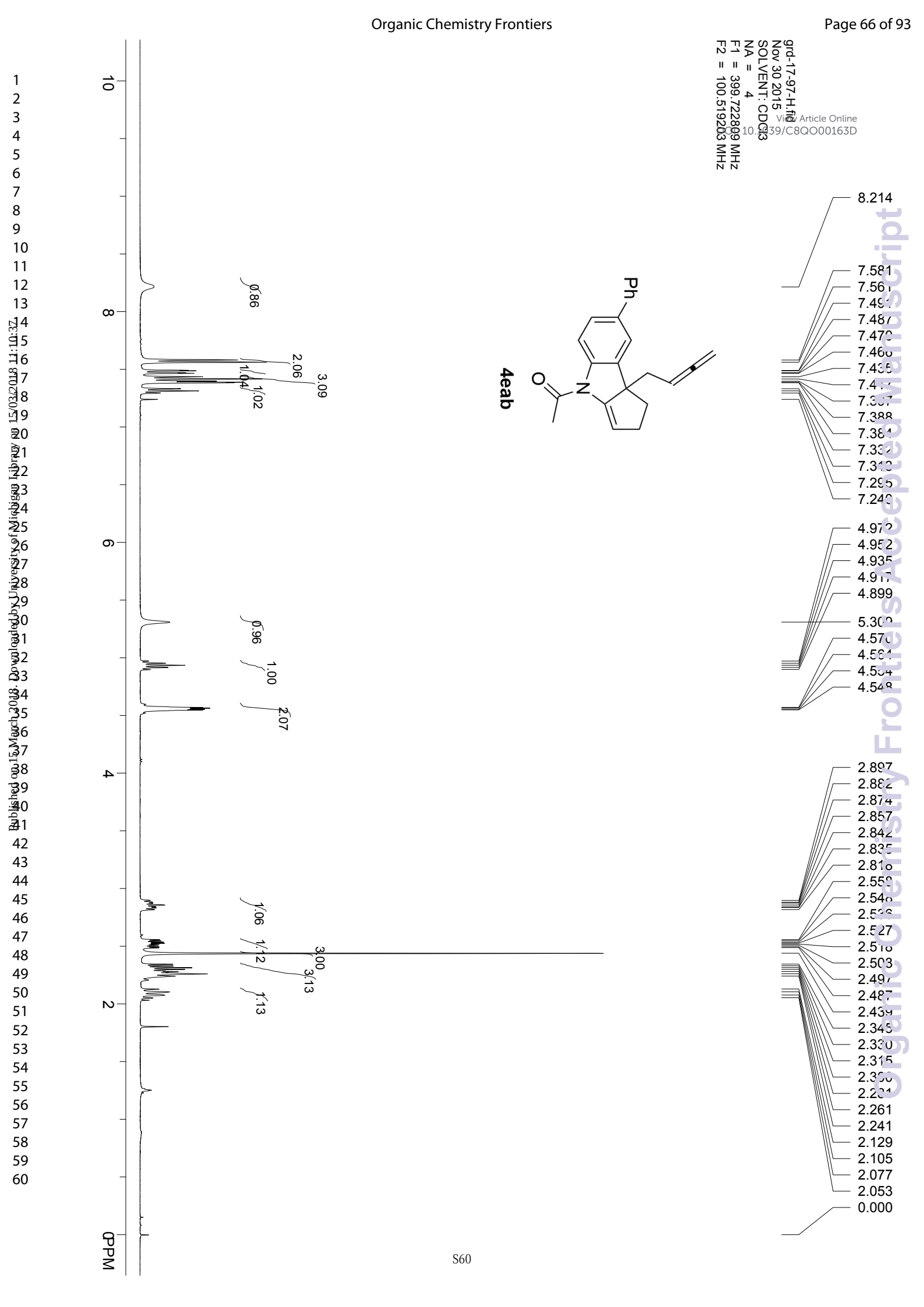


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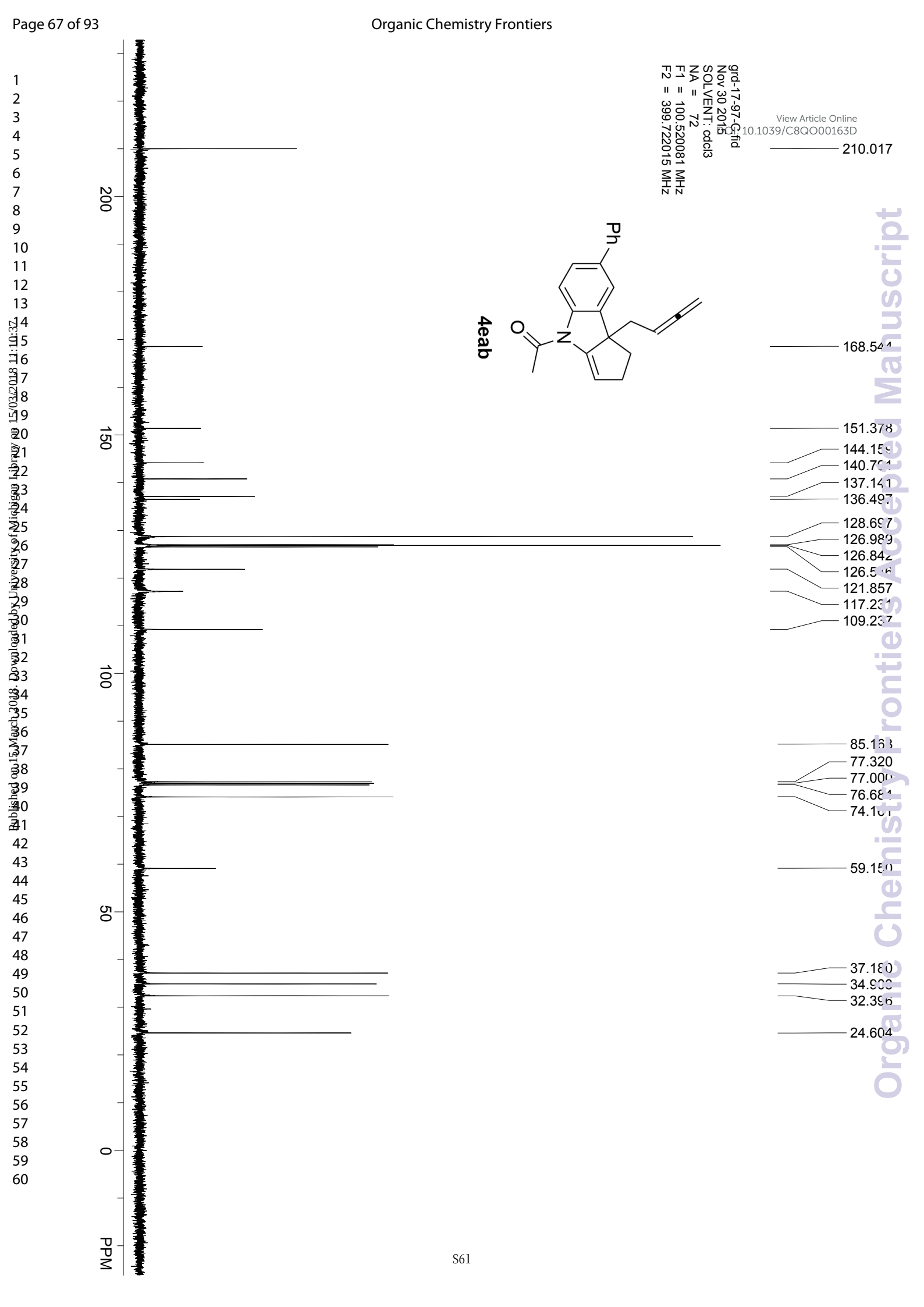
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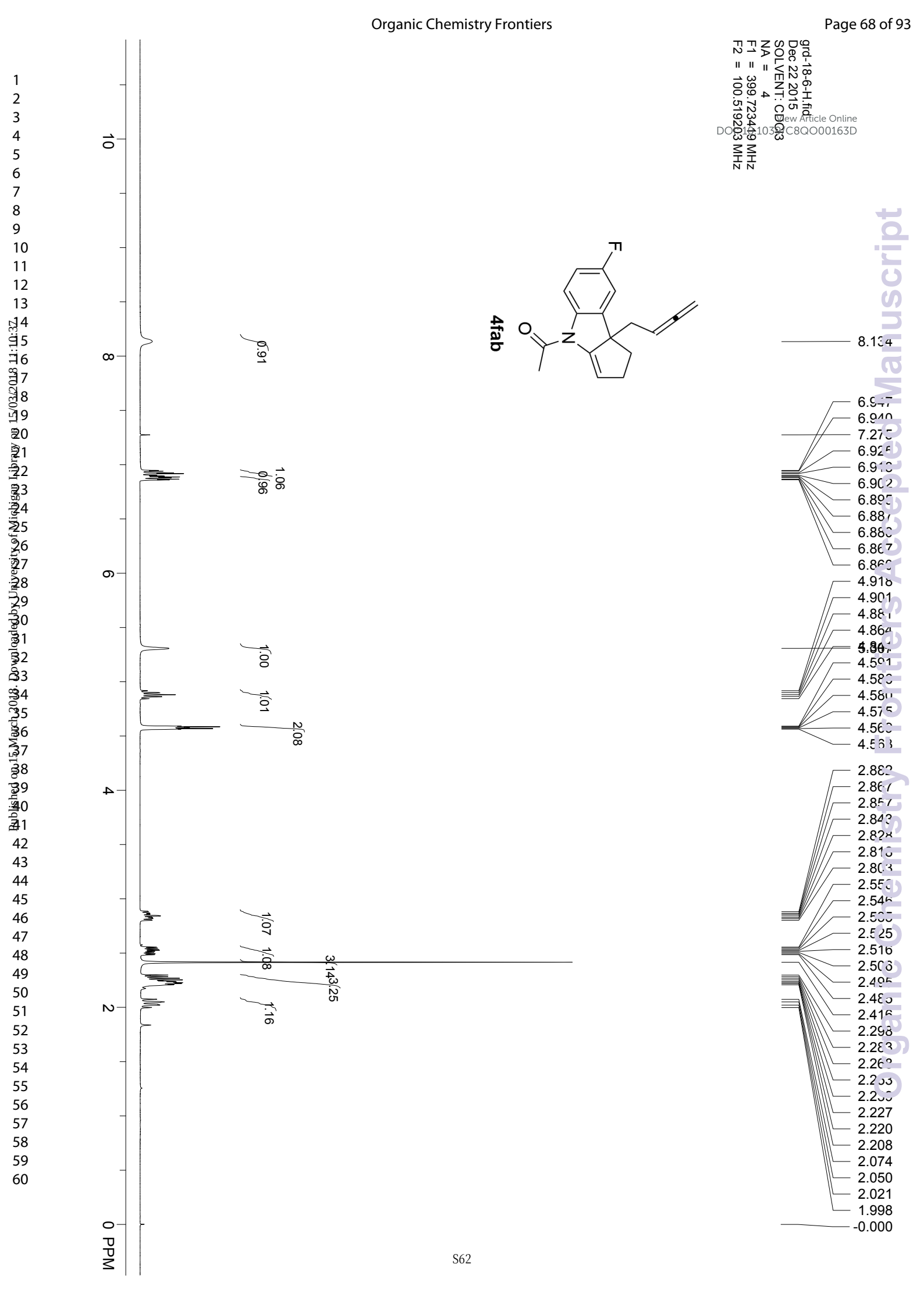
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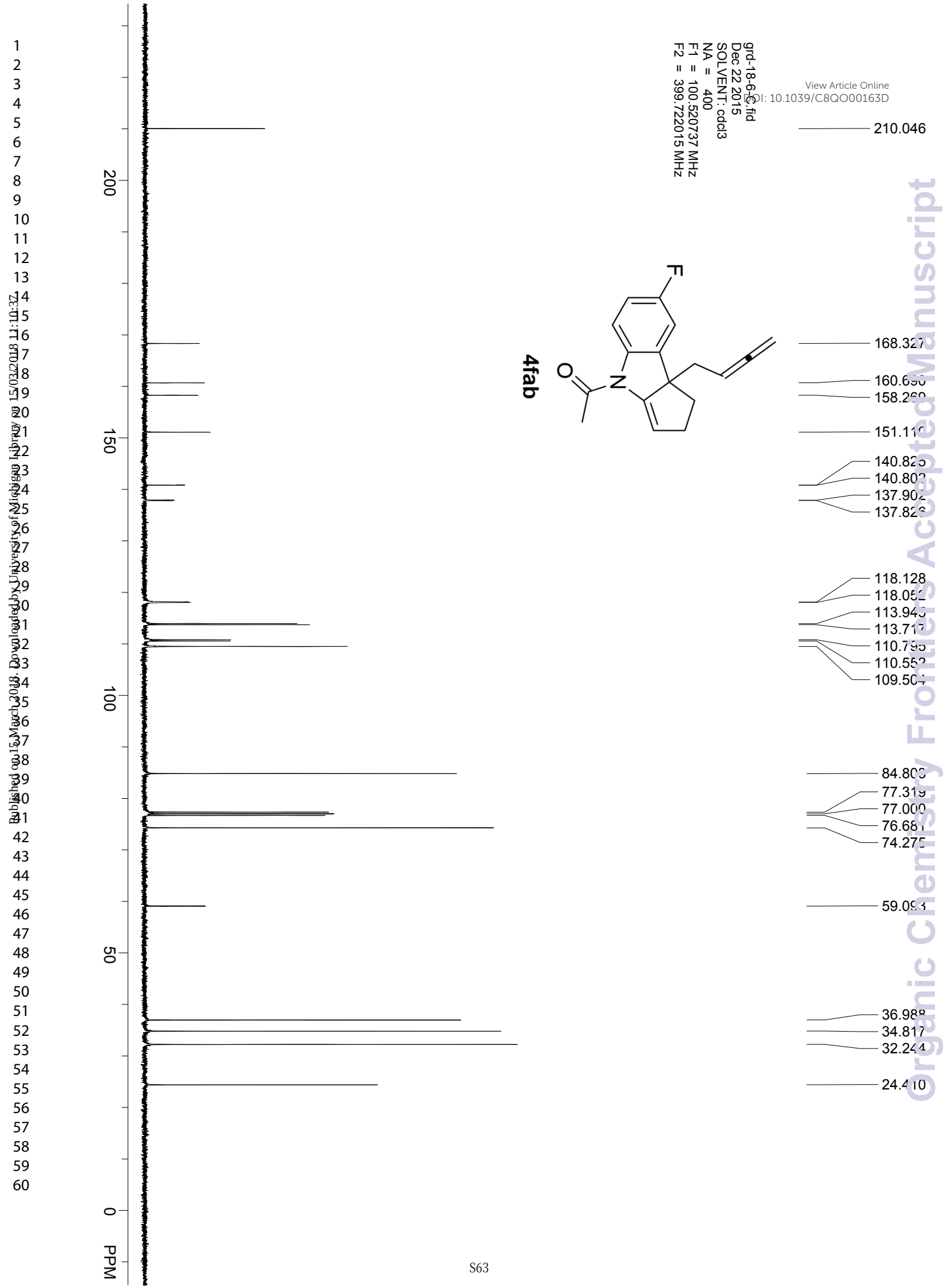
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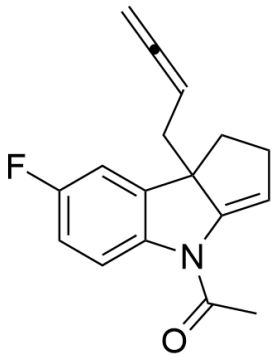


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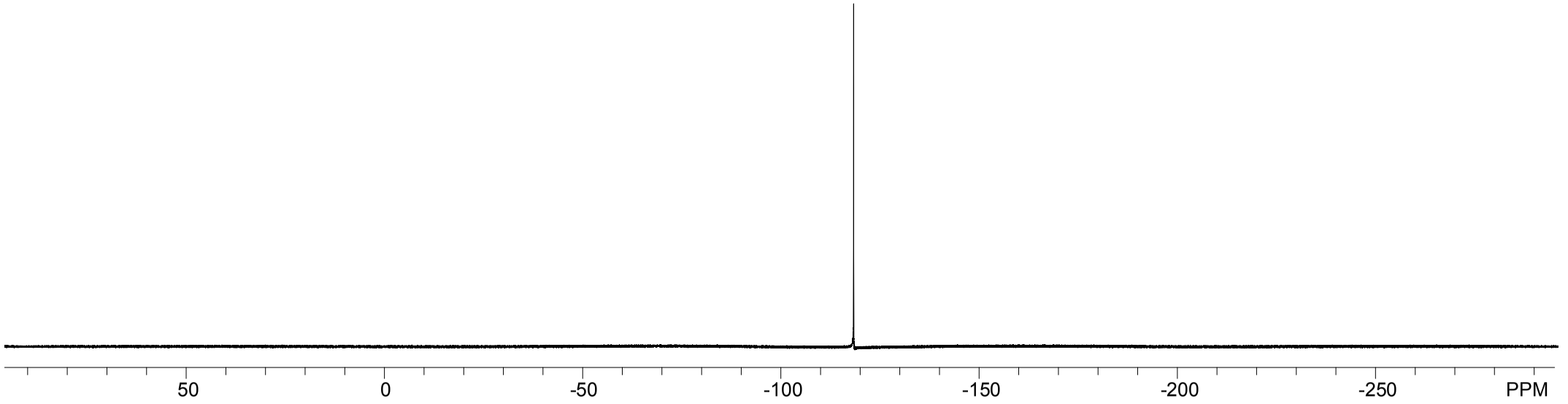
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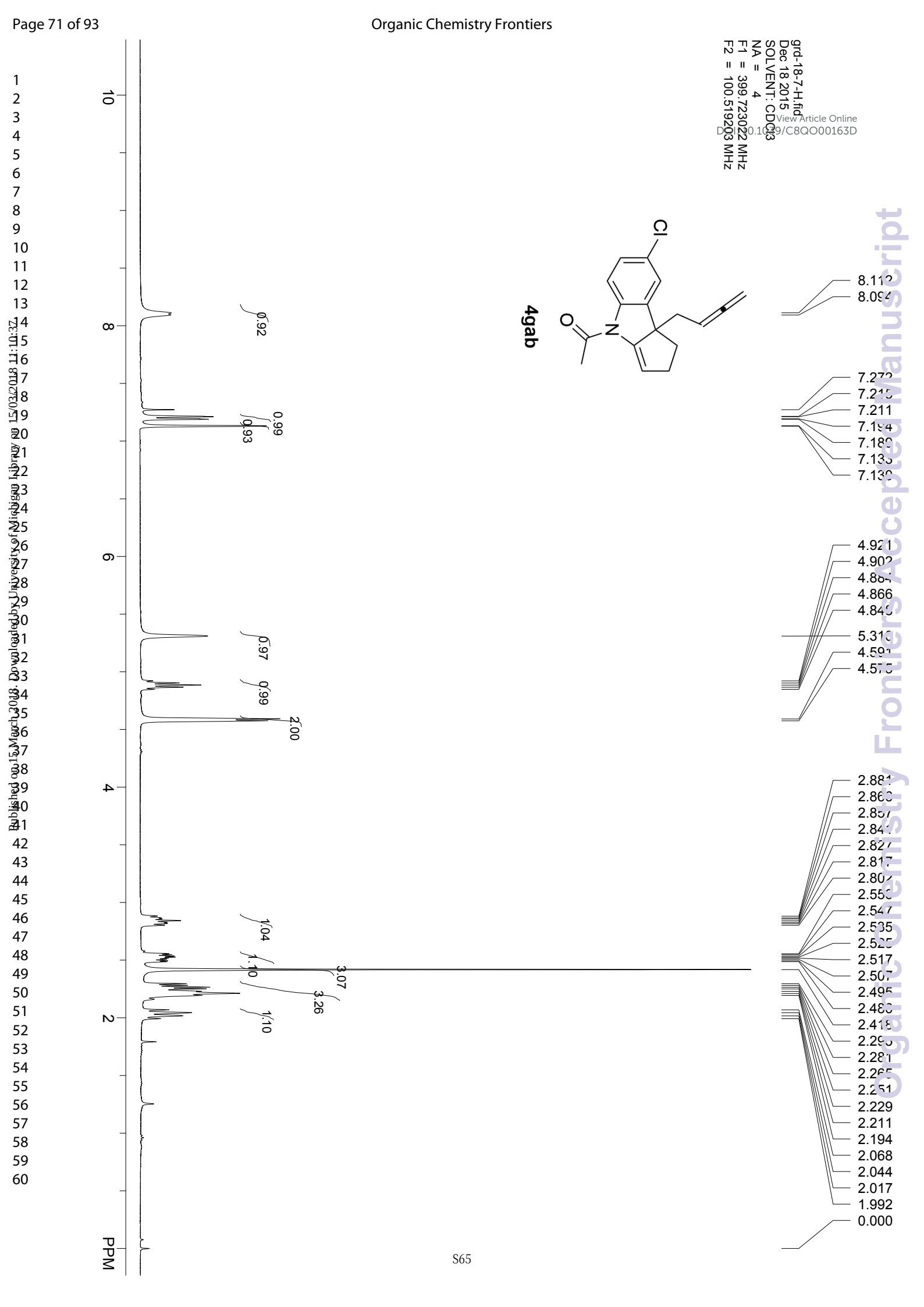
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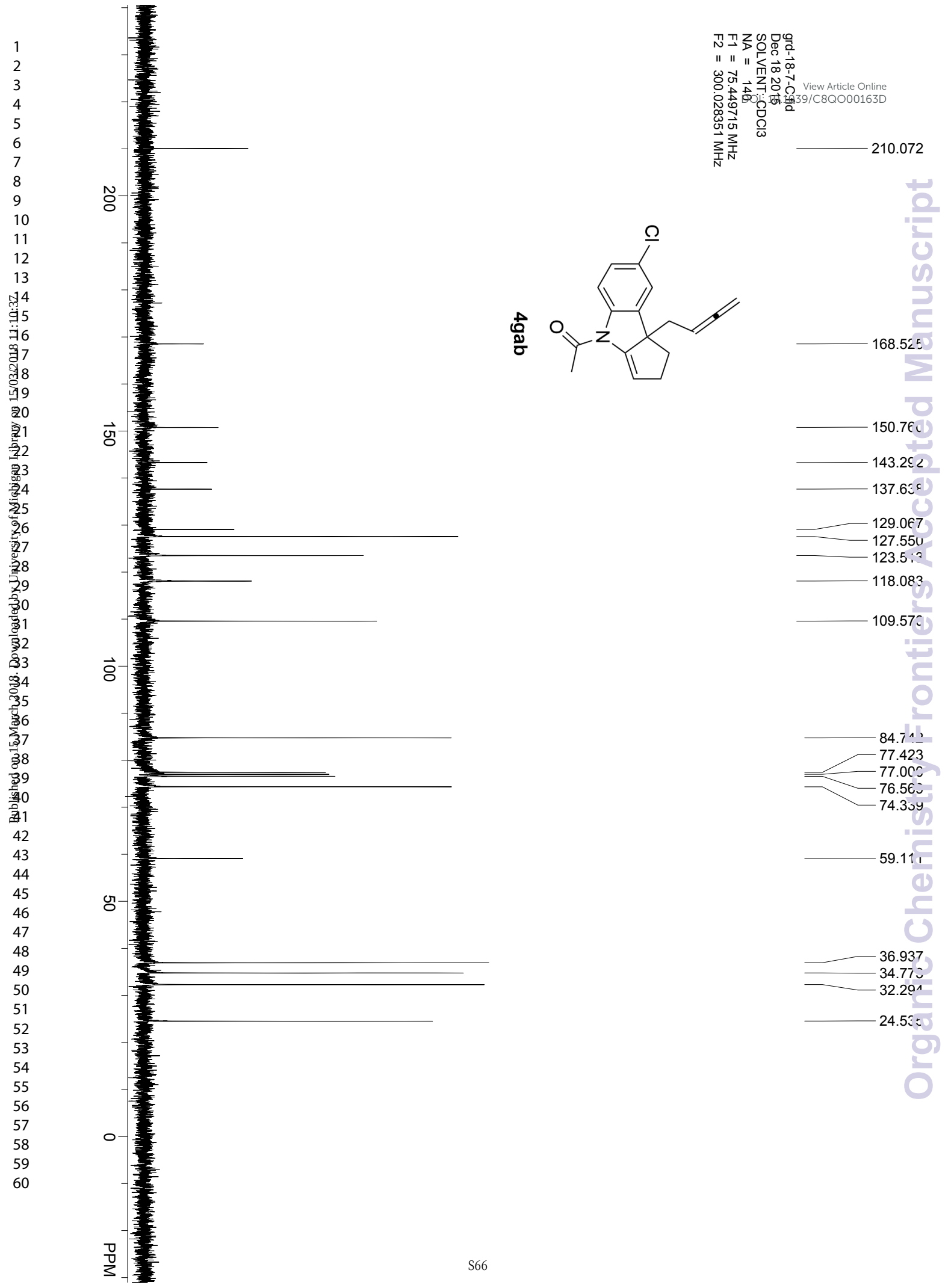
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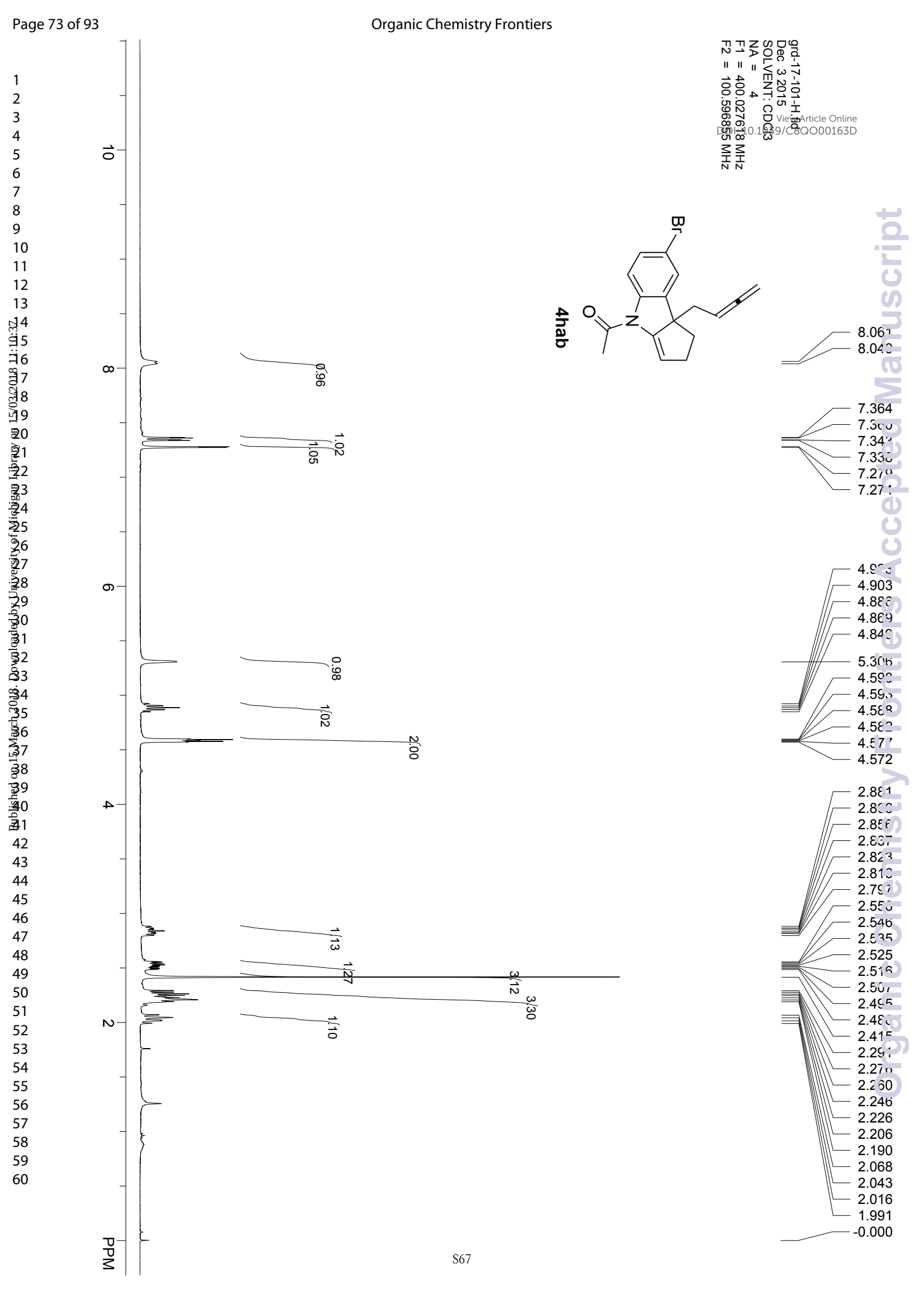
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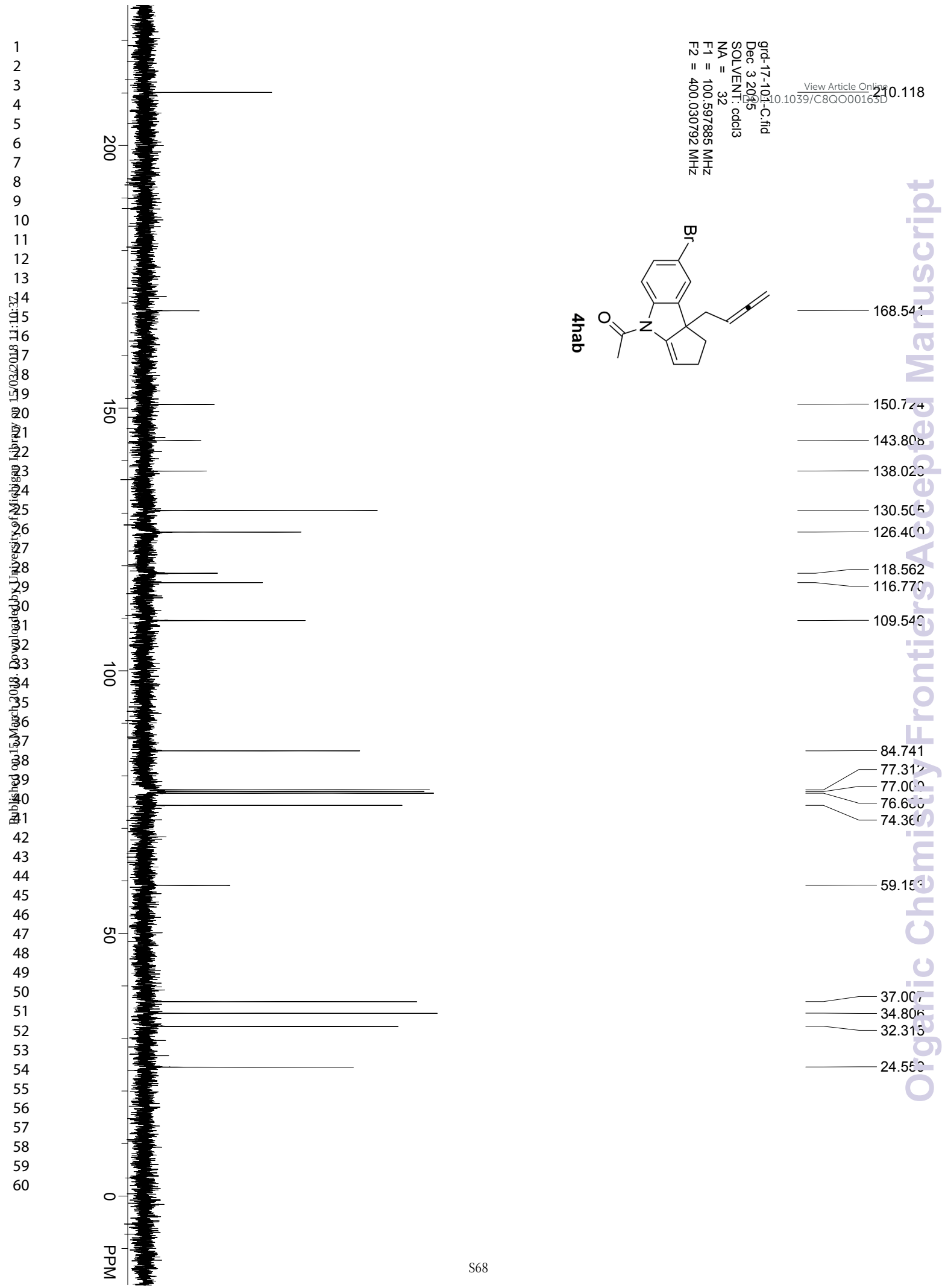
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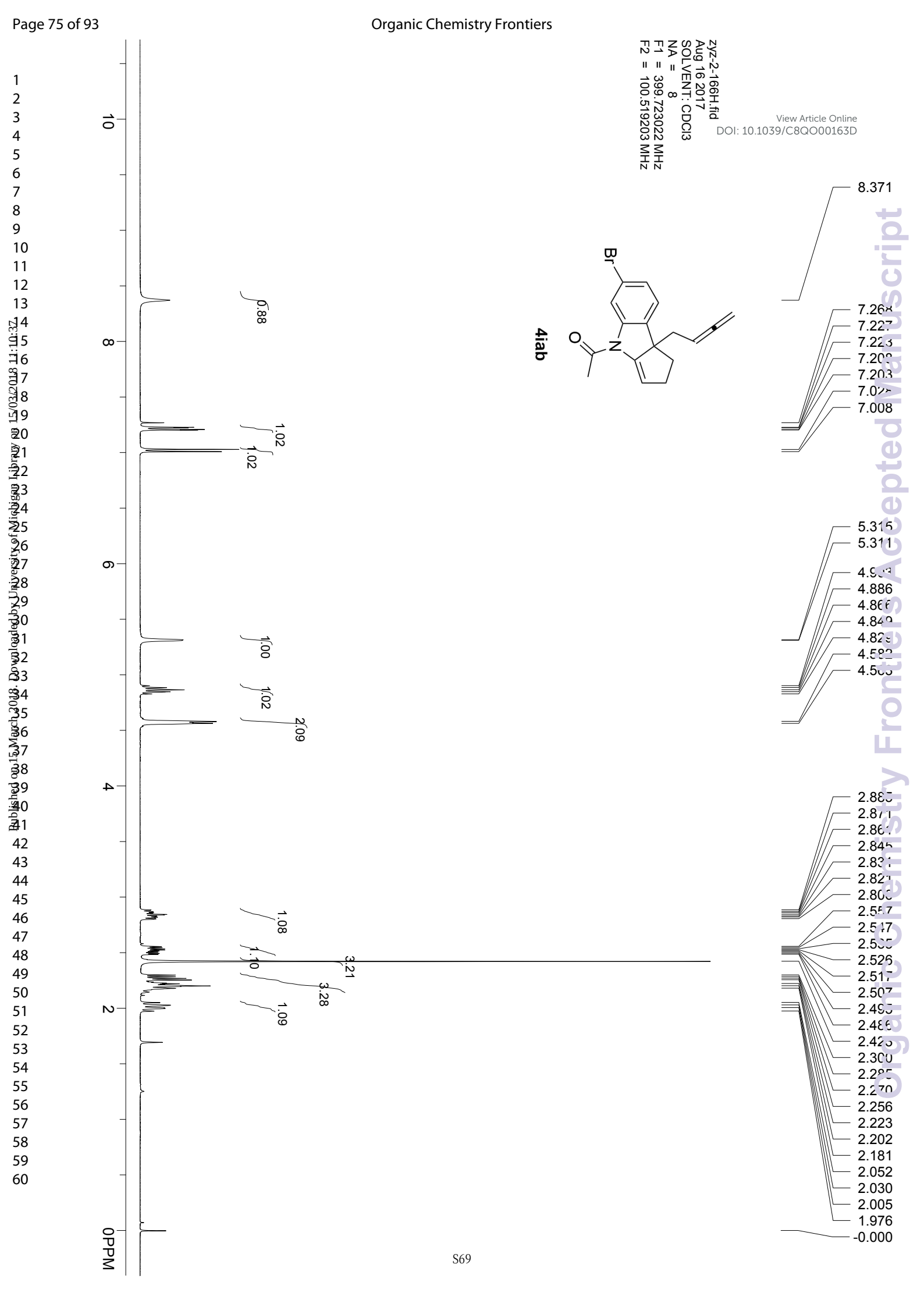


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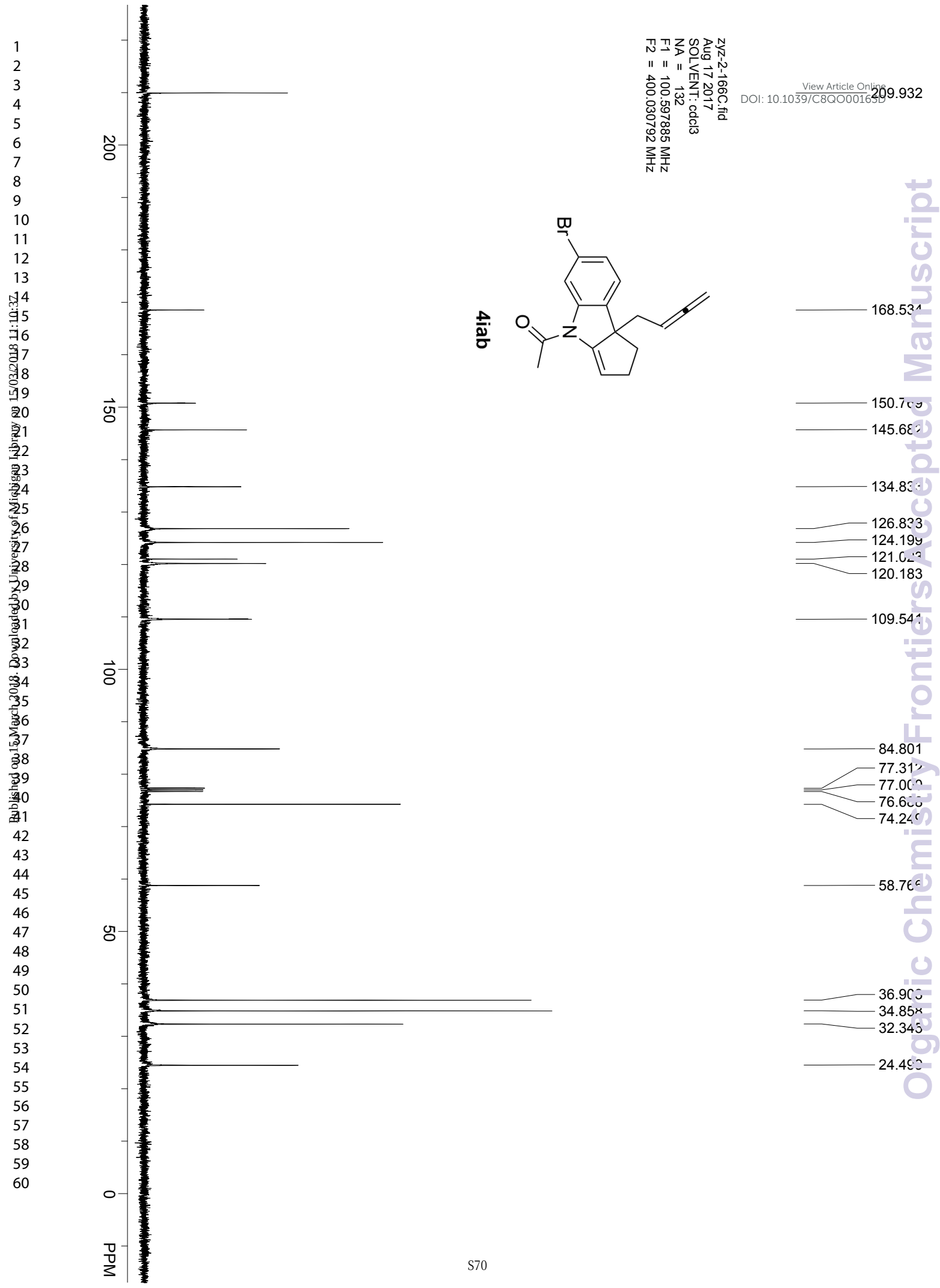
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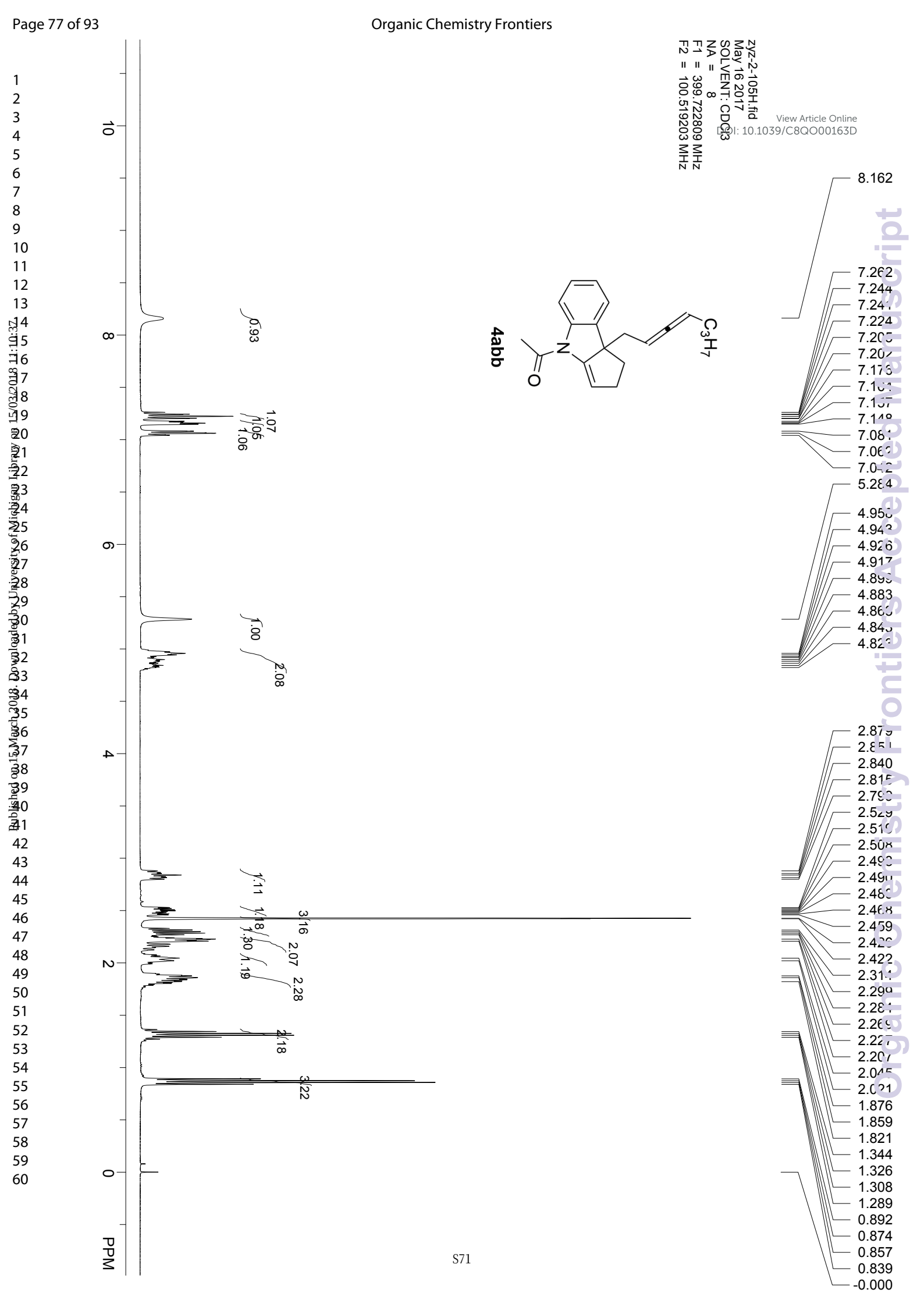
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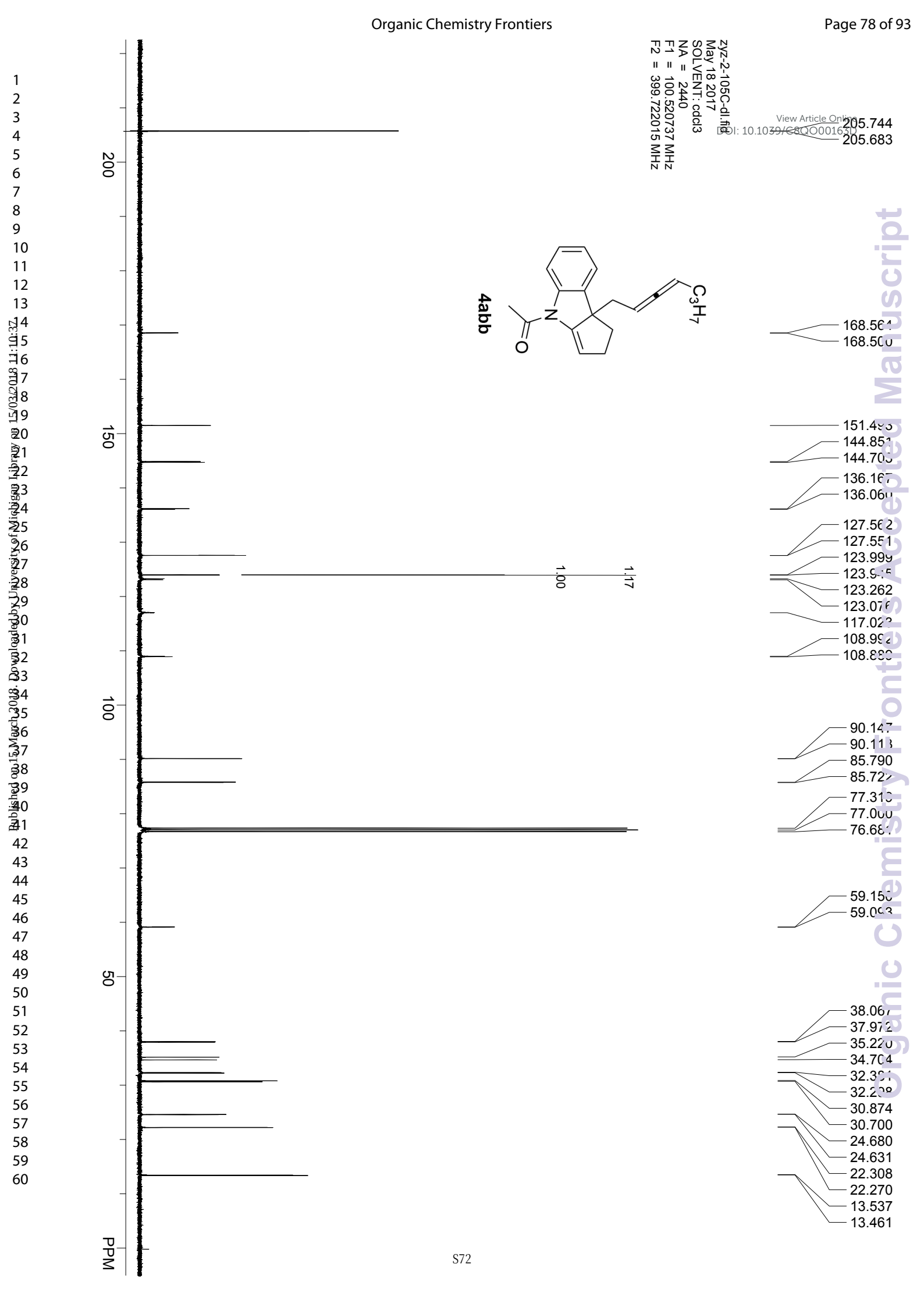
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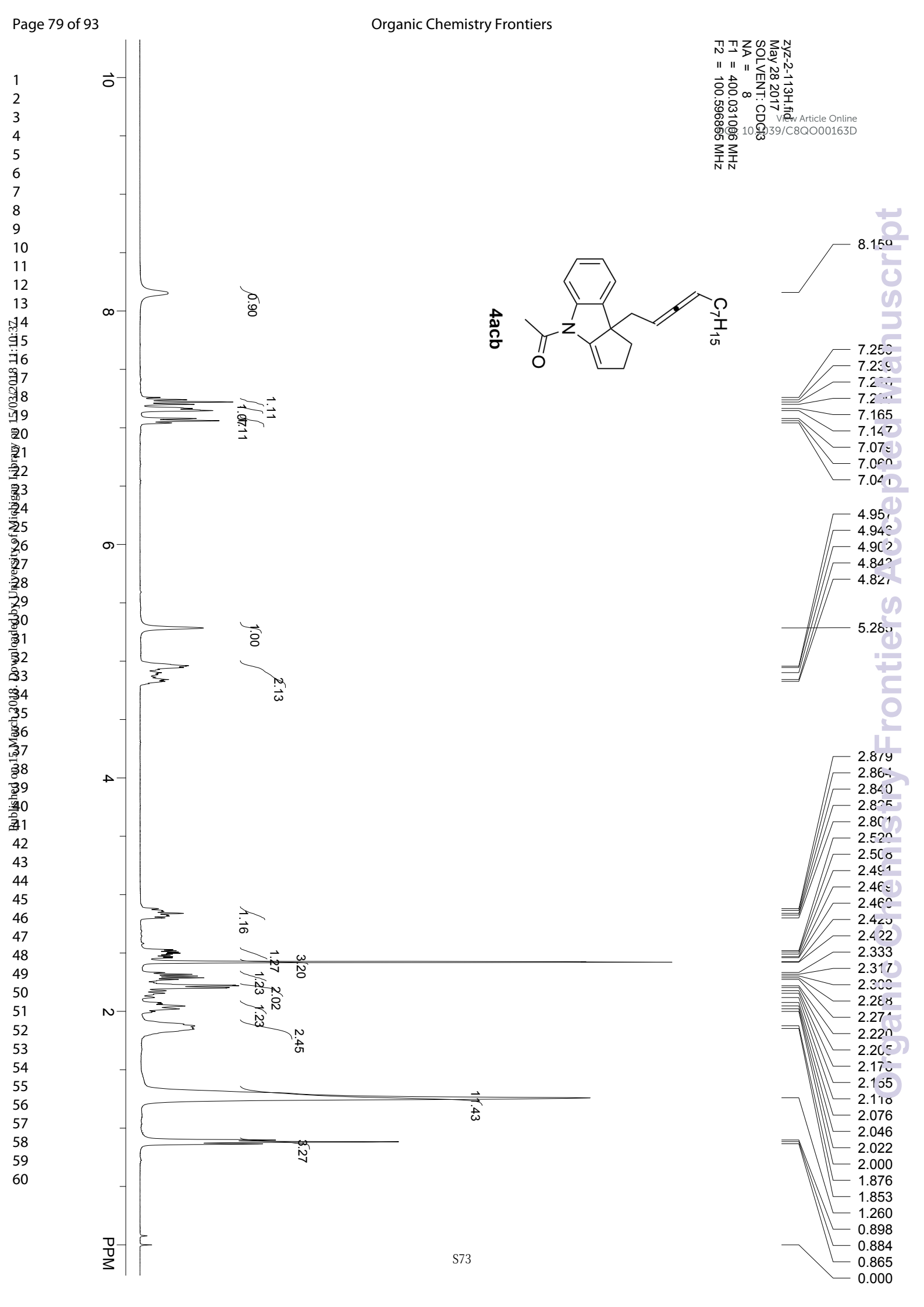
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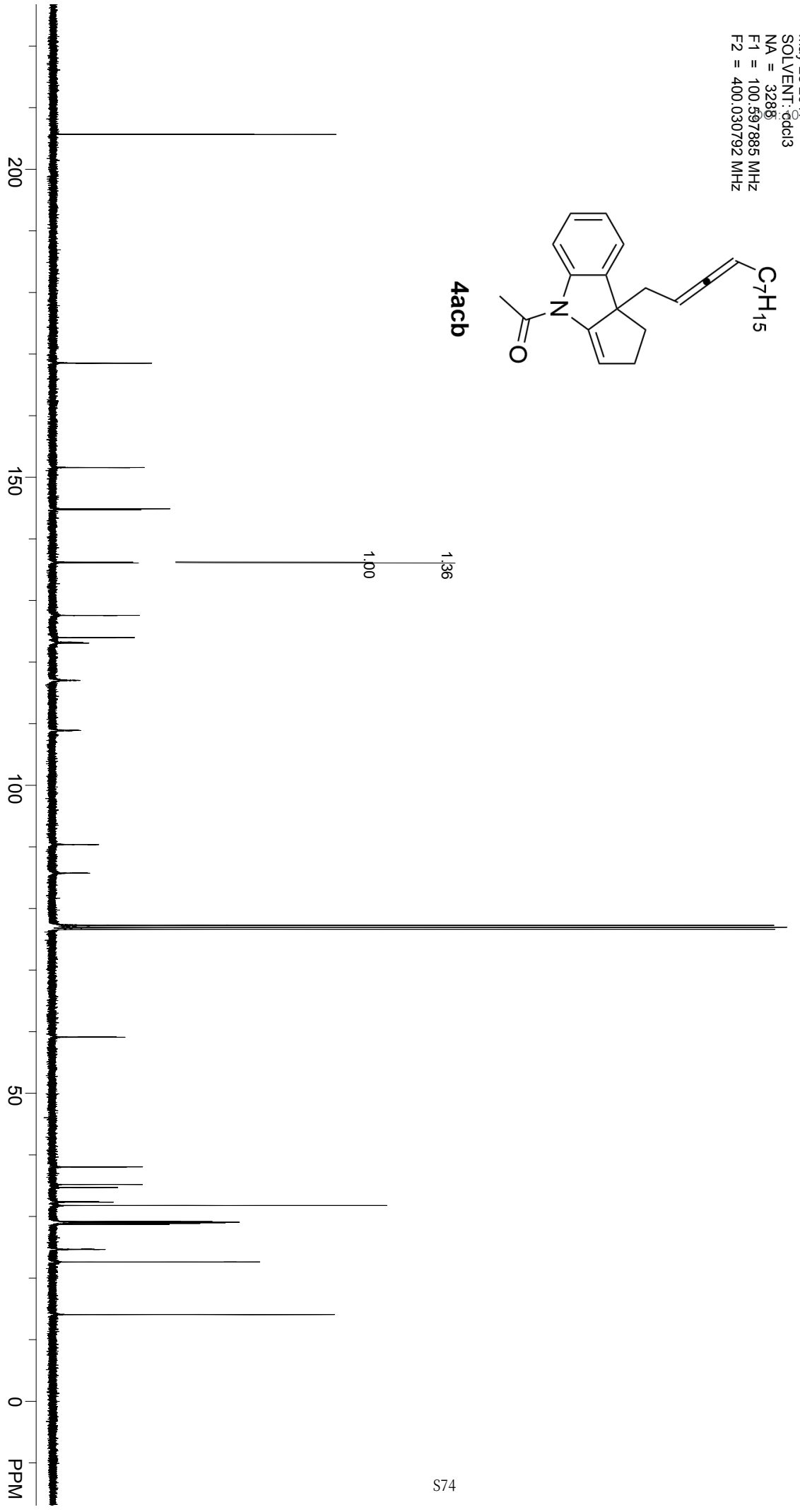
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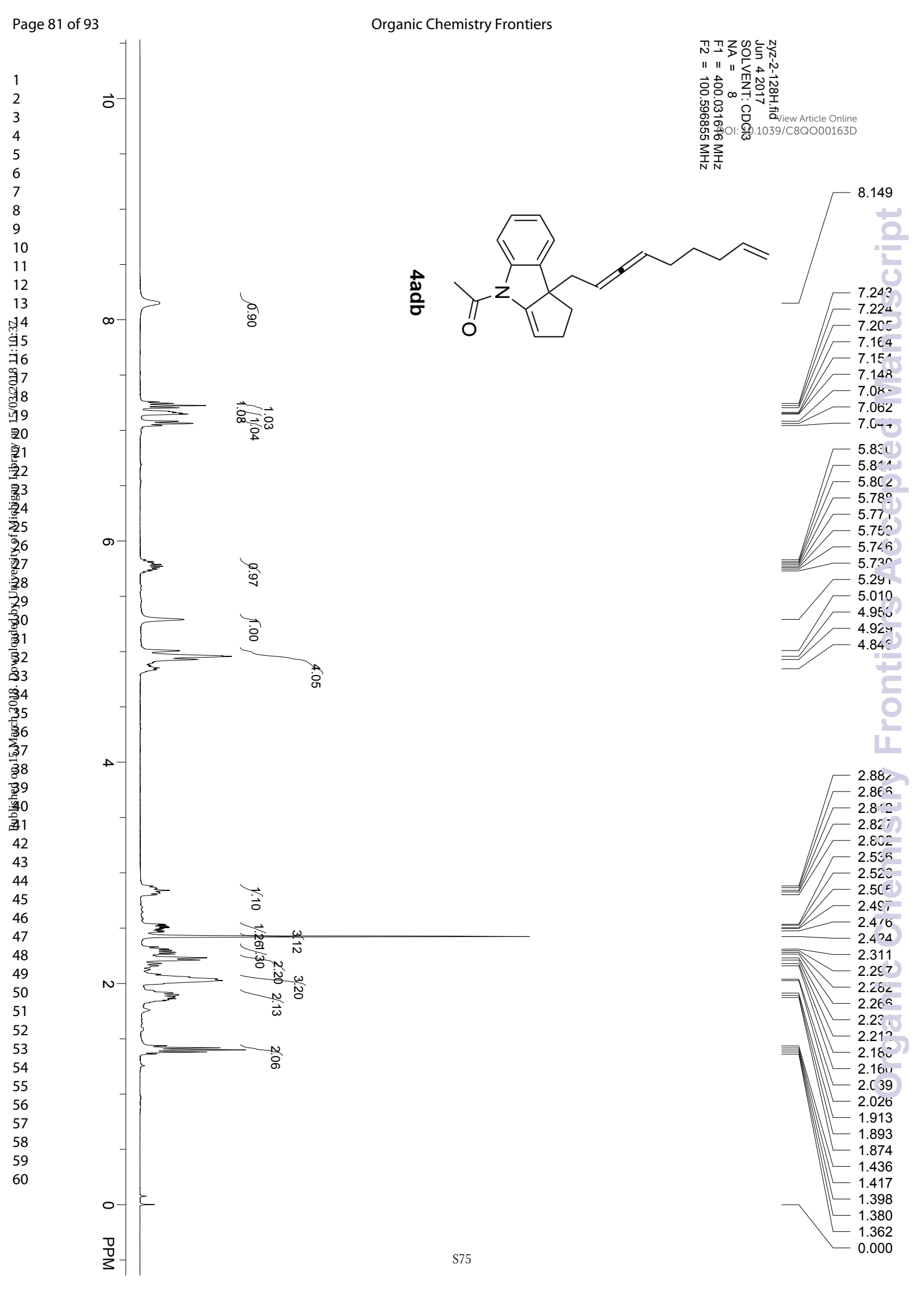
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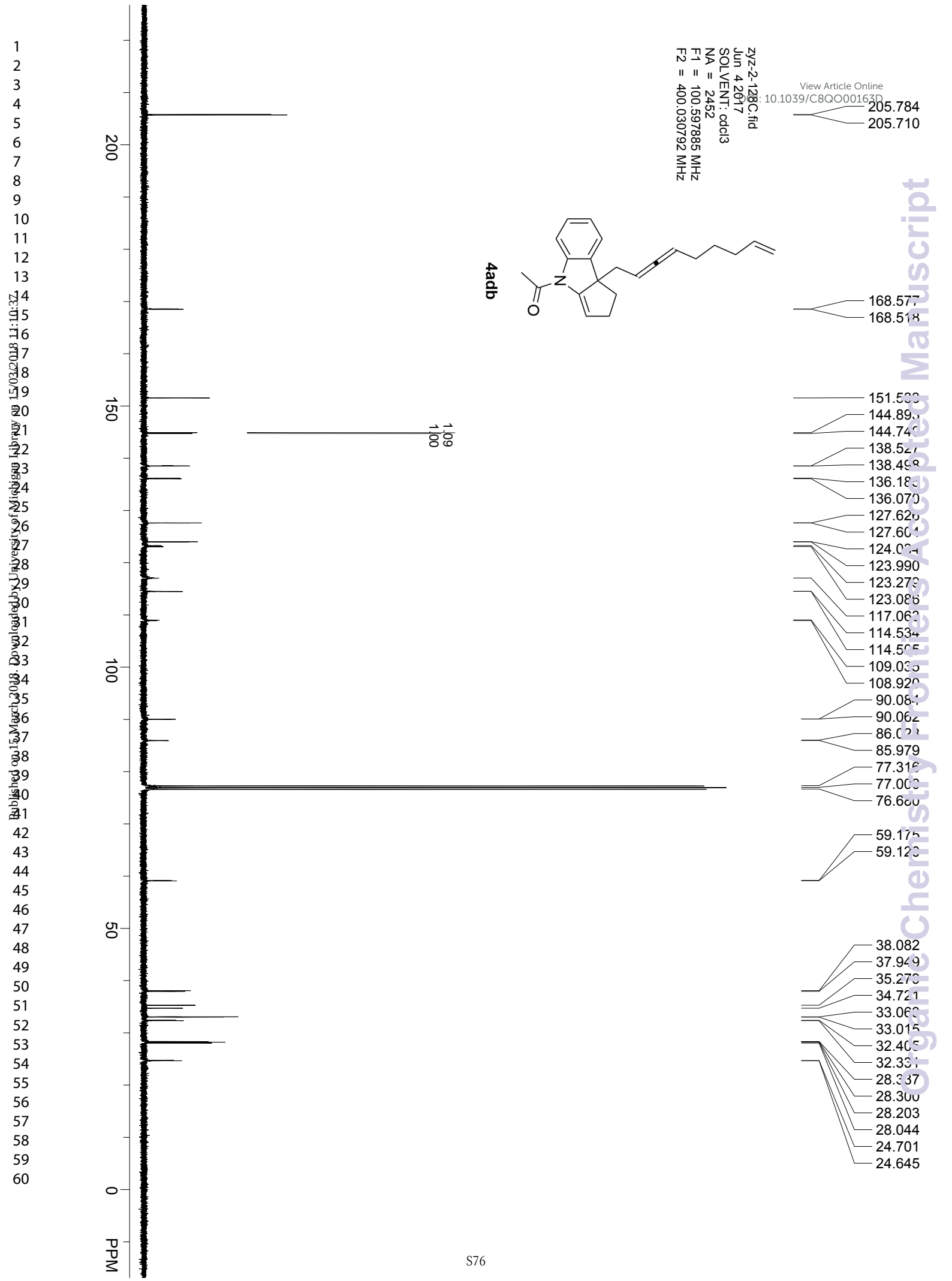
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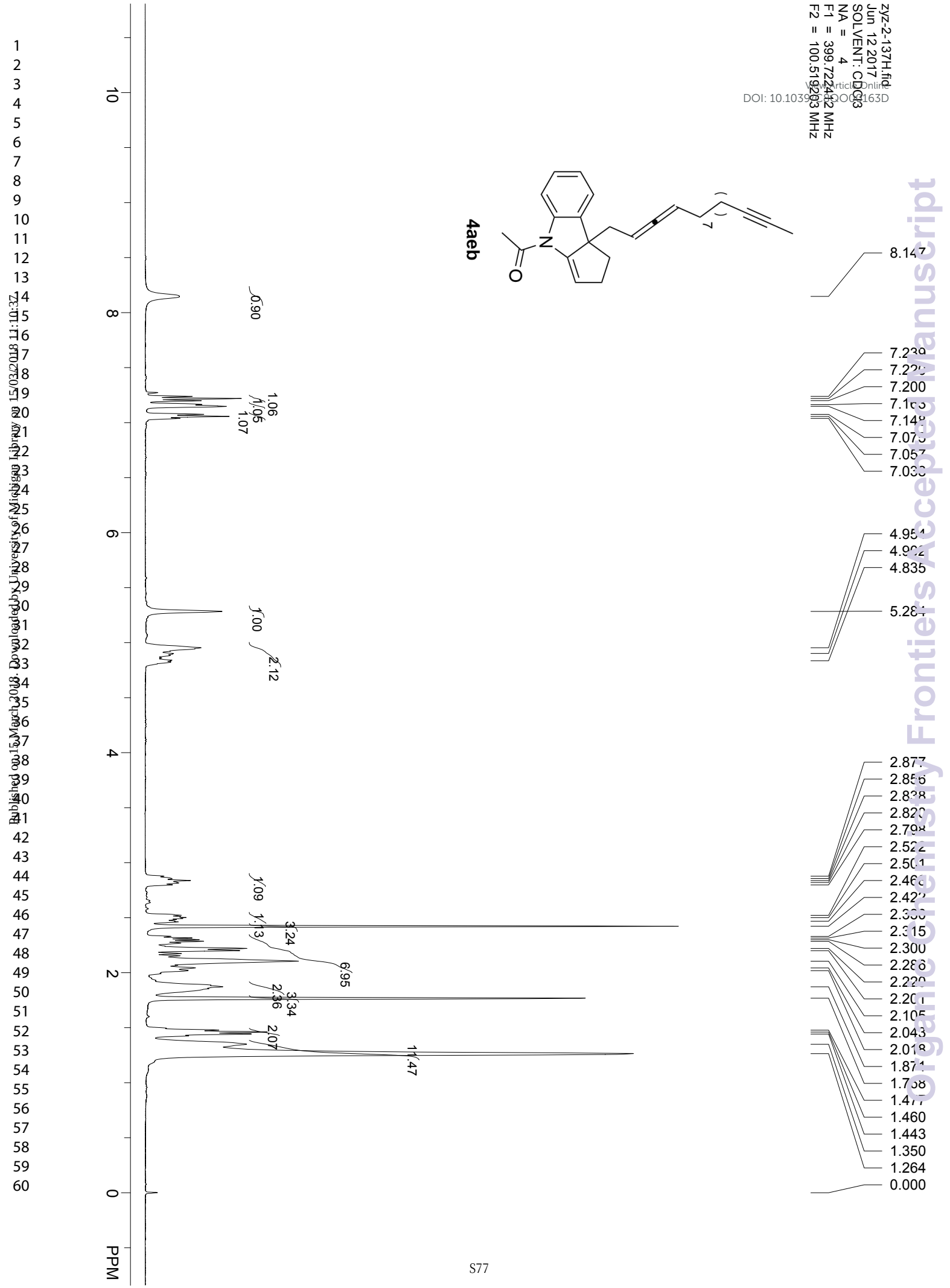
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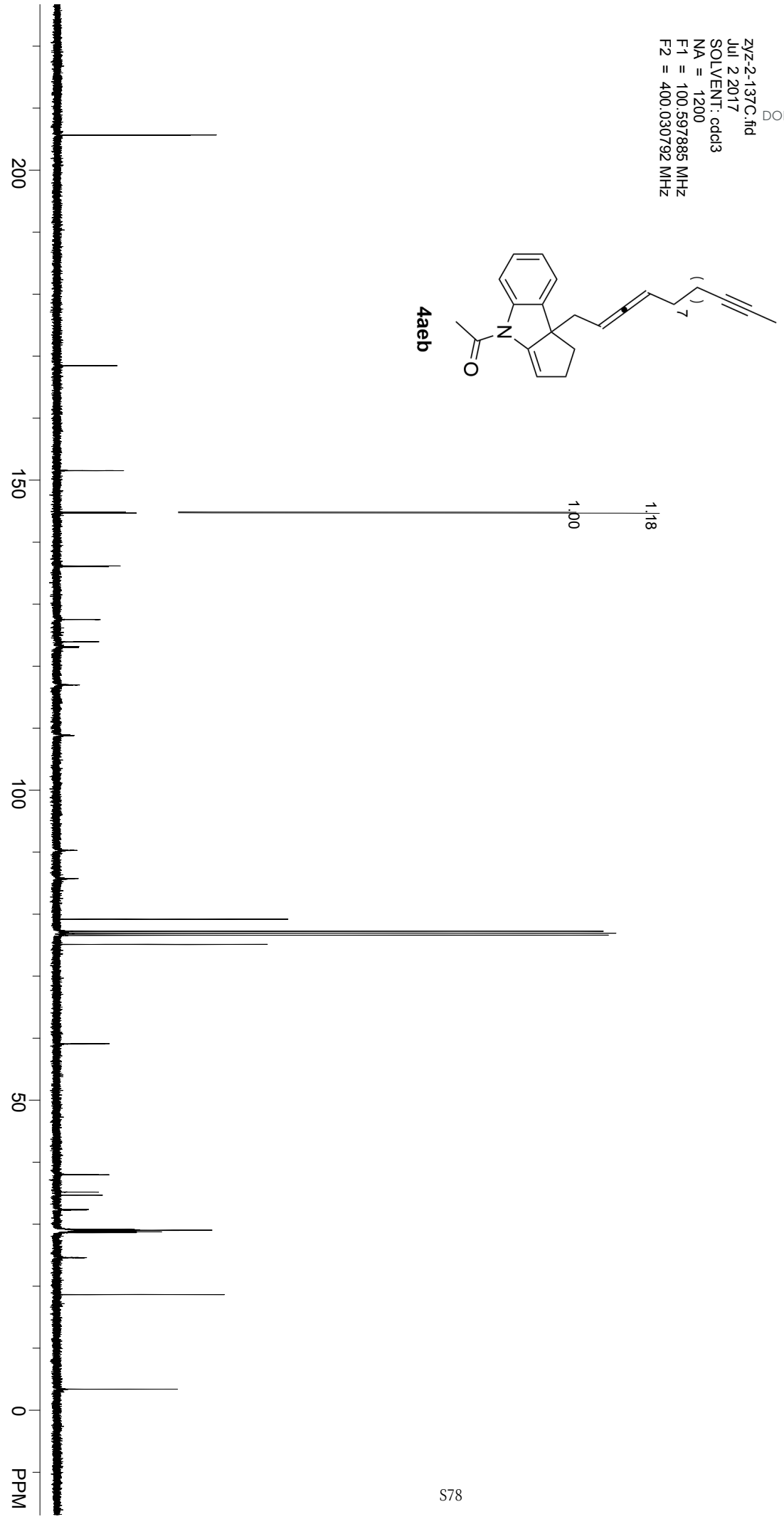


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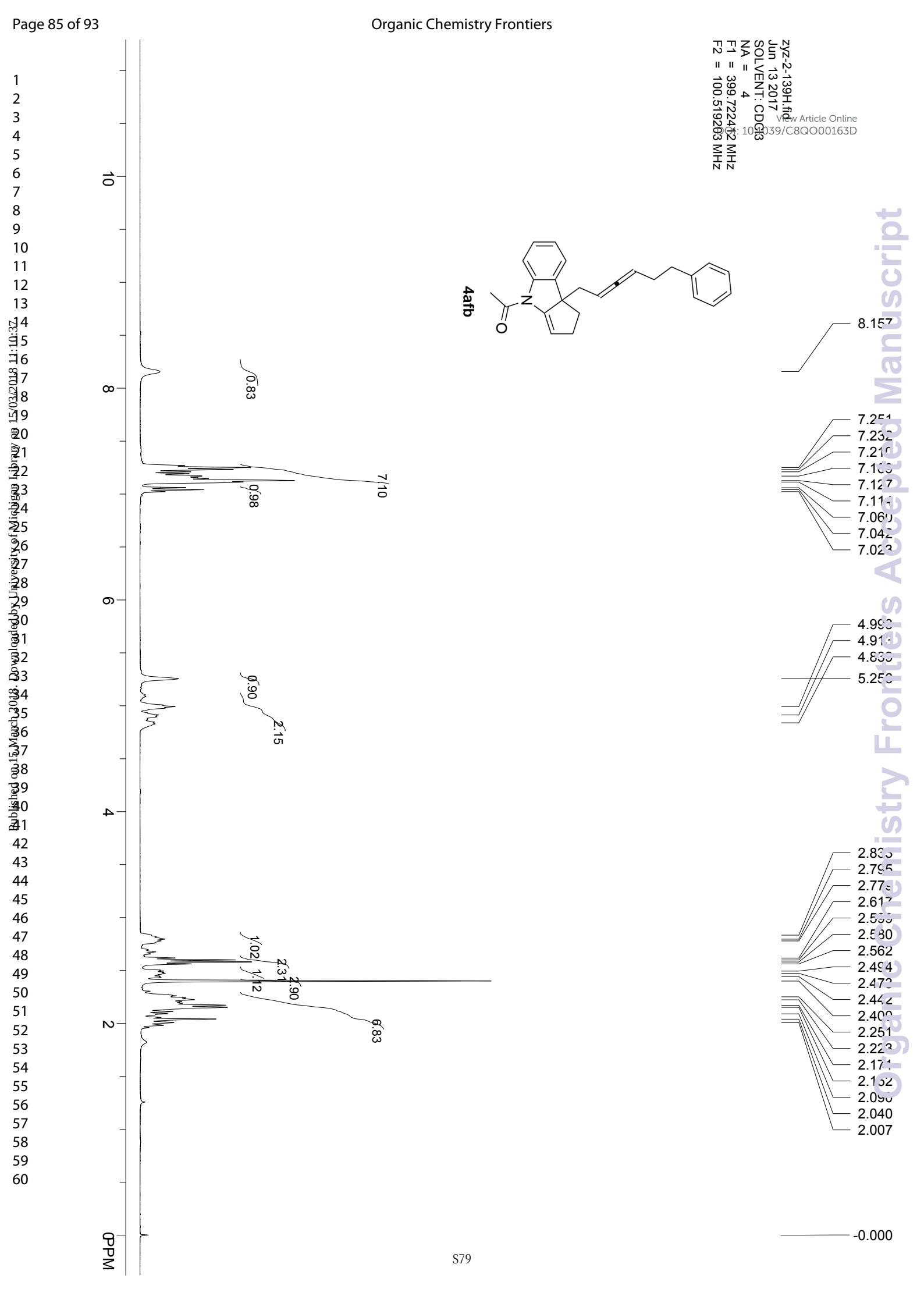
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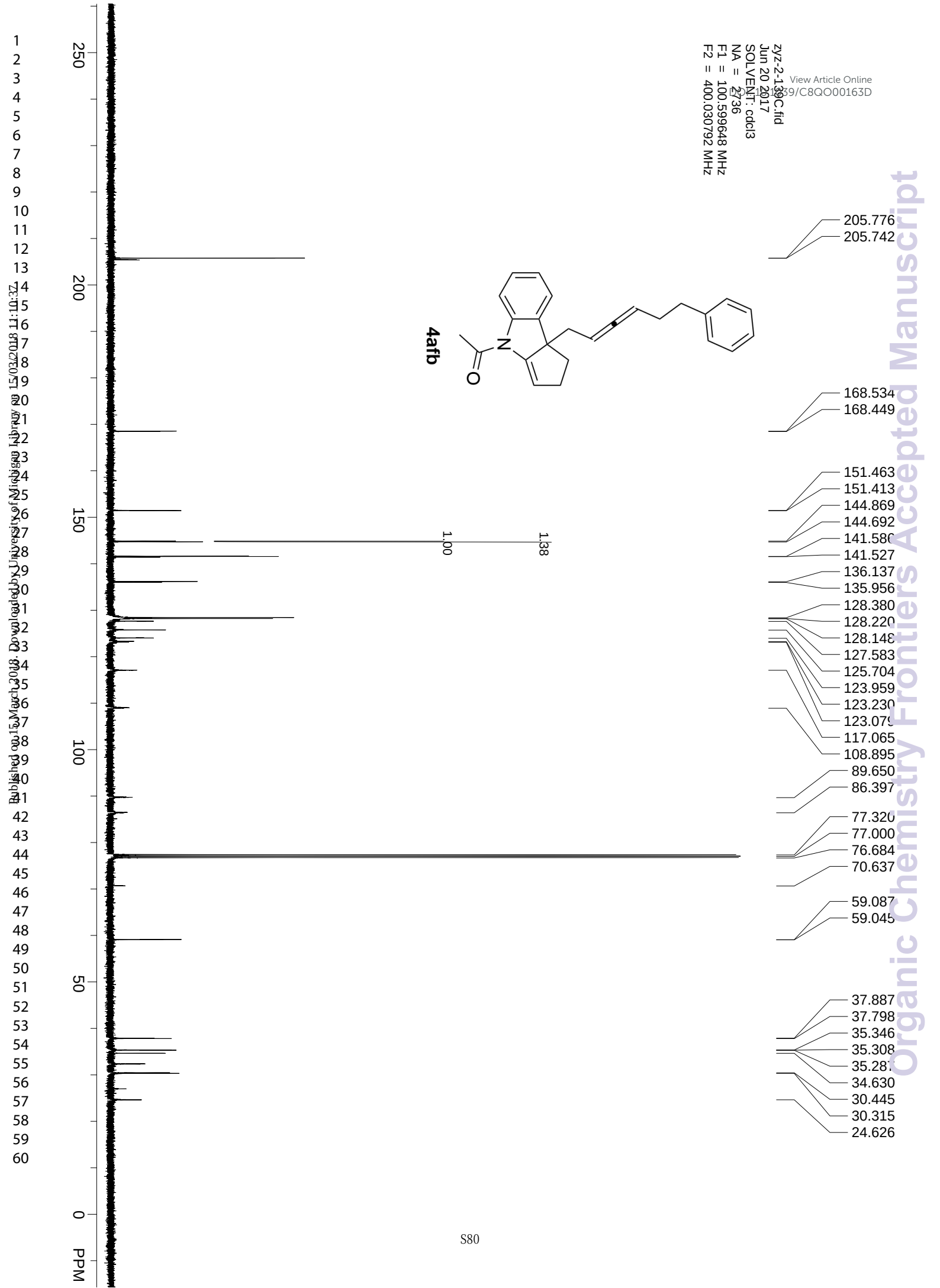
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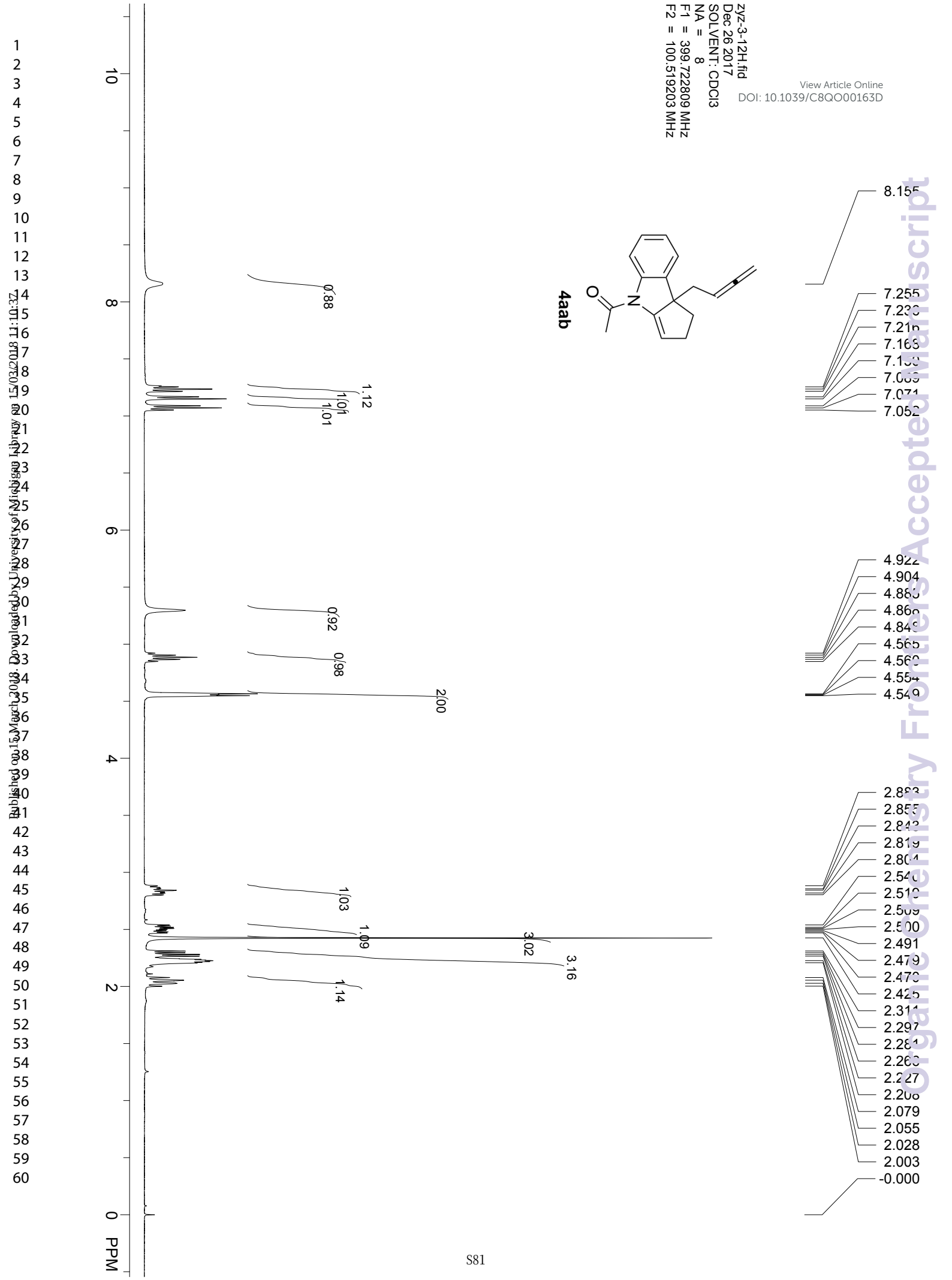


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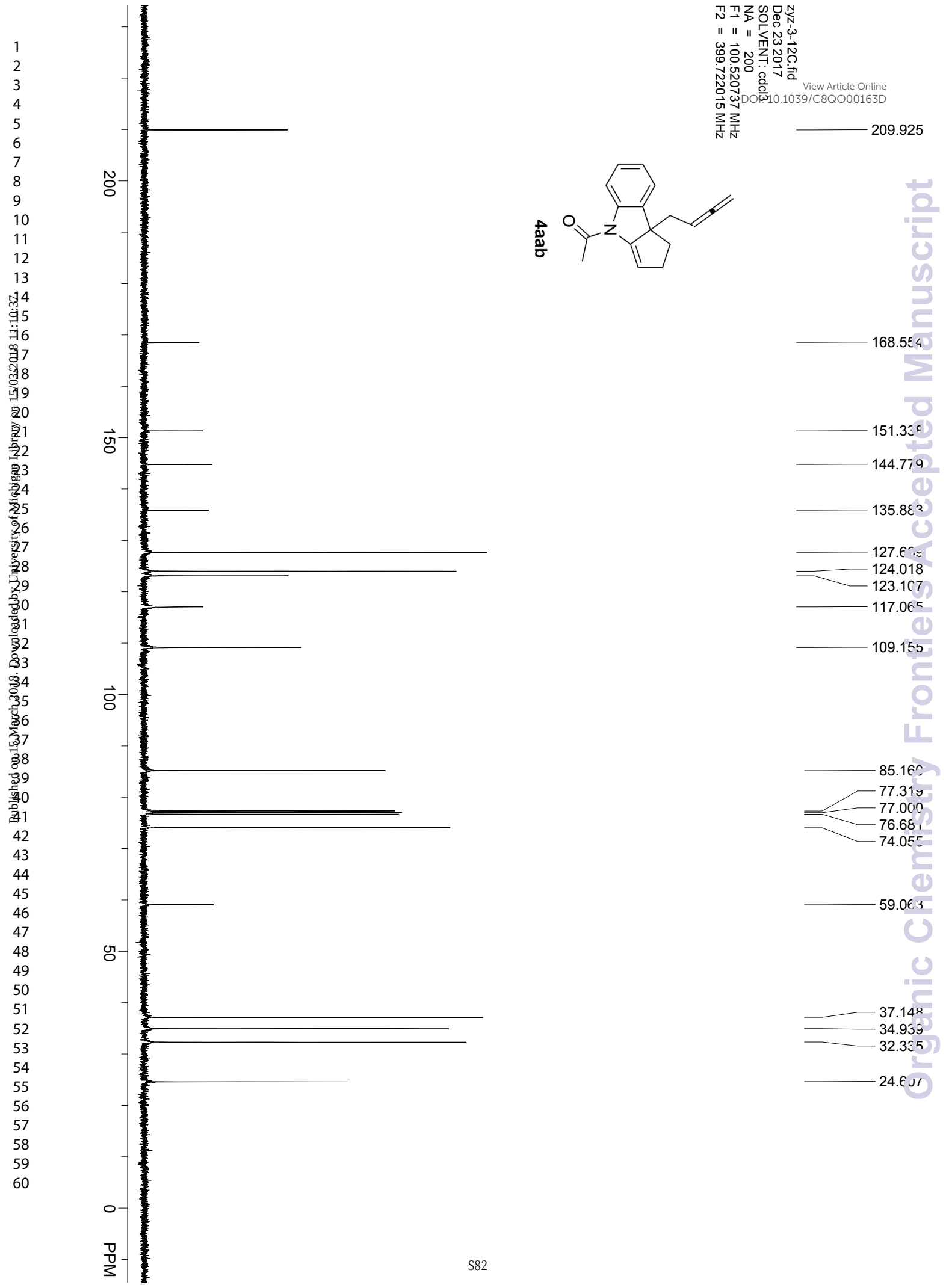
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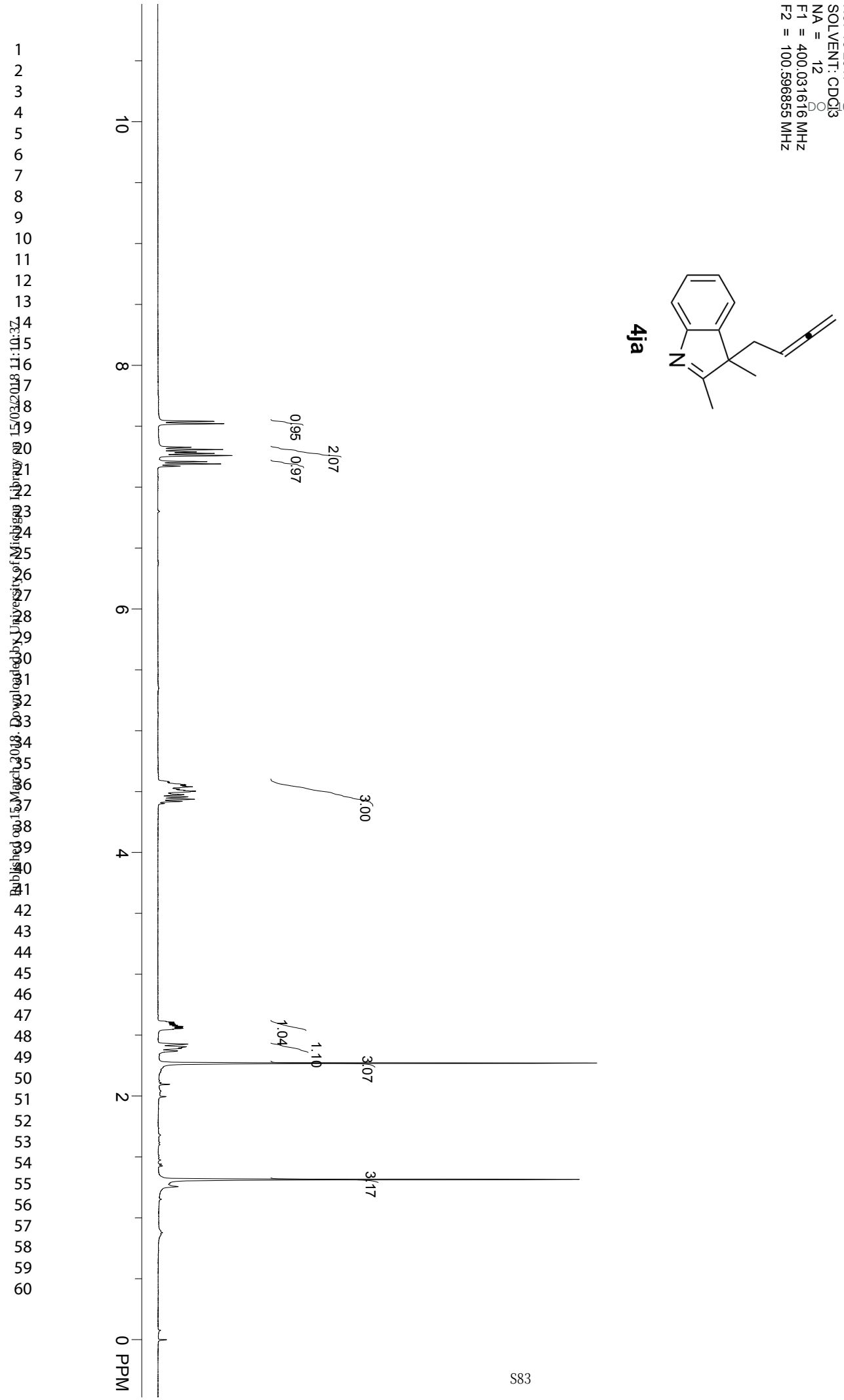


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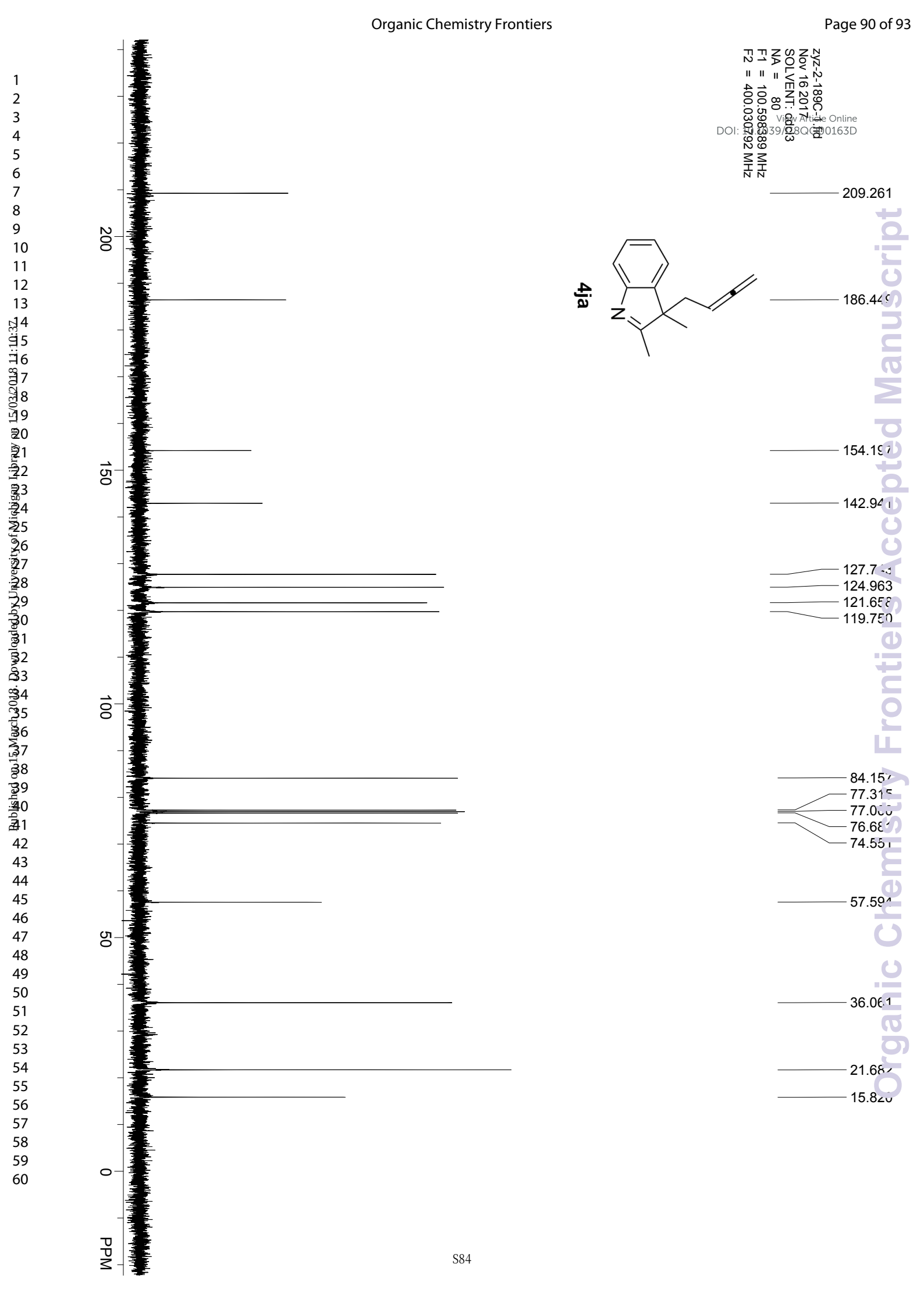
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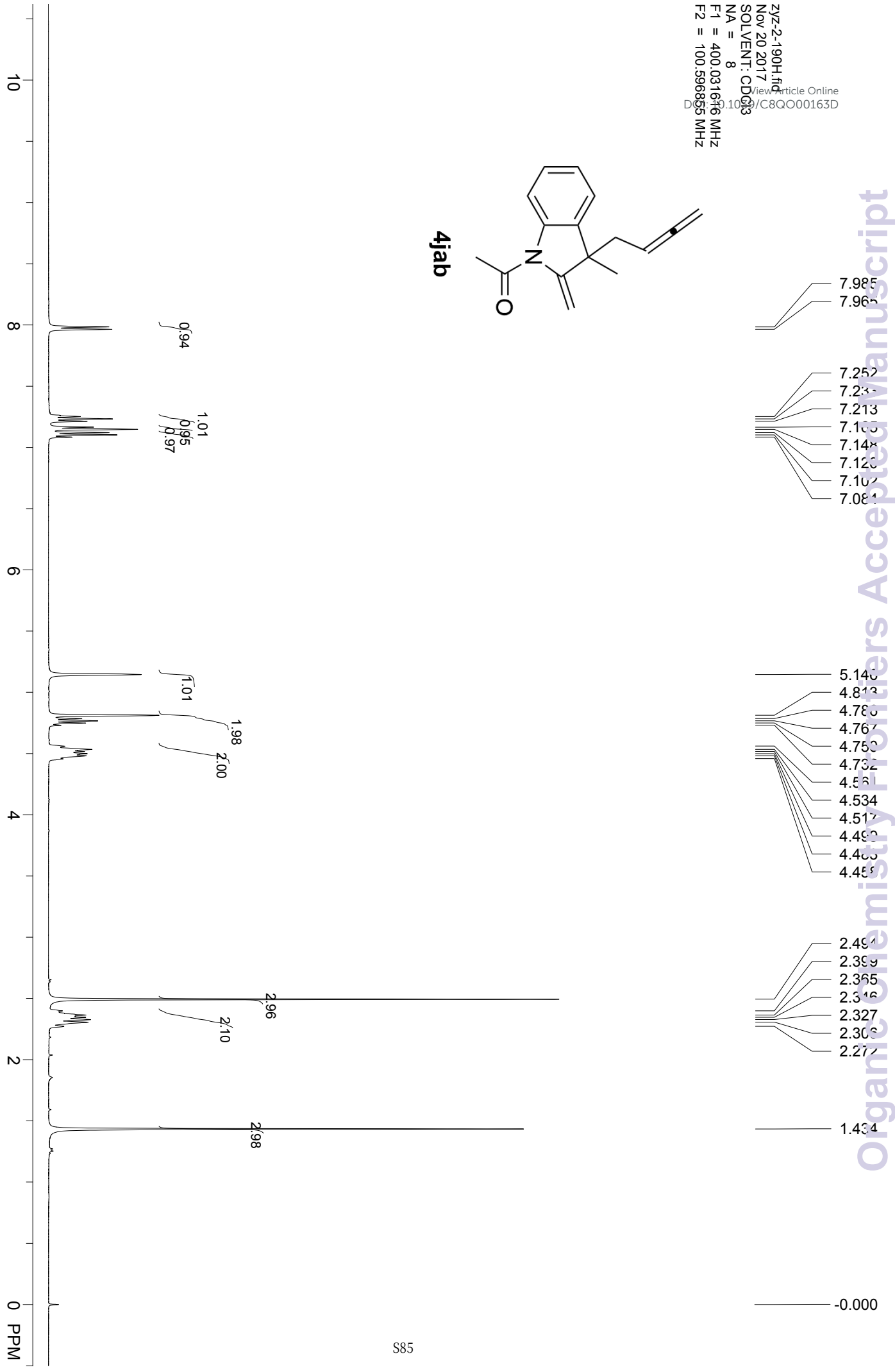
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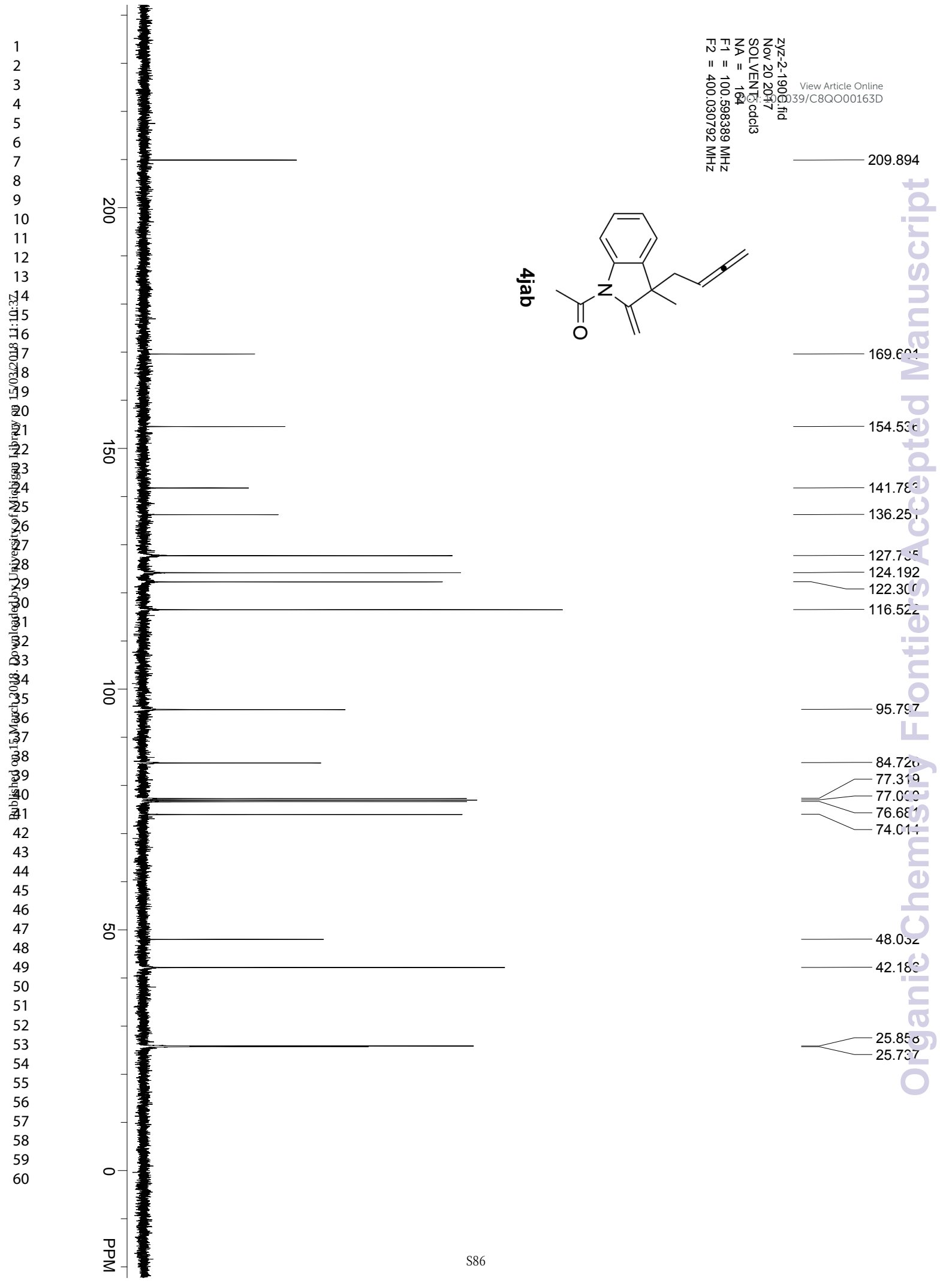
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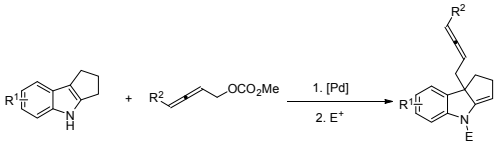


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Graphic Abstract



A palladium-catalyzed allenylation of hydrocycloalk[*b*]indoles with 2,3-allenyl carbonates provides functionalized allenes containing an indole.

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